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DEPARTEMENT OF COMPUTER SCIENCE

GRADUATION PROJECT REPORT

In ordre to obtain :
National Degree of Engineering in :
Computer Science
Option : Software Engineering

Pixel2Sheet Platform: Next-Gen AI-Powered Solution for Flawless 2D Tables Extraction and Data Analysis with an Advanced Chatbot Assistant

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Defended on **07/04/2025** **before the jury :**

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Dedication

I dedicate this work,

To my dear parents,

Thank you for your endless love, support, and encouragement. You have always been there for me, guiding me through every obstacle and cheering me on every step of the way. Your sacrifices have shaped me into the person I am today. I owe everything to you, and no words can ever truly express my gratitude.

To my brothers,

You have been my companions through thick and thin, my best friends and my greatest support. From childhood memories to the present day, we've shared laughs, challenges, and success. I am grateful for your constant belief in me and for always being there to share in my triumphs.

To my friends,

You've been the ones who have made the journey so much more enjoyable. Thank you for the countless moments of joy, laughter, and support. I will forever cherish the memories we've made and look forward to the many more that await us. Your friendship has been a source of strength, and I am deeply thankful for each one of you.

To all those who participated in the development of this work,

Thank you for your guidance, assistance, and support throughout this journey.

To all those I love, all those who love me, and those who are dear to me,

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Amna khemiri

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Finally, I would like to express my most sincere thanks to the members of the jury for having honored us by agreeing to evaluate this work. I hope that they will find the clarity and motivation they expect.

Glossary of Acronyms

AI Artificial Intelligence

CNN Convolutional Neural Network

CRISP-DM Cross-Industry Standard Process for Data Mining

CRNN Convolutional Recurrent Neural Network

CTC Connectionist Temporal Classification

CPU Central Processing Unit

DL Deep Learning

FP False Positive

FN False Negative

GPU Graphics Processing Unit

IoU Intersection over Union

LLM Large Language Model

ML Machine Learning

NMS Non-Maximum Suppression

OCR Optical Character Recognition

TP True Positive

TN True Negative

VLM Vision Language Model

YOLO You Only Look Once

Abstract

This project aims to develop an advanced platform that automates the extraction and structuring of tabular data from unstructured sources such as images and PDFs, which are commonly found in the trading industry. By leveraging advanced Deep Learning technologies such as Optical Character Recognition (OCR), object detection methods , AI generative models, and Vision-Language Models (VLMs), the system transforms complex data into structured, machine readable formats like CSV. This solution addresses the inefficiencies of manual data extraction, enabling Data Analysts to make faster, more accurate decisions in a time critical trading environment. The proposed platform, named Pixel2Sheet Pro, not only automates the extraction process but also allows for natural language querying through our advanced chatbot, data manipulation, and provides comprehensive reporting including visualizations, summaries, and actionable insights, significantly enhancing the decision-making capabilities for Data Analysts.

keywords: AI-powered chatbot, Deep Learning , data extraction, OCR, data recognition, tables detection, tabular data, Generative AI, Vision-Language Models, CSV, reporting, analysis ,decision-making , market data.

Résumé

Ce projet vise à développer une plateforme avancé qui automatise l'extraction et la structuration de données tabulaires à partir de sources non structurées telles que des images et des fichiers PDF, couramment rencontrées dans l'industrie du trading. En utilisant des technologies avancées d'apprentissage profond telles que la reconnaissance optique de caractères (OCR), les méthodes de détection d'objets, les modèles génératifs d'IA et les modèles Vision-Language (VLM), le système transforme des données complexes en formats structurés et lisibles par machine, tels que le CSV. Cette solution répond aux inefficacités de l'extraction manuelle des données, permettant aux analystes de données de prendre des décisions plus rapides et plus précises dans un environnement de trading critique. La plateforme proposée, nommée Pixel2Sheet Pro, automatise non seulement le processus d'extraction, mais permet également des requêtes en langage naturel via notre chatbot avancé, la manipulation des données, et offre des rapports complets incluant des visualisations, des résumés et des informations exploitables, améliorant ainsi de manière significative les capacités de prise de décision des analystes de données.

mots-clés : chatbot alimenté par IA, apprentissage profond, extraction de données, OCR, reconnaissance de données, détection de tableaux, données tabulaires, IA générative, modèles Vision-Language, CSV, rapports, analyse, prise de décision, données de marché.

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General Introduction

In today's fast paced world of scientific and technological innovation, the accelerating pace of progress in numerous fields presents significant challenges for those seeking to stay at the forefront of these developments. The explosion of information technologies has particularly reshaped the way organizations manage, analyze, and leverage data, positioning sophisticated information systems as the cornerstone of modern development. This rapid transformation is most evident in the realm of data science, where the need to convert vast amounts of unstructured data into actionable insights has become a driving force behind innovation.

A prime example of this trend is the increasing demand for the automation of data extraction from images. While a variety of methods, including OCR and other data extraction techniques, have been proposed in the literature, significant challenges persist. Variations in image quality, complex document layouts, and data structures create substantial obstacles for automatic systems aiming to transform image-based data into machine readable formats. These challenges require robust, adaptive solutions capable of addressing these complexities. Typically, such systems involve Three primary phases: The first, a detection phase where regions of interes such as text blocks, tables, or graphs are isolated using characterization features such as color, texture and shape. The second, a recognition phase that recognizes the included objects in these isolated regions. The third, an extraction phase that transform those objects into structured, readable data. To establish this kind of system, we mainly applied CNN-based object detection methods for tables detection , OCR for text object recognition, and Generative AI Vision-Language Models (VLMs) for reviewing, anomaly detection, data correction, and the translation of tabular data .

The present final-year project, conducted as part of my engineering curriculum at the Higher Institute of Applied Science and Technology of Sousse, introduces an innovative solution designed to tackle these challenges: the "Pixel to Sheet" system. This system seeks to automate the extraction of data from images, reducing the reliance on manual data entry and significantly minimizing the risk of human error. The primary objective of this project is to design, implement, and evaluate a comprehensive solution that streamlines the data extraction process from unstructured image and pdfs formats, ultimately improving both efficiency and accuracy.

Our deployed solution must integrate into existing workflows, providing real-time data extraction capabilities. This solution is designed to be continuously updated, ensuring users always benefit from the latest enhancements in the AI model accuracy.

This report offers a detailed exploration of the methodology and developmental process that led to the creation of the "Pixel to Sheet" system. The report is organized into five comprehensive chapters as follows:

— **Chapter 1** introduces the host organization and the problem that led to the suggestion of this project, along with the proposed solution. We then examine existing solutions and conclude the chapter with a description of the used development methodology .

— **Chapter 2** focuses on specifying the requirements, starting with the presentation of actors, followed by a description of both functional and non-functional requirements. The chapter also exposes with a set of use case diagrams modeling the main functionalities required for the solution.

— **Chapter 3** presents both the preliminary and the detailed design, providing an overview of static and dynamic views of the software.

— **Chapter 4** presents the development of our AI-based platform, detailing the complete pipeline for table detection, recognition, extraction, correction, and translation. It also highlights the deep learning methods employed and the specialized solutions applied for various use cases and languages.

— **Chapter 5** describes the experimental protocol, presenting the custom-created datasets, the evaluation metrics adopted and our intelligent system performance results and interpretations.

— **Chapter 6** defines the various technologies used during the production phase,and presents several graphical interfaces of the solution.

We conclude the report with a general summary of our work and discuss potential perspectives for the extension and improvement of our application.

Chapter 1

Project Overview And Context

Introduction

In this chapter, we start by presenting the host company and its services. We then delve into the project's context, including the broader industrial environment and specific challenges that necessitate the project. Lastly, we present the project's objectives, timeline, and offering a comprehensive overview of the project's scope and planned progress.

1.1 The host Company

1.1.1 Overview

DNEXT Intelligence SA, a dynamic and privately-owned Swiss-based company specializing in agriculture commodity expertise, was founded in 2020. With six strategically located offices across Switzerland, the UK, Spain, Tunisia, India, and Brazil, DNEXT has generated around 5 million euros in annual revenue in 2023 and employs more than 100 professionals. As independent leaders in commodities market analysis and agricultural commodity market research, the company offers unparalleled insights across the global agricultural landscape. From world agricultural supply and demand estimates to agricultural commodity market trends, its expertise spans the entire spectrum.

Figure 1.1: Host Company

1.1.2 Professional activities and areas of expertise

The scope of **DNEXT Intelligence SA** covers a wide range of complementary fields, including consulting, assistance, project management, integration of new technologies, and support services related to information systems. The company primarily operates in the following sectors:

- **Commodity Trading : DNEXT** supports key players in the commodity trading market by providing solutions to identify market opportunities and reduce risks associated with price fluctuations and market conditions.
- **Food Manufacturing : DNEXT** stands out for its expertise in understanding market trends and inventory management for food producers, providing analytical tools to optimize production and pricing decisions.
- **Financial Institutions : DNEXT** offers an analytics platform that helps financial institutions identify investment opportunities, assess risks, and manage their portfolios by providing insights into market trends and creditworthiness of agricultural and food production companies.

The proposed solution presented in this report aims to interact with all these modules to enhance their functionality and provide seamless data access through natural language queries.

1.2 Project presentation

In this section, we present the project and its general context. We then outline the problem statement, and the project timeline. This comprehensive overview provides a clear understanding of the project's scope and direction.

1.2.1 General context

In the fast-evolving world of trading, vast amounts of market data are generated daily, often stored in unstructured formats such as images, PDFs, and scanned documents. These documents contain essential information, including market prices, financial reports, and trading summaries, which are critical for data analysts to make informed decisions. However, the unstructured nature of this data poses significant challenges for efficient extraction and utilization. In **agriculture trading**, this data can include crop prices, weather forecasts, market trends, and commodity reports, all of which impact decisions on buying, selling, and risk management. Additionally, data related to **ship tracking**, **logistics**, and **product shipment** are vital for managing the flow of goods, ensuring timely deliveries, and optimizing supply chains. The challenge lies in extracting actionable insights from these diverse, often fragmented data sources, making automation and advanced data extraction techniques essential for improving efficiency, accuracy, and decision-making in a dynamic, global market environment.

This project aims to develop an advanced system that automates the extraction and structuring of complex tabular data from images and PDFs. By employing powerful techniques such as the object detection (CNN-based YOLO), characters recognition methods (PaddleOcr) and AI generative models(VLMs) ,the system will convert unstructured data into editable formats (CSV).

This project is conducted as part of a final-year internship, intended to fulfill the requirements for the engineering degree in Software Engineering at Higher Institute of Applied Science and Technology of Sousse.

1.2.2 Problematic

Data analysts in trading companies face significant challenges due to the time consuming and error prone process of manually extracting data from unstructured formats such as images, PDFs containing complex tables and trading reports. In the fast paced trading industry, the ability to access accurate and timely data is critical. However, the current manual extraction methods create inefficiencies and delays that can complicate decision making processes.

Several factors contribute to the difficulty of this task: the complexity and variability of document formats, the high volume of data to be processed, and the potential for human errors during data retrieval. Analysts often find themselves overwhelmed with repetitive data processing tasks, which prevents them from focusing on higher value activities such as market trend analysis, strategy development, and rapid decision making.

Given the sensitive and time critical nature of trading, even minor delays or inaccuracies in data handling can lead to missed opportunities and financial losses. Moreover, the pressure to process large amounts of data quickly increases the risk of errors, which can compromise the reliability of insights derived from the data.

Therefore, there is a clear and urgent need for a robust, efficient, and reliable method to manage and extract data from unstructured sources to support timely and accurate decision-making in trading environments.

1.2.3 Proposed Solution

Our proposed solution is the development of an AI-powered platform named **Pixel2Sheet Pro** designed to streamline the extraction and analysis of tabular data from PDFs and images. This platform empowers data analysts by automating the conversion of complex tables into editable CSV formats and offering a versatile interface for interacting with the data.

Key features of the platform include:

- **Table Extraction:** Automatically extract complex tables from PDFs and images, converting them into structured, machine readable CSV files.
- **AI-Powered Conversational Query Interface:** Our chatbot empower analysts to engage in dynamic, real-time conversations with the system, querying the extracted CSV data using natural language. This feature enables intuitive access to specific insights, eliminating the need for manual searching and streamlining the exploration of complex datasets.
- **Intelligent Data Manipulation Engine:** Leverage advanced AI capabilities to allow analysts to perform sophisticated data operations seamlessly through natural language. Whether it's adding or removing columns, applying complex aggregations, merging multiple datasets, or regenerating tables based on tailored prompts (e.g: "Insert a new row labeled 'totals' that sums each column"), this intelligent engine enhances data manipulation with ease. Additionally, it automatically handles the mapping and melting of tables to match the desired format for integration with other platforms used in ETL (Extract, Transform, Load) processes and dashboarding, streamlining data workflows and ensuring compatibility across systems.
- **Comprehensive Reporting:** Our solution not only generates detailed reports but also analyzes the extracted data to provide an in-depth understanding of key statistics, data quality, and valuable insights. These reports include a summary of the data, highlight potential anomalies or inconsistencies, and present visualizations such as total sums by category, boxplots, correlation heatmaps, pairplots, KDE (Kernel Density

Estimation), and the cumulative contributions of categories. Additionally, a table of statistics is generated, showcasing key metrics like sum, mean, min, max, std, and more. These features enhance the analyst's ability to draw meaningful conclusions and make data-driven decisions quickly.

This integrated platform significantly reduces the time and effort required for manual data processing, minimizes human error, and equips analysts with advanced tools to better interpret and utilize market data. By delivering fast, accurate, and actionable data insights, **Pixel2Sheet Pro** supports more informed decision making in the competitive trading environment.

1.2.4 Objectives

The primary objectives of this project are as follows:

- Develop an AI-powered platform capable of accurately extracting complex tabular data from unstructured documents such as PDFs and images, converting them into clean, editable CSV files to facilitate seamless data analysis.
- Provide users with the ability to preview and modify the extracted data directly within the platform, before downloading, ensuring that the final output is tailored to their specific needs. This feature sets our platform apart from other solutions, which often lack the flexibility for real-time data modification and refinement.
- Enable intuitive interaction with extracted data by allowing analysts to query the content through an advanced chatbot, reducing the need for manual data searches and improving accessibility.
- Implement flexible advanced automatic data manipulation features including but not limited to adding , updating or deleting columns/rows/cells, performing arithmetic calculations (e.g: summations), merging multiple datasets, and melting and reshaping tables based on specific user instructions(prompts).
- Provide comprehensive analytical reporting, offering detailed summarizations, data quality assessments, and insightful visualizations that help analysts understand better their data and make more informed decisions.
- Reduce manual workload and minimize errors by automating repetitive and error sensitive data extraction and processing tasks, thereby increasing overall operational efficiency.
- Ensure the platform is user friendly, responsive, and secure, promoting wide adoption among trading analysts and supporting their workflow in dynamic market environments.

1.3 Study of the Existing Solutions

This phase involves analyzing existing solutions for recognizing tabular data and extracting information from unstructured documents currently available in the market. This study will help us identify the strengths of these solutions to leverage them, as well as their weaknesses, which we aim to address through the development of our own platform. Among the existing solutions examined are Python libraries like **PyMuPDF** and **Camelot**, and large lanuage models such as **ChatGPT** and DeepSeek.

1.3.1 Definition of Existing Solutions

- **PyMuPDF** [6]

  PyMuPDF is a powerful Python library designed for extracting text, images, and metadata from PDF documents. It is widely used for its efficiency and accuracy in processing complex PDFs, making it a valuable tool for converting unstructured document data into machine-readable formats.

  Figure 1.2: PyMuPDF logo

- **Camelot** [7]

  Camelot is a Python library specialized in extracting tables from PDF files. It provides robust methods to detect and parse tabular data, enabling users to convert complex tables embedded in PDFs into clean, structured CSV or Excel files for further analysis.

  Figure 1.3: Camelot logo

- **ChatGPT** [8]

  ChatGPT is an advanced Large-Language Model developed by OpenAI. It enables natural language interaction with data and documents, allowing users to query, summarize, and generate insights based on the extracted content.

Figure 1.4: ChatGPT logo

- **DeepSeek** [9]

  DeepSeek LLM facilitate semantic search and content understanding across images and documents. It supports sophisticated extraction of relevant information by combining visual and textual data analysis, improving data accessibility and decision making efficiency.

Figure 1.5: DeepSeek logo

1.3.2 Comparative Study

Table 1.1 presents a comparative analysis of the advantages and disadvantages of existing tabular data extraction and analysis solutions currently available in the market. The goal is to compare their functionalities, identify their limitations, and propose our solution to overcome these issues.

Table 1.1: Comparative Study of Existing Applications

Application	**Advantages**	**Disadvantages**
PyMuPDF	Efficient text and metadata extraction from PDFs. Open source and widely supported.	Limited to PDF documents. Requires programming knowledge to use effectively.
Camelot	Specialized in extracting tables from PDFs. Generates clean, structured files for classic tables.	Can struggle with complex or irregular table formats. Limited support for non-PDF formats.
ChatGPT (Pro version)	Natural language interaction with data. Powerful generative capabilities for summarization and insights.	Has difficulty accurately preserving complex table structures even with its Pro version. May occasionally omit certain textual elements. Performance can degrade on very large documents. May require fine-tuning for domain-specific accuracy. Does not provide the possibility to preview and modify tables directly before downloading.
DeepSeek	Combines visual and textual analysis. Enables semantic search across documents and images.	Struggles with extracting complex tables, sometimes losing certain details. Occasionally fails to detect all textual content within documents. Not optimized for handling documents with large page sizes.

Our solution is carefully designed to combine the key strengths of the existing tools while effectively addressing their limitations. By integrating advanced technologies into a unified and user-friendly platform, we ensure robust and accurate data extraction and analysis. Moreover, our system undergoes a rigorous testing phase followed by continuous improvement cycles. This approach guarantees that each deployed version is optimized for performance, reliability, and usability, providing analysts with a dependable and efficient tool to meet their evolving needs.

1.4 Development Process

The development process for deep learning projects often involves multiple stages, from understanding the business problem to deploying a solution. To ensure a systematic, efficient, and reproducible approach, we followed the Cross Industry Standard Process for Data Mining (CRISP-DM).

1.4.1 Cross Industry Standard Prcess for Data Mining CRISP DM

CRISP-DM [10], developed by IBM in 1996, is a widely adopted methodology for organizing data mining and data science projects. It provides a clear, structured framework applicable across many industries, including engineering, healthcare, and marketing. The methodology consists of six main phases: business understanding, data understanding, data preparation, modeling, evaluation, and deployment.

Each phase builds on the previous one in an iterative and flexible manner, allowing the process to adapt as the project progresses. This approach ensures that both the technical and business objectives are aligned throughout the project lifecycle. The six phases are illustrated in the software lifecycle model shown in Figure 1.8.

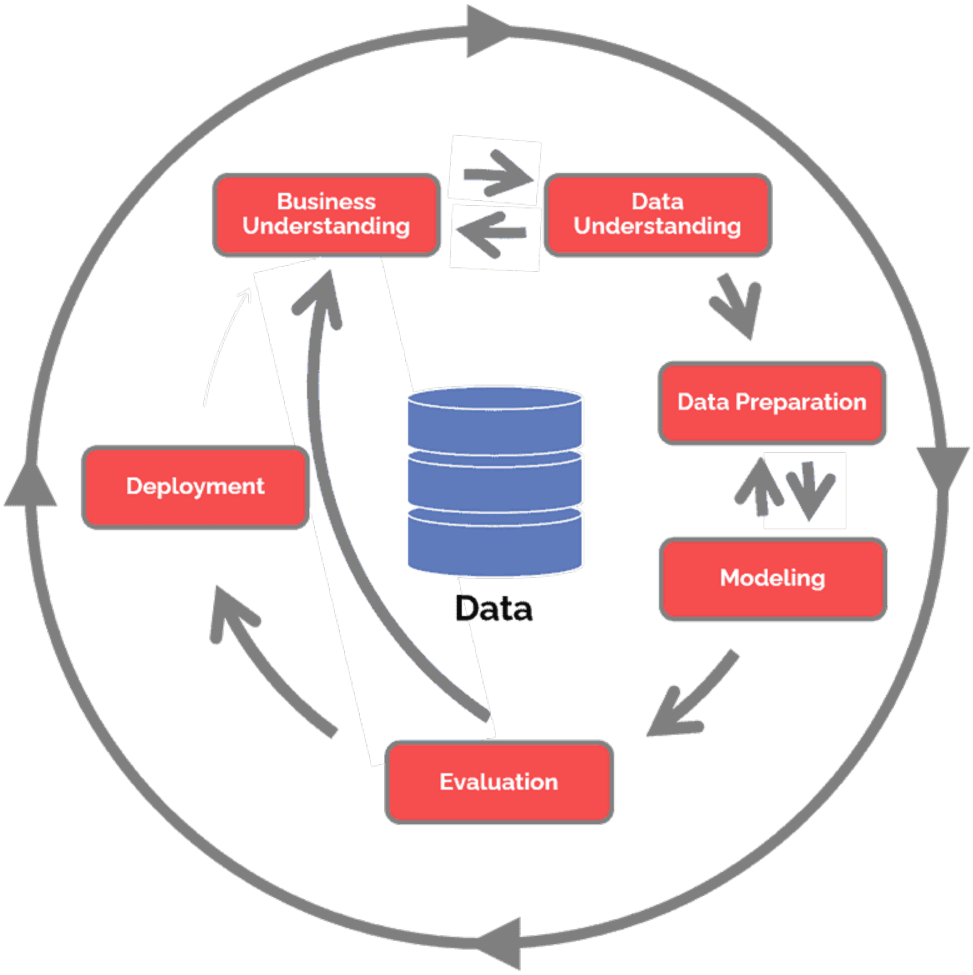


Figure 1.6: CRISP-DM Development Process [1]

- **Business Understanding:** This initial phase focuses on comprehending the project's objectives and the business context. In a trading business, this means identifying key challenges, defining success criteria, and determining how AI can enhance processes like trading efficiency, risk management, and decision-making.

- **Data Understanding:** In this phase, the available data is collected and examined in depth.We assess the quality, completeness, and relevance of the data, looking for patterns, trends, or anomalies. This helps uncover insights that will guide the data preparation and modeling stages of the project.

- **Data Preparation:** This crucial phase involves cleaning the raw data, handling missing values, transforming and integrating datasets, adding data augmentations, and formatting the data for modeling phase. The goal is to ensure the data is accurate, consistent, and ready for analysis.

- **Modeling:** Here, we select the appropriate modeling techniques and apply algorithms to the prepared data.We experiment with different parameter settings and iteratively refine the models to optimize performance and improve recognition accuracy.

- **Evaluation:** After modeling, the results are rigorously assessed to verify that the models satisfy the initial business objectives. This phase includes validating model accuracy, checking for overfitting, and ensuring the solution is practical and reliable.

- **Deployment:** The final phase involves implementing the validated model in a production environment. This includes integrating the model into business processes, monitoring its performance, and providing end users with actionable insights or automated decision support.

1.4.2 Project timeline and Gantt chart

To effectively manage and track the progress of our project, we developed a detailed project timeline and Gantt chart. The Gantt chart, shown in Figure 1.7, outlines the key tasks, their durations, start and end dates, and dependencies. This visual representation provides a clear roadmap for the project's execution and ensures that all team members are aligned with the project's objectives and deadlines.

Tasks	Start Date	End Date	February	March	April	May	June
Research and Documentation	04-02-2025	12-05-2025					
Study of Business Concepts	14-02-2025	24-02-2025					
English Tables solution	20-02-2025	28-02-2025					
Chinese Tables solution	28-02-2025	14-03-2025					
Spanish Tables solution	15-03-2025	10-04-2025					
Chatbot Development	11-04-2025	28-05-2025					
Customer Acceptance	29-05-2025	06-06-2025					
Model Optimization	31-06-2025	11-06-2025					
Solution Deployment	15-06-2025						
Report Writing	01-03-2025	15-06-2025					

Figure 1.7: Gantt chart for project timeline

Our project will be carried out according to the following phases:

- **Research and Documentation**: Conduct research and gather documentation according to the project's requirements.

- **Study of Business Concepts**: Study various business concepts, such as understanding the business models, system structures, and key concepts relevant to our project.
- **English Tables Solution**: Develop the first solution for handling English table data.
- **Chinese Tables Solution**: Develop the second solution for handling Chinese table data.
- **Spanish Tables Solution**: Develop the third solution for handling Spanish table data.
- **Chatbot Development**: Implement the final and main solution, the chatbot, which serves as the core of the project.
- **Customer acceptance** : After delivering the chatbot to the end user for testing, we gathered their feedback and evaluate the system's performance.
- **Model Optimization**: Optimize the system based on the feedback received during the customer acceptance phase.
- **Solution Deployment**: In this phase the proposed solution must be deployed on the company's server and integrated with their other platforms to ensure seamless functionality and accessibility across the organization
- **Report Writing**: Document the entire process with preliminary user requirements specification and solution architecture design steps.

Conclusion

In this chapter, we introduced the host company Dnext, its services, and its services. We discussed the critical need for accessible data access in Trading industry and highlighted our project's goal to create a user-friendly, natural language based solution for accessing industrial data. A detailed project timeline and Gantt chart were presented, outlining key tasks and milestones. This structured approach ensures systematic progress and alignment with Dnext's objectives.

As we move forward, the subsequent chapter will delve deeper into the Requirements specification. By following the outlined plan, we aim to contribute significantly to the digital transformation efforts of Dnext and the broader manufacturing sector.

Chapter 2

Requirements specification

Introduction

This chapter focuses on identifying the system's actors, analyzing its functional requirements, and representing them using use case diagrams. Additionally, we will outline the system's non-functional requirements , where functional requirements analysis helps identify the main features that the software must provide, while non-functional requirements analysis focuses on determining the key quality attributes the solution must possess.

2.1 Actors

An actor represents a role played by an external entity that interacts with the system. In our case, the system interacts with a single actor:

Data Analyst : The main user of the system, using it to perform a variety of tasks based on their needs. These tasks include extracting tables, manipulating datasets, analyzing trends, generating reports, and valuable insights to support decision-making and business strategies. Essentially, the Data Analyst uses the system to transform raw data into meaningful information for better decision-making.

2.2 Functional Requirements

The system must allow the **Data Analyst** to:

- **Import unstructured documents:** Upload files in PDF or image format (standard or scanned) that contain tabular data to be extracted.

- **Extract tables:** The main functionality is to extract tables. The following sub-functionalities support this extraction process:
  - **Detect tables within documents:** Automatically identify the location and boundaries of tables in unstructured documents.
  - **Detect and recognize text:** Apply text detection and optical character recognition (OCR) techniques to extract textual content from tables, even in complex layouts or scanned images.
  - **Parse table layout:** Analyze the structure of each table, including rows, columns, and cell positions, to accurately reconstruct the tabular format.
  - **Structure tabular data:** Extract the parsed data and convert it into structured, editable formats such as CSV.
  - **Review tables:** The system automatically compares and analyzes the extracted table with the original input to detect issues such as missing data, misaligned values, and calculation errors.This process eliminates the need for manual intervention, ensuring that the data is reliable and ready for further analysis.
  - **Correct table structures and data:** The system automatically corrects any issues found during the review phase, such as removing redundancies, correcting multi-level headers, ensuring proper alignment , also identifies and adds any missing data that may have been overlooked during the OCR phase. This process improves the integrity of the data.
  - **Translate extracted data:** Translate the content of the extracted data into a target language to support multilingual analysis or reporting.

- **Preview and Edit Extracted Tables:** The analyst can preview the tables extracted from the document before downloading. This feature allows them to review the content, make direct edits, and ensure the data meets their expectations. By providing the ability to modify the tables immediately, analysts can correct any discrepancies or refine the data before saving or exporting it.

- **Interact with extracted data using an AI-Powered Conversational Query Interface:** Effortlessly engage with your data by asking complex questions through an intuitive, AI-driven conversational interface (Advanced chatbot). Whether it's something simple like "What is the average price?" or more advanced queries like "What are the trends in sugar prices over the last 6 months?" the system provides accurate and relevant responses in real time. This powerful interaction allows users to explore and analyze their data dynamically, making it easy to extract valuable insights without needing advanced technical knowledge.

- **Provide custom extraction instructions/prompts:** Enter a custom prompt to precisely describe the analyst's intent or guide the extraction process,This includes specifying which columns to include or exclude, any new columns to be added, or any transformations required (e.g: : Extract the data for the last 3 months, including the columns 'Date', 'Sugar Type', 'Quantity Sold', and 'Revenue'. Exclude the 'Discount' column and add a new column for 'Net Revenue' (Revenue - Discount)).

- **Select specific pages for processing:** Choose individual pages or a range of pages within a PDF document to focus extraction on relevant sections only.

- **Manipulate the data:** Perform automatic operations such as:

  - Adding , deleting or modifying columns
  - Adding , deleting or modifying rows/cells
  - Merging multiple datasets
  - Filtering, sorting, or renaming columns
  - Performing automatic calculations (sum, average, std, etc)

- **Regenerate Tables**: Analysts don't typically use the extracted tables in their raw form. They require the data to be regenerated and restructured into a specific format. Our solution intelligently handles this process by **"melting"** and **"reshaping"** the tables to match the desired structure. It analyzes the content of the data, identifies key columns, and maps them to the required format, ensuring that the table fits seamlessly into the analyst's workflow. This approach saves the analyst significant time, as it automates a process that would otherwise require substantial intelligence and focus to manually identify and map the columns. Instead of spending hours restructuring tables, the analyst can now focus on higher-level insights and decision-making.

- **Generate automated insights:** By automatically generating statistical summaries, comprehensive data quality reports, and dynamic visualizations such as charts and graphs. This process not only highlights key trends and patterns but also provides a deeper understanding of the data, making it easier to uncover actionable insights and drive informed decision-making.

- **Download outputs and reports:** Download the results produced by the system, including extracted CSV files, visualizations, and detailed analysis reports.

To support the above functional needs, the system must:

- Automatically identify table structures in various PDFs and images, even when layouts are complex or noisy.
- Convert extracted tables into standard formats (CSV) while preserving key information.
- Accurately interpret natural language instructions using our powerful Chatbot, including ambiguous or incomplete queries, based on the data context.

2.3 Non-Functional Requirements

Non-functional requirements describe the performance-related goals of the system and the constraints of its operating environment. The non functional requirements that the proposed solution must meet are as follows:

- **Robustness:** The system must consistently deliver accurate and reliable outputs, even when processing noisy, poorly formatted, or complex unstructured documents.
- **Reliability:** The chatbot platform must maintain high availability and provide consistent response times regardless of the size or complexity of the uploaded files and queries.
- **Scalability:** The solution must be designed to scale, allowing easy integration of additional features such as support for new file formats, advanced analytics modules, or extended natural language capabilities.
- **Usability:** The interface should be intuitive and user friendly, enabling users with minimal technical background to interact efficiently with the system.
- **Maintainability and Technical Documentation:** The application code must be well documented to ensure readability and maintainability. The documentation should include detailed explanations of APIs, data structures, algorithms, and system workflows.

2.4 Use Case Diagrams

Use case diagrams provide a static functional overview of the system, focusing on how users interact with it without detailing the internal architecture.

In this section, we present both a high level and detailed modeling of the system's required functionalities using use case diagrams. These diagrams illustrate the different services the application must offer and highlight the interactions between the user and the system to ensure that all identified needs are addressed effectively.

2.4.1 General Use Case Diagram

The general use case diagram, illustrated in Figure 2.1, brings together all the fundamental use cases to provide an overall view of how the system operates.

This diagram presents the various use cases associated with the Analyst actor.

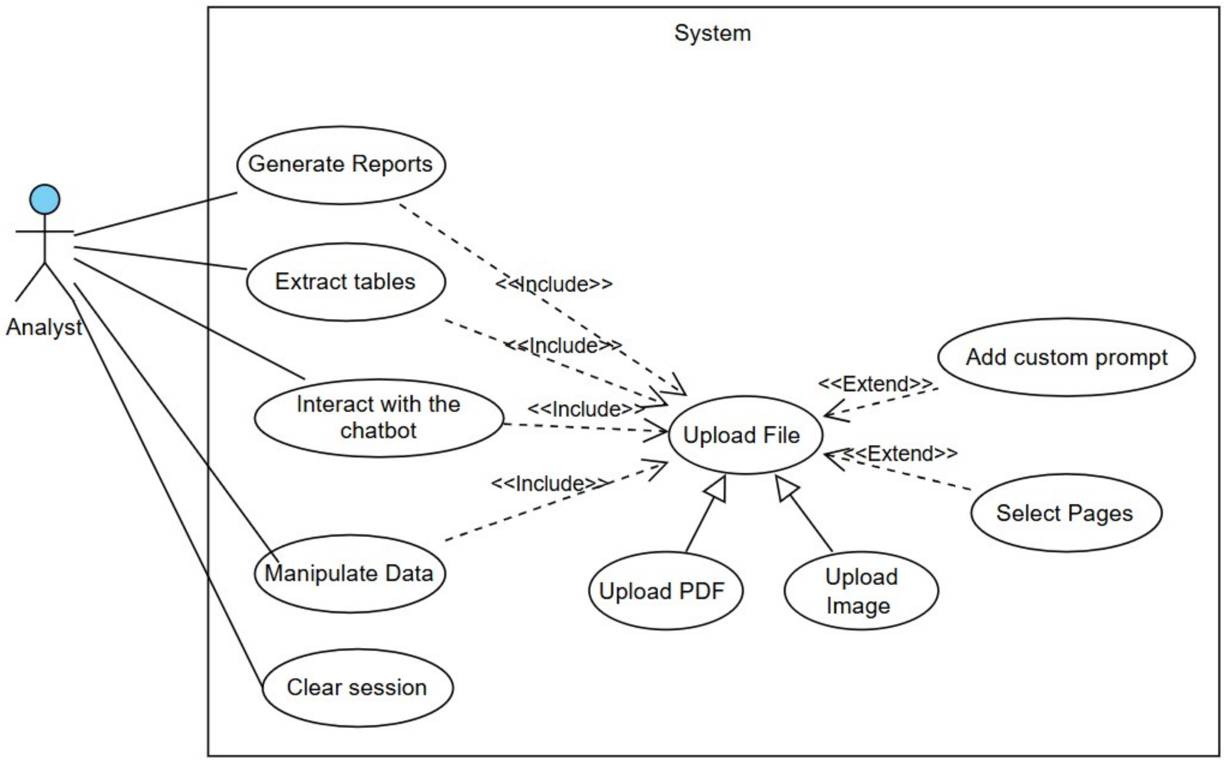


Figure 2.1: General Use Case diagram

2.4.2 Refinement of the Use Case: "Extract Tables"

This use case provides users with the ability to extract tables from PDFs and images. The extraction process involves detecting tables within the uploaded files, including steps such as detecting tables, recognizing data, structuring the tables, and correcting the data to ensure the data quality.Additionally, users are able to preview the extracted tables and make edits as needed, providing greater flexibility and control over the final output.

Figure 2.2 illustrates the refined use case diagram of the "Extract Tables" use case.

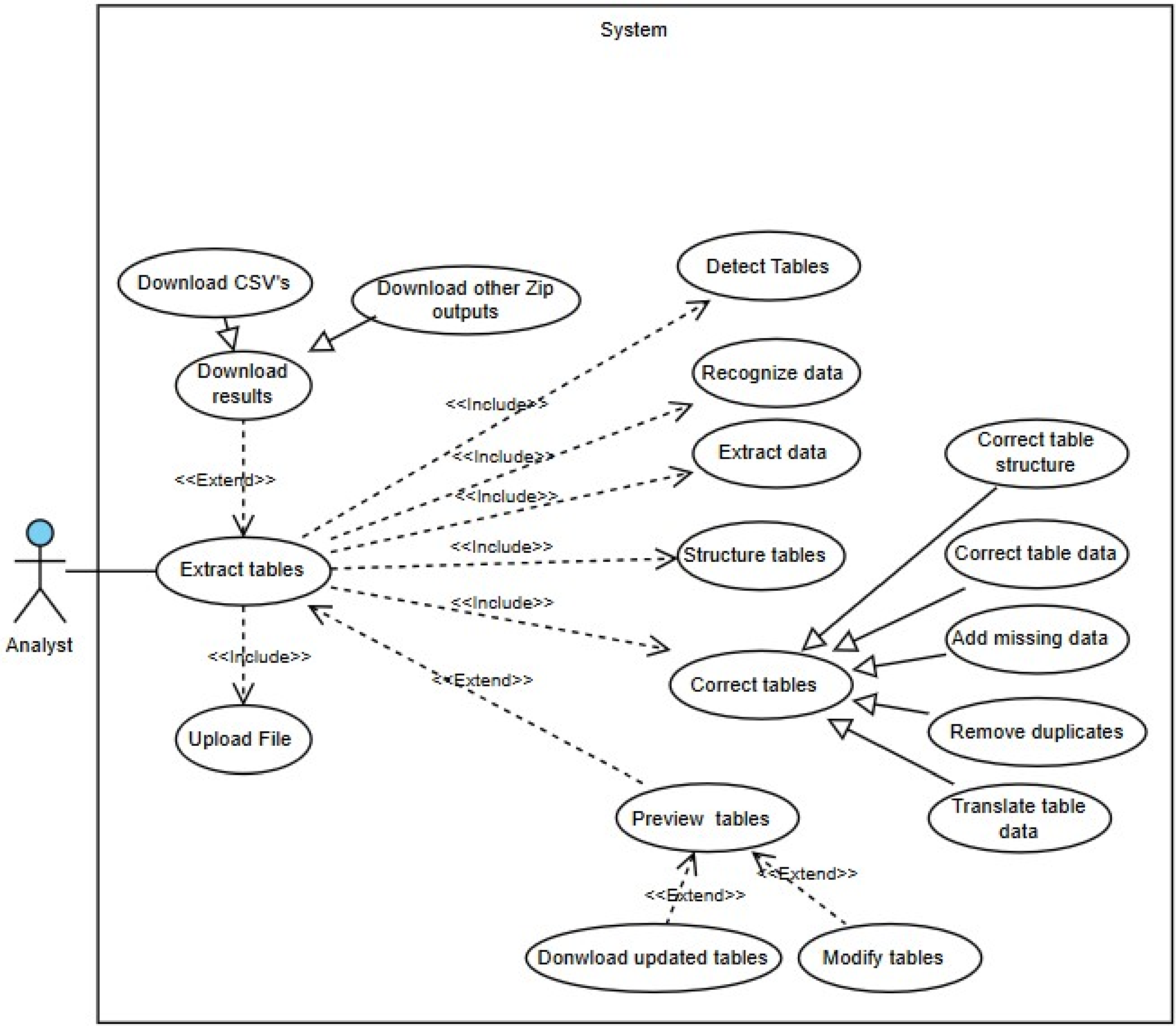


Figure 2.2: Extract tables Use Case

At the end of this refined diagram, and to better understand the functioning of this use case, we will describe it textually in Table 2.1.

Table 2.1: Textual Description of the Use Case « Extract Tables »

Title	Extract Tables
Actors	Analyst
Summary	This use case allows the analyst to extract and structure table(s) from uploaded file(s).
Pre-conditions	File(s) uploaded, with optional custom prompts and specific pages selection.
Post-conditions	Extracted CSV table(s) , along with ZIP archives containing all generated outputs ready for download.
Main Scenario	1. Analyst uploads the file(s). 2. Analyst may add a custom prompt and select specific pages (optional). 3. Analyst initiates file processing. 4. System detects table(s) within the files. 5. System recognizes data within table(s). 6. System extract(s) data from tables. 7. System structures data into initial table(s). 8. System review the initial table(s). 9. System performs the needed corrections (structure,wrong data , missing data, duplicates, translations). 10. System transfers final outputs to storage. 11. System returns final outputs to Analyst. 12. Analyst preview the CSVs before the download . 13. Analyst download the final CSVs.
Alternative Scenarios	**9.a Table(s) are already correct :** 1. System transfers final outputs directly to storage. 2. System returns the outputs to Analyst. 3. Analyst preview the CSVs before the download. 4. Analyst download the final CSVs. **12.a Analyst preview and modify the previewed table(s) :** 1. System save the new updated tables 2. The Analyst downloads the final tables

2.4.3 Refinement of the Use Case: "Manipulate data"

This use case must provide analysts with the ability to manipulate data by choosing among several options such as correcting data, merging tables, or regenerating tables with specific instructions. The analyst can freely select and perform any combination of these operations to refine the data according to their needs. The goal is to offer flexibility in data handling to ensure accurate and useful outputs. Figure 2.3 illustrates the refined use case diagram of "Manipulate Data".

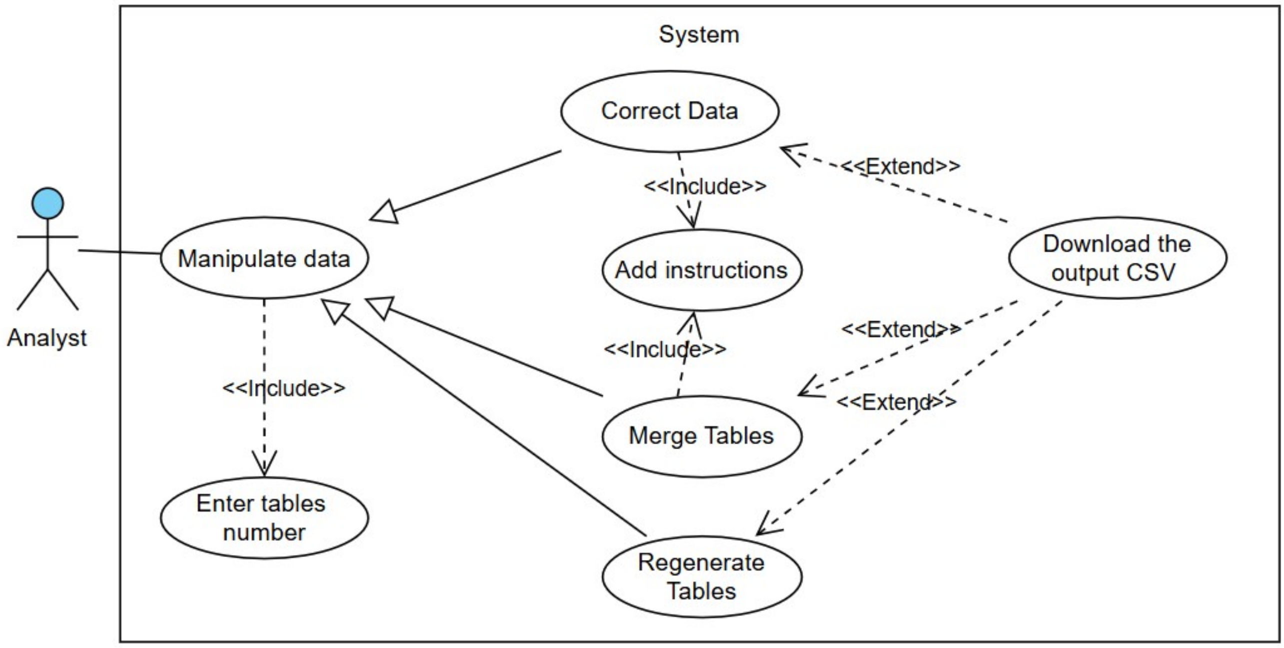


Figure 2.3: Manipulate data Use Case

At the end of this refined diagram, and to better understand the functioning of this use case, we will describe it textually in Table 2.2.

Table 2.2: Textual Description of the Use Case "Manipulate Data"

Title	Manipulate Data
Actor	Analyst
Pre-conditions	Tables already extracted.
Post-conditions	Data tables are corrected, merged, or regenerated as needed, with output files available for download.
Main Scenario	1. Analyst clicks on the "Correct Data" tab in the Data Operations section. 2. The analyst enters the tables number to correct. 3. Analyst adds correction instructions and clicks "Apply Corrections." 4. The system analyzes the instructions, generates corrected tables, and transfers them for storage. 5. Analyst can download the corrected tables in CSV format.
Alternative Scenarios	**1.a Analyst clicks on the "Merge Tables" tab in the Data Operations section.** 1. The analyst enters the tables number to merge. 2. Analyst clicks "Merge" to initiate the merge operation. 3. Analyst adds merge instructions and clicks "Merge Tables". 4. The system analyzes the merge instructions, merges the tables, and stores the merged tables. 5. Analyst can download the merged tables in CSV format. **1.b Analyst clicks on the "Regenerate Tables" tab in the Data Operations section.** 1. The analyst enters the tables number to regenerate. 2. The analyst clicks "Regenerate" to initiate the table regeneration process. 3. Analyzes the selected tables. 4. Detects the required columns. 5. Applies a melting operation to restructure the tables into the required format. 6. Stores the regenerated tables. 7. Analyst can then download the regenerated tables in CSV format.

Conclusion

In this chapter, we presented the functional and non-functional requirements of our solution. Then, we detailed the different use cases.

The next chapter will focus on the static and dynamic internal modeling of our solution.

Chapter 3

Conceptual Study

Introduction

In this chapter, we will specify the structural and behavioral views of our solution, applying architectural patterns and UML diagrams.

3.1 Static View of the Application

This section starts with an overview of the general architectural design of our software, followed by a detailed description of the internal structure of each module within the system.

3.1.1 Software General Architecture

Figure 3.1 shows a package diagram modeling the main modules composing our system.

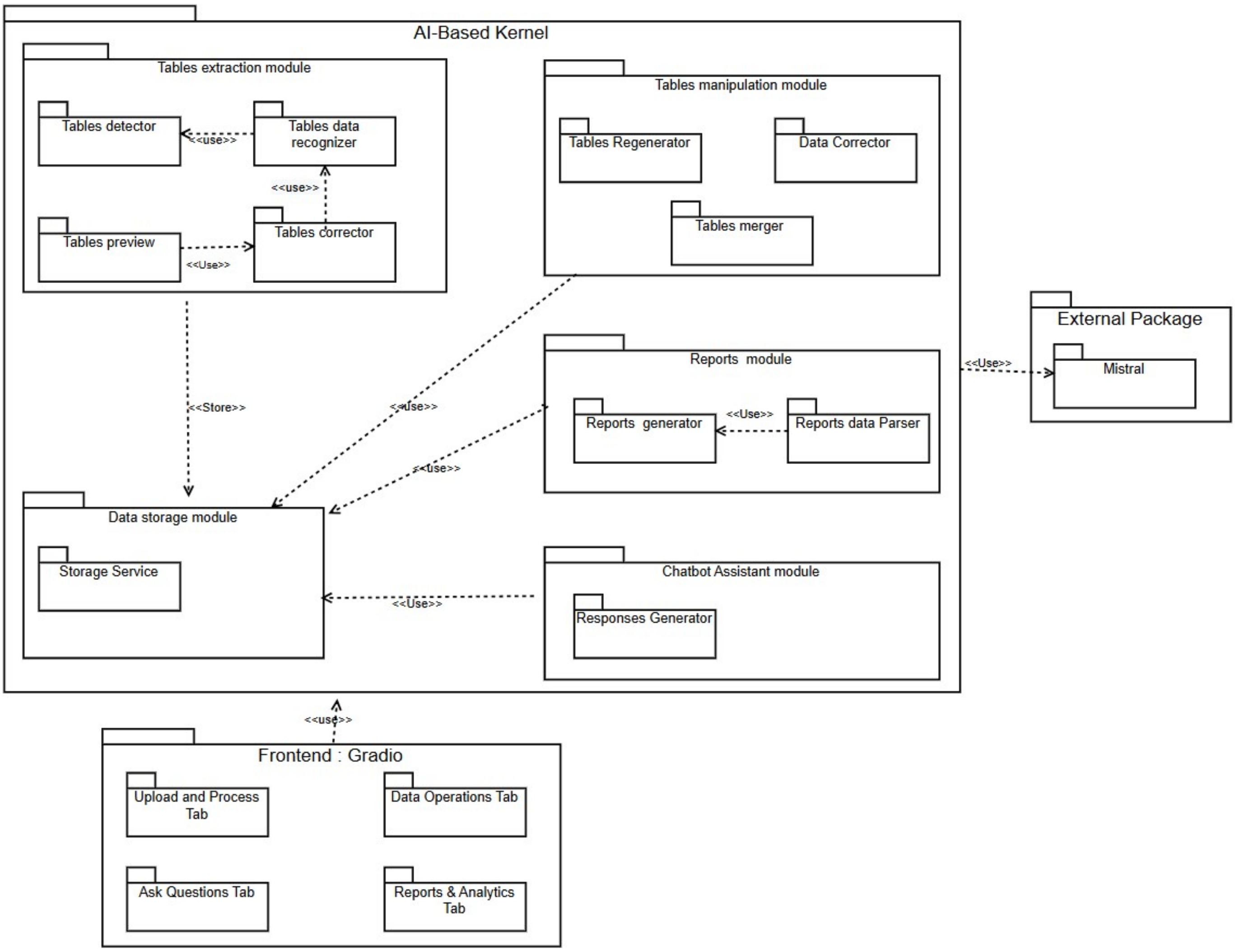


Figure 3.1: Package diagram

The main modules composing our solution are:

- **AI-Based kernel**: This includes the core processing, analysis, storage, and assistance modules that enable table extraction, manipulation, report generation, and chatbot interaction.

  - **Tables Extraction Module**: Responsible for detecting tables within documents, extracting data using OCR, processing and correcting detected tables, and formatting them to improve data structure and usability. The **Tables Detector** detects the tables, and the **Tables Data Recognizer** extracts the data. If any errors are found, the **Tables Corrector** is used to correct the detected tables. After correction, the **Tables Preview** module allows users to view the extracted tables and directly modify any elements within the tables as needed.

- **Tables Manipulation Module**: This module enables the manipulation and transformation of tables based on extracted CSV files, applying specific data corrections, merging multiple tables as needed, and ensuring the integrity and structure of the data. It includes the **Tables Regenerator**, which automatically restructures and pivots tables to match the required format, the **Data Corrector**, which identifies and corrects issues such as missing values, redundancies, and inconsistencies based on user prompts, and the **Tables Merger**, which combines data from multiple tables into a single, unified dataset for seamless analysis based on user prompts.
- **Reports Module**: This module analyzes extracted data to generate detailed reports, including key metrics, trends, and visualizations. It consists of the **Reports Generator**, which creates customized valuable reports based on the tables content, and the **Reports Data Parser**, which structures the information for easy interpretation.
- **Chatbot Assistant Module**: Consists of a natural language interpreter to understand user prompts and a **Responses Generator** that provides answers or analyses based on the extracted data.
- **Data Storage Module**: Manages the storage, organization, and preservation of extracted table data, as well as intermediate pipeline outputs and final results, through the **Storage Service**.
- **External Package**: Includes the **Mistral** package, which is used by various modules for processing.

• **Frontend**: The user interface, developed with Gradio, is divided into multiple tabs corresponding to the key functionalities offered to the user.

- **Upload and Process Tab**: Allows document upload, selection of processing options, and displays status along with the option to download results.
- **Ask Questions Tab**: Features an AI-powered chatbot that enables users to ask natural language queries (prompts) about the data and receive accurate, real-time responses, simplifying data exploration and analysis.
- **Data Operations Tab**: Dedicated area for operations on extracted data, including CSV corrections, table merging, and data regeneration.

- **Reports & Analytics Tab**: Enables users to generate comprehensive analytical reports, select tables for analysis, and interactively view results.

3.1.2 Tables Extraction Package

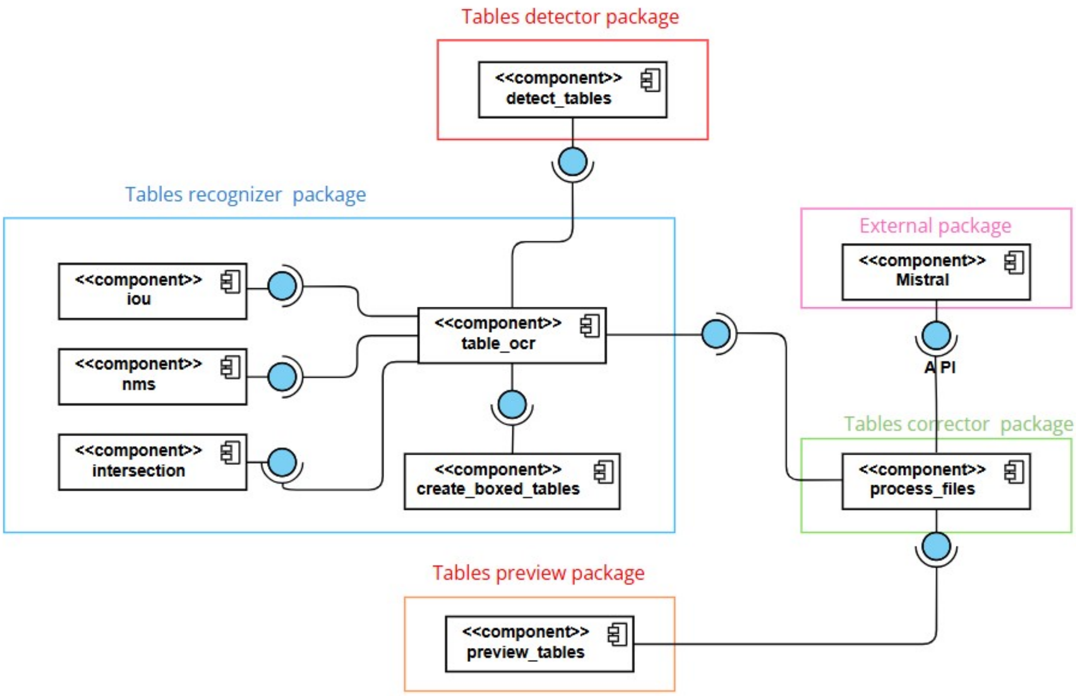


Figure 3.2: Tables Extraction component diagram

Figure 3.2 illustrates the component diagram of the "Tables Extraction" module, highlighting the service relationships.

- **Tables Detector Package:** Detects tables in the input data and provides the detected tables to the **Tables Recognizer Package**, specifically to `table_ocr` for further processing.

- **Tables Recognizer Package:** Recognizes and structures the detected tables using the following components:

  - `table_ocr`: Extracts and structures tables using services from `detect_tables`, `iou`, `intersection`, `create_boxed_tables`, and `nms`.

  - `iou`: Calculates the Intersection over Union to assess the accuracy of table detection. **see section 5.2.2**

  - `intersection`: Handles overlapping tables to improve extraction accuracy.

- `create_boxed_tables`: Generates visual boxed representations of tables.
- `nms`: Applies the Non-Maximum Suppression method to eliminate redundant bounding boxes and improve the accuracy of table detection **see section 4.1.3.3**.

- **Tables Corrector Package:** Finalizes and corrects the extracted tables.
  - `Process_files`: Processes and corrects tables using the structured data from `table_ocr`.
- **Tables preview Package:**
  - `preview_tables`: Provides an interactive preview of the extracted tables, allowing users to review the data and make any necessary modifications directly before finalizing and downloading the output.
- **External Package:** Provides external processing services:
  - `Mistral`: An external Vision-Language-Model (VLM) **(see section 4.1.3.4)** service used by the `Tables Corrector Package` to correct and enhance the final table output through the `API`.
  - `API`: Serves as the interface for communication.

3.1.3 Tables Manipulation Package

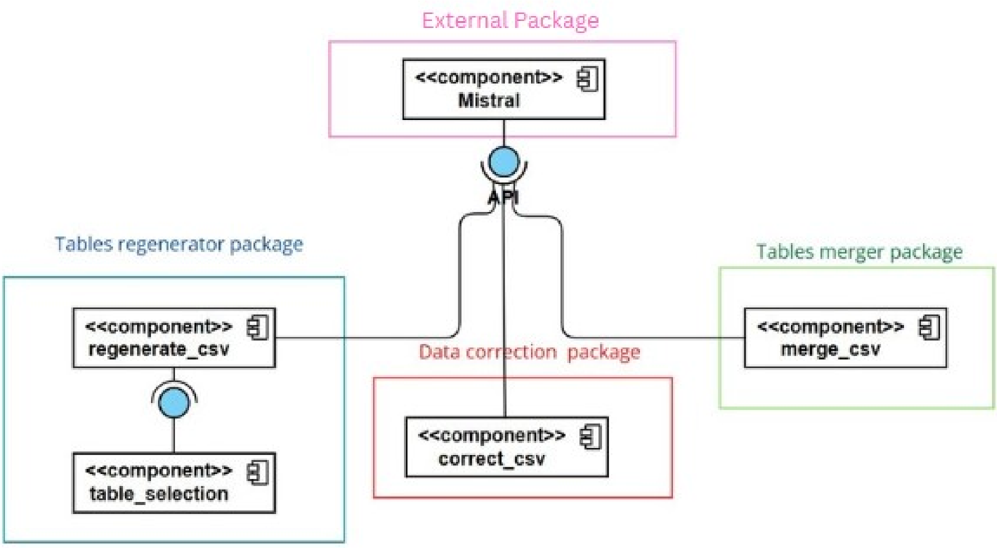


Figure 3.3: Tables Manipulation component diagram

Figure 3.3 illustrates the component diagram of the "Tables Manipulation" module, highlighting the service relationships.

- **Tables Regenerator Package:** This package is designed to assist analysts by automatically regenerating and restructuring raw data tables into a format that fits their workflow. Instead of manually reformatting tables.
  - `regenerate_csv`: This component handles the process of regenerating and reshaping/melting CSV tables to the desired structure for further analysis.
  - `table_selection`: This component enables the analyst to select the specific table they wish to regenerate, ensuring the right data is targeted for transformation.
- **Data Correction Package:** This package is responsible for correcting and preparing the data (CSV files) for further processing.
- **Tables Merger Package:** This package is responsible for merging the CSV files into one unified output by following the instructions provided by the user.
- **External Package:** `Mistral` An external VLM service that processes natural language prompts and tables, correcting or further processing them as needed.

3.1.4 Reports Generator Package

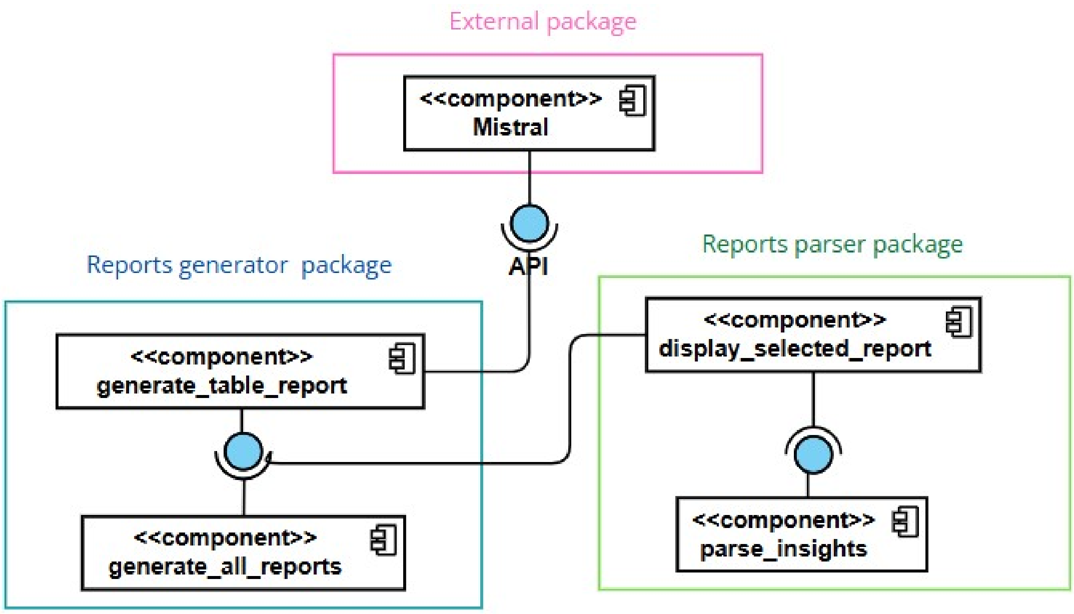


Figure 3.4: Reports Generator component diagram

Figure 3.4 illustrates the component diagram of the "Reports Generator" module, highlighting the service relationships.

- **Reports Generator Package:** This package is responsible for generating tables reports.
  - `generate_table_report`: This component generates detailed report for a table by analyzing it.It communicates with the `Mistral` VLM service to create the reports.
  - `generate_all_reports`: This component aggregates all the individual reports generated by **generate_table_report** into a final, consolidated report.
- **Reports Parser Package:** This package handles the parsing and display of the generated reports.
  - `display_selected_report`: This component displays the selected report to the user for further analysis or review.
  - `parse_insights`: This component processes the content of the generated reports and extracts meaningful insights from the data.
- **External Package:**
  - `Mistral`: An external service that processes the reports and provides advanced analysis and insights **see section 6.2.8**.

3.2 Dynamic View of the Application

Sequence diagrams and activity diagrams are visual tools for modeling the dynamic behavior of software systems. Sequence diagrams focus on depicting the interactions between software objects. Meanwhile, activity diagrams provide a complementary view by illustrating the flow of actions, organized into partitions that show the specific tasks performed by each object.

3.2.1 Sequence Diagram: «Extract tables Use Case »

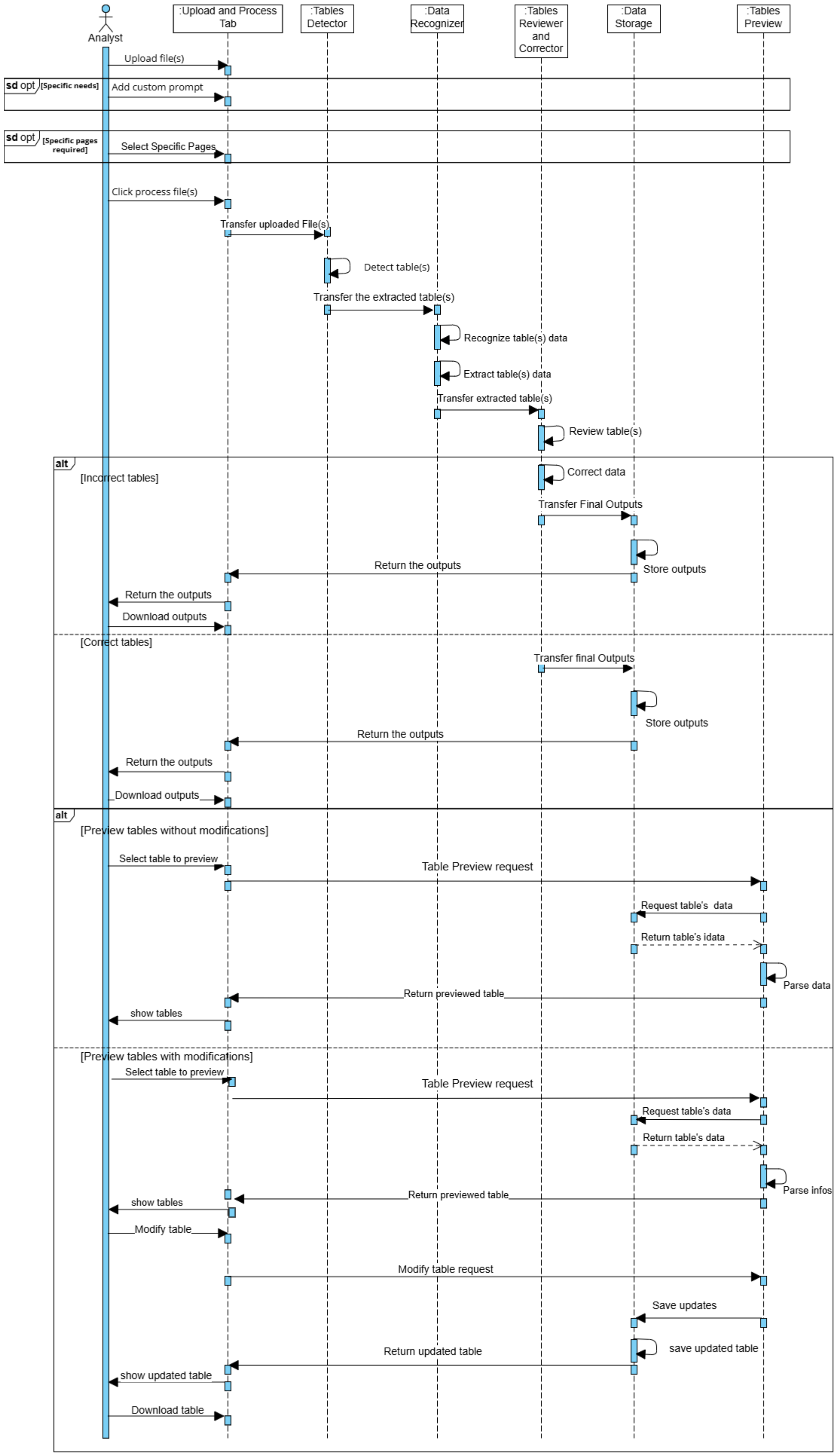


Figure 3.5: Extract Tables UC Sequence diagram

The **Analyst** begins by opening the interface and uploading the file to be processed. If specific requirements are needed, he can optionally add a custom prompt. Additionally, if only certain pages of a PDF are required, the analyst can select specific pages or define a page range.

Once the configuration is complete, the analyst clicks the **Process File** button to initiate the processing pipeline. The uploaded file is transferred to the **Tables Detector** module, where the tables within the document are detected.

The detected tables are then passed to the **Data Recognizer** module, where the content and structure of the tables are recognized. This step involves identifying the table structure and extracting the data while ensuring the table's format is consistent and properly organized.

After the tables have been recognized and formatted, they are transferred to the **Tables Reviewer and Corrector** module, where the tables undergo a review process. If the tables are accurate, the final outputs are sent to the **Data Storage** module. If any errors are identified, the tables are corrected within the **Tables Reviewer and Corrector** module, and the corrected versions are then transferred to the **Data Storage** module.

Once the tables are extracted, the **Tables Preview** module allows the analyst to preview the tables' content before downloading them. If the analyst wants to make changes, they can directly modify the tables in the preview view. After making the necessary updates, the modified tables are returned, and the changes can be saved for download.

Finally, the system returns the final outputs to the analyst, who can view and download the updated results.

3.2.2 Sequence Diagram: «Manipulate Data Use Case »

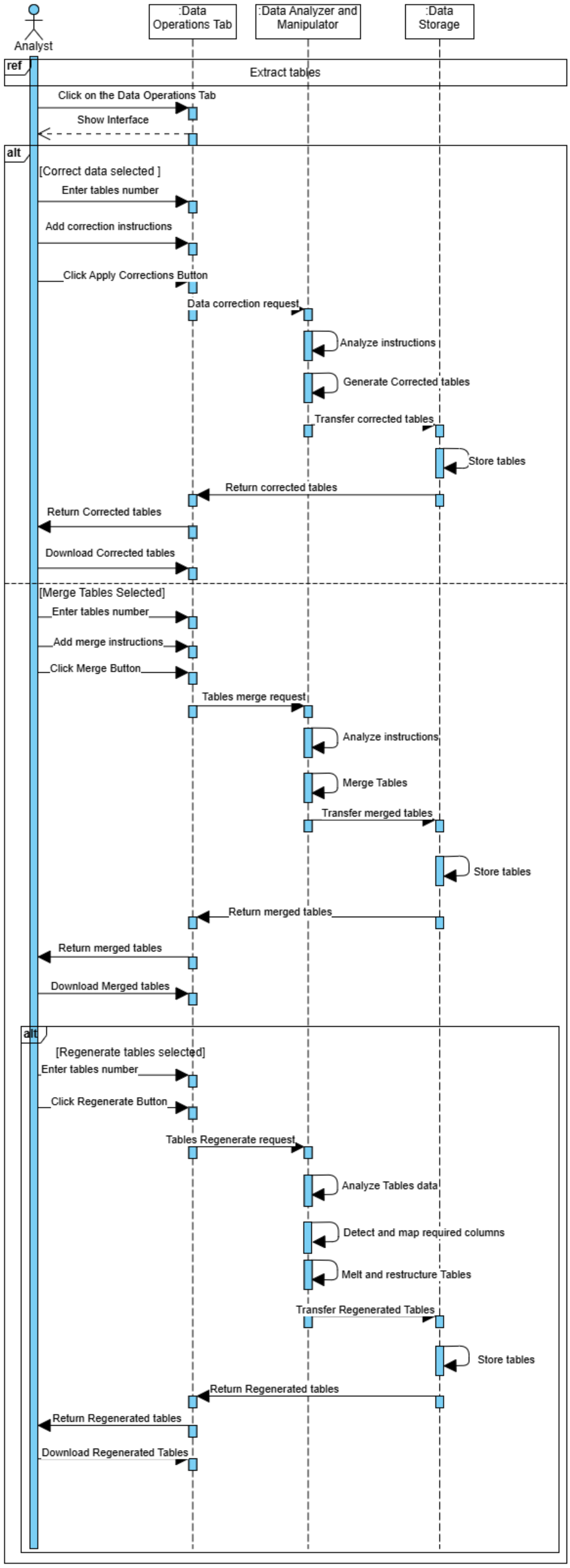


Figure 3.6: Manipulate Data UC Sequence Diagram

The analyst begins by accessing the interface in the Data Operations Tab. After selecting the desired task (correction, merging, or regeneration of tables), the analyst enters the tables number and adds any necessary instructions for the selected task then the system processes the tables as follows:

- For corrections, the system analyzes the data, applies the corrections, and generates the corrected tables.
- For table merging, the system merges the selected tables based on the provided merge instructions.
- For regeneration, the system regenerates the tables by analyzing the data, detecting and mapping required columns, and restructuring the tables accordingly.

The processed tables are then transferred to storage. Once stored, the final output is returned to the analyst. The analyst can then download the processed tables or have them returned for further use.

3.2.3 Partitioned Activity Diagram: «Extract Tables »

Figure 3.8 presents the partitioned activity diagram describing the table extraction workflow. The process begins when the analyst uploads a document through the **Frontend Module**, with the option to provide a custom prompt or use the default configuration. For PDF documents, the system offers additional processing options, allowing the analyst to either extract tables from the entire document or select specific pages for processing.

The **Table Extraction Module** then automatically performs several key operations: detecting tabular structures within the document, extracting the table data, and formatting the tables while preserving their structural integrity. The system conducts an automatic quality review, detecting any potential issues for correction while validating properly extracted tables.

After the system completes the review, the extracted tables are stored in the **Storage Module** if correct, or corrected and then stored if incorrect. The analyst can then preview the tables. If satisfied, the analyst proceeds to download the tables. Alternatively, the analyst can modify the tables (e.g: adding ,removing or modifying rows/columns ...) before saving and downloading the new updated version.

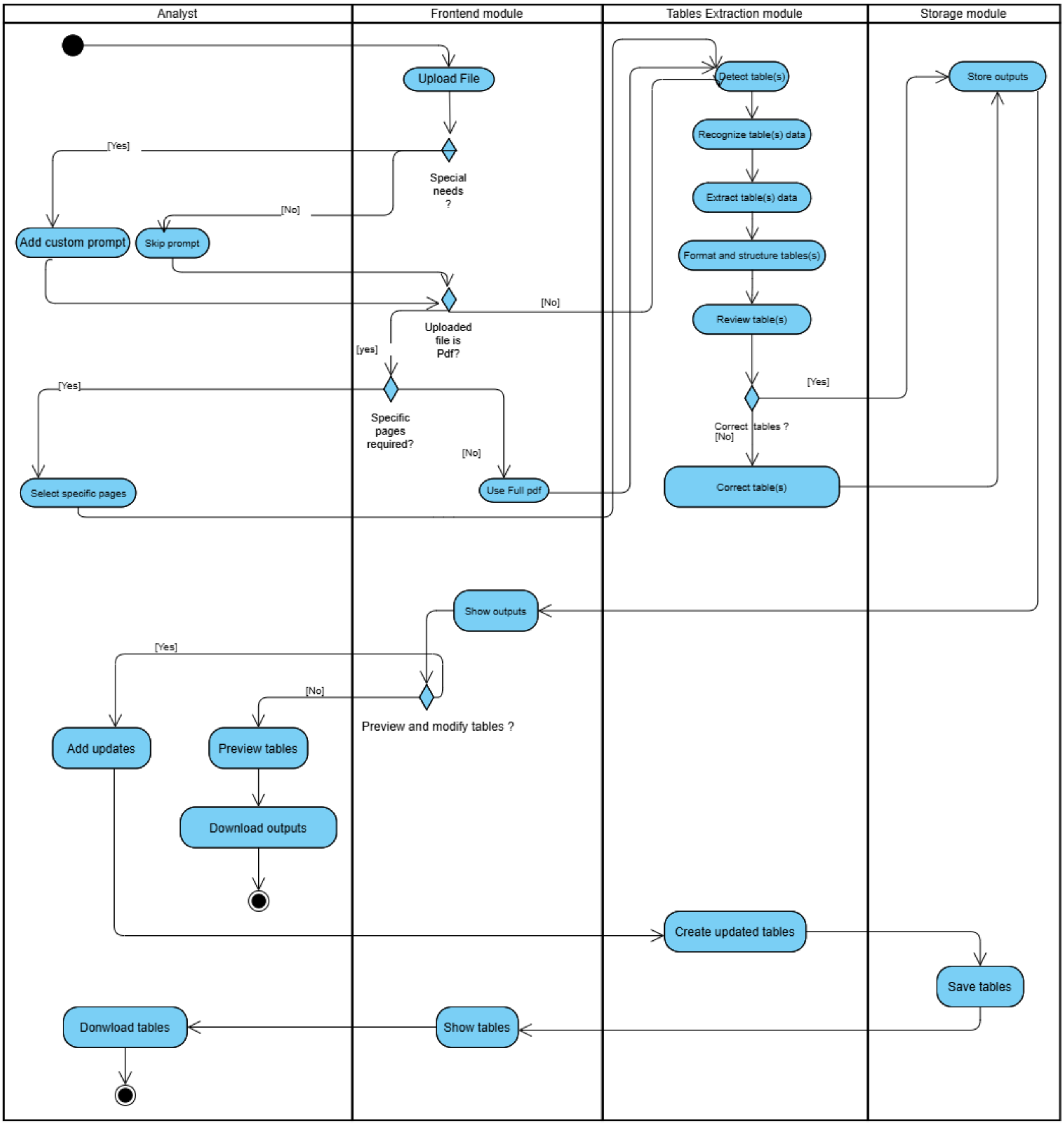


Figure 3.7: Partitioned Activity Diagram: «Extract Tables »

Conclusion

In this chapter, we covered the design phase of the project, highlighting the system's internal composition and the communication flow between its components. The next chapter will focus on the intelligent module, detailing the solution process.

Chapter 4

AI-Based Platform

Introduction

In this chapter, we provide a more detailed explanation of our intelligent solutions.Where the core functionalities are the detection, recognition, extraction ,correction and understanding of complex tabular data from uploaded documents. We define the recognition process and outline its main stages.

4.1 Background

4.1.1 Computer Vision

Computer vision is a field of Artificial Intelligence that enables machines to interpret and understand visual information from the world, much like human vision. By leveraging techniques from image processing, machine learning, and deep learning, computer vision systems can analyze images and videos to detect objects, recognize patterns, extract features, and make decisions based on visual input. In recent years, advancements in neural networks particularly convolutional and transformer-based architectures have significantly improved the accuracy and scalability of vision-based tasks across industries such as healthcare, manufacturing, autonomous driving, and document analysis.

In our project, computer vision plays a central role in enabling the automatic recognition and extraction of tables from document images. It allows the system to visually identify the structure of a table, localize its boundaries, and interpret the spatial layout of cells and content within scanned or digital documents. By using computer vision techniques such as OCR and data processing modules, we built a pipeline that transforms unstructured visual data into clean, structured information suitable for downstream analysis or integration.

4.1.2 Recognition Approaches

Object recognition is a common application of computer vision, a subfield of Artificial Intelligence (AI) that aims to equip machines with perception capabilities that mimic human vision.

Our solution focuses on developing an intelligent system capable of recognizing and extracting tables from document files. This process involves several key stages, including table detection, content recognition, data extraction, formatting, and correction.

There are two main approaches to implementing such a system:

- **End-to-End Approach:** This approach employs a single, unified model often a Vision-Language Model (VLM) to handle the entire recognition pipeline. VLMs, such as those based on the Transformer architecture , integrate visual and textual features to jointly perform tasks like layout analysis, entity recognition, and structured output generation. These models are trained on large-scale document datasets with multi-modal supervision, enabling them to infer complex spatial and semantic relationships within documents. The primary advantage of this approach is efficiency: the document is processed in one pass, offering fast inference and simplified deployment.

  However, relying solely on VLMs for end-to-end table recognition has several limitations:

  - **Opacity and Interpretability:** VLMs function as black boxes, making it difficult to debug or isolate errors in specific subtasks such as structure recognition or OCR.
  - **Generalization Limits:** Despite extensive pretraining, VLMs can struggle with complex layouts, unusual table formats, or noisy scans that differ from the training data.
  - **Customization Constraints:** Fine-tuning large VLMs for domain specific use cases is resource-intensive and often impractical for iterative development.
  - **Error Propagation:** Errors in early stages (e.g., misidentifying a table region) cannot be easily corrected later, as all tasks are tightly coupled in the unified model.

- **Modular (Sequential) Approach:** This approach follows the **Single Responsibility in SOLID Design Principle** [11], assigning a specific task to each component in the pipeline. The system is broken down into well defined stages such as table detection, text recognition, structure parsing, formatting, reviewing and correction each handled by a dedicated module or model. For example, a CNN-based detector might be used for locating tables, followed by a transformer based OCR engine for textual content.

  While this decomposition increases architectural complexity and may reduce inference speed, it brings significant benefits:

  – **Modular Tuning:** Each module can be trained, evaluated, and optimized independently, allowing fine-grained control over performance.

  – **Transparency and Debugging:** Intermediate outputs can be inspected, making it easier to trace and correct errors at specific stages.

  – **Flexibility:** Modules can be updated or swapped without affecting the entire pipeline, enabling better adaptation to specific domains or evolving requirements.

In our case, we adopted the **modular approach**, prioritizing recognition accuracy and maintainability to meet the quality requirements of our solution. This design enables continuous improvement of individual components and supports long-term scalability.

4.1.3 Applied Deep Learning (DL) Methods

4.1.3.1 Concept

Deep learning is a subset of Artificial Intelligence (AI), built upon machine learning principles, where systems learn to identify patterns and make decisions directly from data. Unlike traditional programming, which relies on explicitly defined rules, deep learning models are capable of improving their performance through experience, without being manually programmed for each task.

At the core of many deep learning applications in computer vision are **Convolutional Neural Networks (CNNs)**, inspired by the way the human brain processes visual information. These networks consist of dozens to even hundreds of layers of interconnected neurons (as shown in figure 4.1), where each layer processes and transforms the output from the previous one. This hierarchical structure allows the network to learn increasingly abstract features at deeper levels. The final layer typically computes the probability of the input belonging to each predefined class. Through training, the system learns to recognize specific visual patterns such as traffic signs or table structures before ever encountering them in real-world scenarios.

CNNs are particularly effective at extracting and learning hierarchical features from images, enabling tasks such as image classification, object detection, and video recognition. Their ability to automatically identify visual patterns has made them the dominant approach in a wide range of vision-based problems, driving significant progress in the field over the past decade.

CNN Architecture Overview

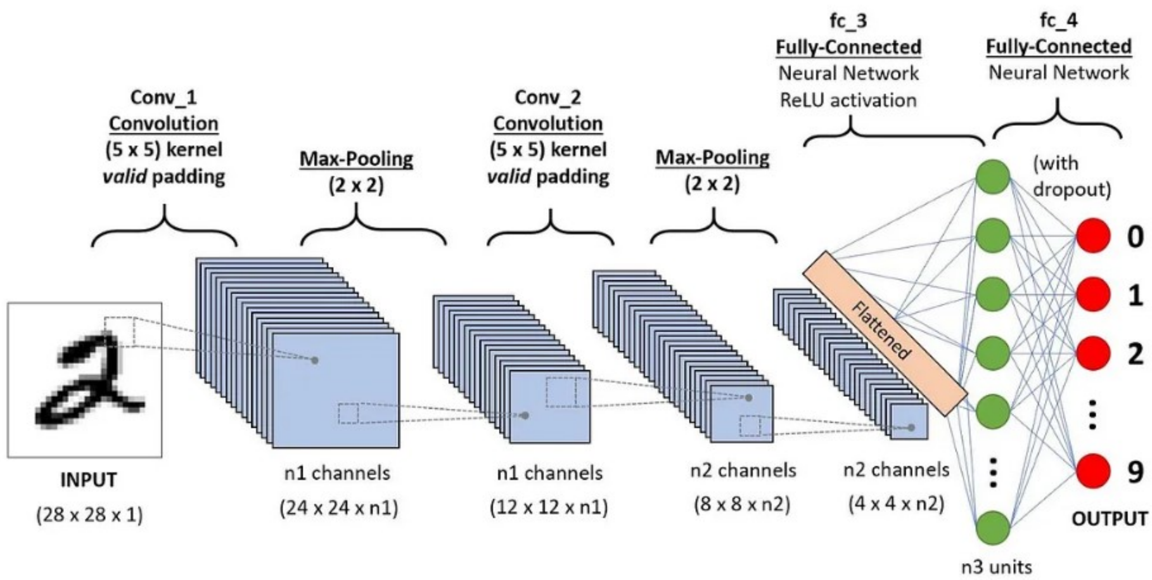


Figure 4.1: CNN Architecture [2]

A typical CNN architecture [2] includes:

- **Convolutional Layers (Conv):** These layers apply convolution operations to the input, using filters (kernels) to extract features like edges, textures, and patterns.
- **Pooling Layers:** These layers (e.g., max pooling) reduce the spatial dimensions of the feature maps, helping reduce computation and control overfitting.
- **Fully Connected (FC) Layers:** After feature extraction, the network usually includes fully connected layers to make final predictions.The flattened output from pre-

vious layers is passed through these dense layers.

4.1.3.2 Deep Learning Methods for Detection

Object detection is a crucial step in our document recognition pipeline, as it determines the accurate localization of tables within document images. Errors in detection can negatively impact subsequent modules like OCR, parsing, and formatting. To ensure high precision and real-time performance, we focused on deep learning-based models from the **YOLO (You Only Look Once)** family, which are renowned for their efficiency and accuracy in object detection tasks.

Selection Criteria:

Models were selected based on the following criteria:

- **Fast inference**: Essential for real-time or batch document processing.
- **High detection accuracy**: Needed to handle varied document layouts and table structures.
- **Scalability**: Compatibility with lightweight and scalable deployment environments.
- **Community support**: Availability of implementation tools and strong community resources for troubleshooting and updates.

Why YOLO ?

While there are other object detection models available, such as Faster R-CNN, RetinaNet, and SSD, we chose YOLO for the following reasons:

- **Real-time performance**: YOLO performs end-to-end detection in a single pass, providing faster inference than models like Faster R-CNN, which rely on computationally expensive region proposal networks.
- **Accuracy with speed**: YOLO offers a balance between speed and accuracy, unlike RetinaNet, which sacrifices speed for improved accuracy.
- **Scalability**: YOLO models come in different sizes allowing for flexibility in deployment on different hardware platforms without significant performance loss.
- **Proven success**: YOLO has been widely adopted in both research and industry for various object detection tasks, ensuring its reliability for detecting tables in diverse document structures.

Based on these factors, we evaluated multiple versions of YOLO [12], including YOLOv4, YOLOv8, YOLOv11, and YOLOv12, as the primary models for our table detection task. After testing several versions, we selected YOLOv12 for its superior performance. **(You can find the details of these YOLO verions in the following annex section 1)**.

YOLOv12

YOLOv12 [13] was chosen over previous versions, for the table detection due to several key advantages:

- **Advanced Attention Mechanisms (A2)**: YOLOv12's A2 attention mechanism improves the model's ability to focus on important regions in complex document structures, such as table cells, enhancing detection accuracy.
- **Residual-Efficient Layer Aggregation Networks (R-ELAN)**: R-ELAN optimizes feature aggregation, improving detection of complex objects like tables by effectively combining multi-scale features.
- **FlashAttention**: YOLOv12's FlashAttention reduces memory overhead during attention computations, enabling faster inference without sacrificing accuracy, ideal for real-time document processing.
- **Hybrid CNN-Transformer Framework**: This framework combines the strengths of CNNs and Transformers, enabling YOLOv12 to better understand both local and global contexts in document layouts, improving table detection in diverse scenarios.

These features make YOLOv12 the optimal choice for detecting complex table layouts in documents, offering superior accuracy, speed, and scalability compared to previous YOLO versions. **(You can see the YOLOv8 detection results compared to other versions in this section 5.3.1).**

The Figure 4.2 illustrates the YOLOv12 architecture.

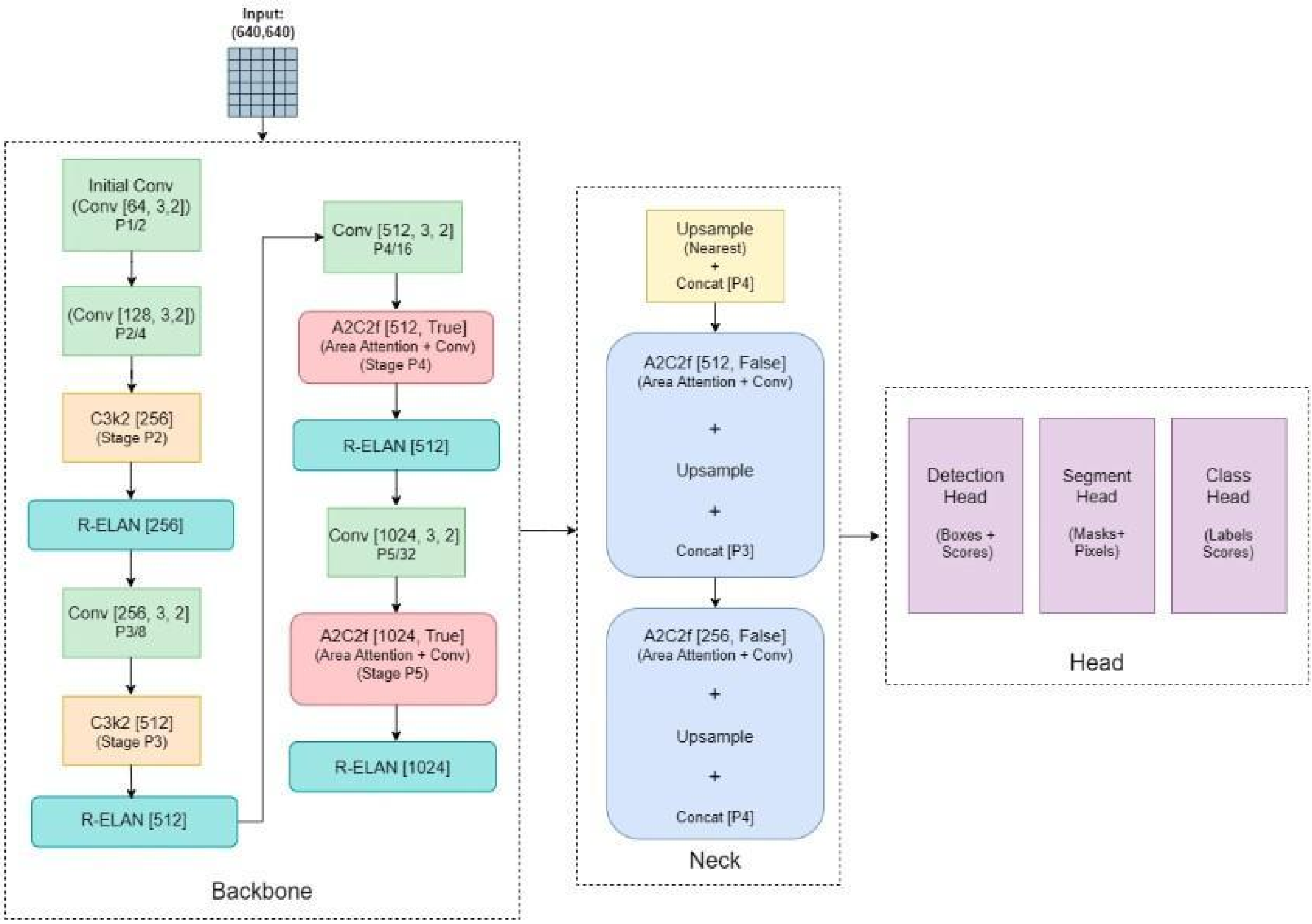


Figure 4.2: YOLOv12 Architecture [3]

4.1.3.3 Deep Learning Methods for characters recognition

Characters recognition is a key part of our pipeline, following the table detection stage. During the research phase, several OCR tools and frameworks were evaluated to determine the most effective solution for extracting structured tabular data. After testing various options **(see annex section 2)**, we ultimately opted for **PaddleOCR** due to its superior performance in detecting and recognizing text within tables, especially in cases involving curved, rotated, or low resolution text regions. **PaddleOCR's** modular architecture, integration of state-of-the-art models, and high accuracy in handling complex layouts made it the most effective choice for our structured data extraction needs.

Text Detection

PaddleOCR employs the **Differentiable Binarization (DB)** text detection model, which offers several advantages:

- It is based on a Fully Convolutional Network (FCN),
- It produces a probability map indicating the presence of text regions.
- It utilizes differentiable binarization and morphological post-processing to refine bounding boxes.
- It is robust in handling curved, rotated, or irregular text layouts, which are common in scanned documents.

This method enables the extraction of precise bounding boxes even in noisy or degraded document conditions, enhancing the accuracy of downstream text recognition. The figure 4.3 illustrates the text detection stages used by PaddleOCR.

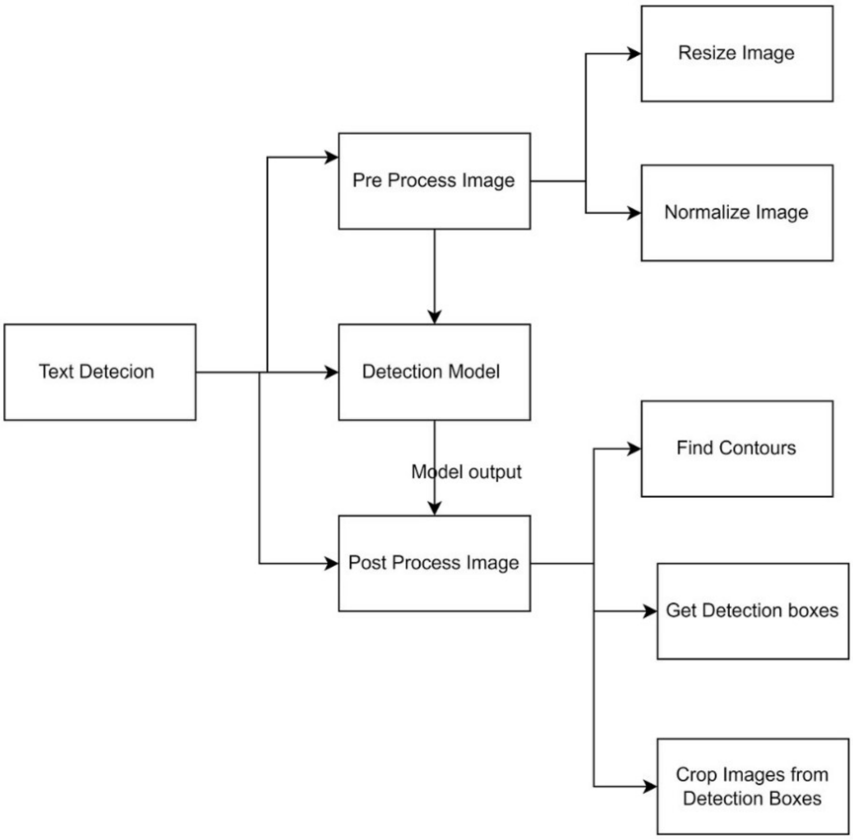


Figure 4.3: Text Detection Stages in PaddleOCR [4]

Text Recognition

Once regions are detected, PaddleOCR applies a deep learning-based recognizer using the **Convolutional Recurrent Neural Network (CRNN)** architecture:

- **CNN layers** [2] extract visual features from cropped word images.
- **RNN layers** [14] model the sequential structure of characters.

- A **CTC (Connectionist Temporal Classification)** [15] decoder is used to map visual features to character sequences without requiring character-level alignment.

This model is robust to text distortion, scale variations, and missing or connected characters.

Text Extraction and Structuring

After detecting and recognizing text in table images, the system must organize this raw information into a structured grid. This step is essential to reconstruct tables in a way that makes the content readable, searchable, and usable. Our approach includes the following key stages:

- **Line Projection:** Each detected word box is used to estimate where the rows and columns of the table should be. This is done by projecting the text boxes horizontally to guess row positions and vertically to guess column positions. These projections help us identify where the table lines would logically be, even if the original table has no visible gridlines.

- **Line Refinement with Non-Maximum Suppression (NMS):** Because the projection step may produce multiple overlapping or closely positioned lines ,to eliminate them the NMS is applied. It retains only the most confident lines, reducing duplicates and improving the clarity and accuracy of the table structure.

- **Grid Construction and Text Placement:** Once the clean set of horizontal and vertical lines is obtained, we intersect them to create a grid that represents the structure of the table. Then, each recognized text element is placed into the correct cell of the grid. This is done by calculating the overlap between each text box and the table cells using the Intersection over Union (IoU) metric **(see section 5.2.2)** the cell with the highest overlap is chosen as the correct position for the text.

In conclusion, after thorough experimentation and analysis, PaddleOCR emerged as the most reliable and robust OCR solution for extracting structured data from tables (see table 5.12). It plays a central role in our document processing .

4.1.3.4 Generative Vision-Language AI

Generative Artificial Intelligence refers to models capable of producing new content such as text, images, audio, or structured data based on learned patterns from vast training corpora. Unlike discriminative models that merely classify or detect, generative models can synthesize, complete, or transform data. They use neural architectures such as **Transformers**, which rely heavily on attention mechanisms to model long range dependencies and context.

In our document understanding system, we leverage a class of generative AI models called **Vision-Language Models (VLMs)**, which are capable of understanding both visual content and textual input. These models can reason over documents and correct noisy OCR outputs by jointly analyzing layout, image cues, and extracted text.

Vision Language Model Architecture

VLMs extend the Transformer to jointly process visual and textual inputs. A typical architecture includes:

- **Visual Encoder:** Extracts spatial features from the table image. This encoder transforms the visual input into dense image embeddings that preserve layout and cell positioning. These features serve as the foundation for downstream cross-modal reasoning.

- **Text Encoder:** The textual stream begins by tokenizing OCR-recognized content and converting it into embeddings. These embeddings represent semantic meaning and serve as the basis for integrating linguistic information with visual cues.

- **Multimodal Fusion via Attention:** The model processes both modalities through a Transformer architecture composed of stacked layers. Within each layer, self-attention allows tokens to interact and build context within their own stream (text-to-text), while cross-attention enables tokens to attend selectively to relevant regions in the visual representation. This mechanism is essential for aligning headers with data cells, resolving ambiguities, and grounding words spatially.

- **Feedforward Processing:** After attention fusion, a feedforward network applies transformations that consolidate information and produce refined token representations. These outputs reflect both the visual layout and textual semantics of the original document.

- **Output Decoder :** In generative VLMs, a decoder is employed to produce structured outputs, such as cleaned table text, markdown format, or JSON, depending on the application.

The figure 4.4 illustrates the Vision Language Models architecture.

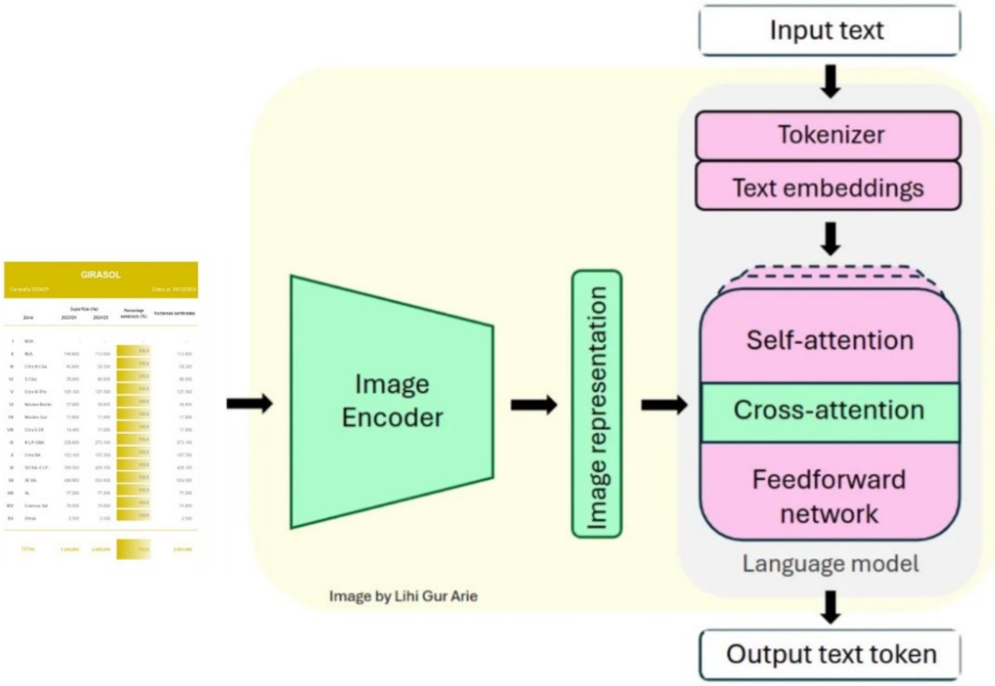


Figure 4.4: VLM architecture [5]

Attention Mechanisms

The Transformer backbone in VLMs relies on **self-attention** and **cross-attention**:

- **Self-Attention:** Enables each token (or image patch) to attend to all others in the same modality. This is crucial for understanding relationships between headers, rows, or grouped content.

- **Cross-Attention:** Allows information to flow between visual and textual tokens. For example, the model can attend to a specific part of the image (like a header box) while interpreting the corresponding text token, improving semantic alignment.

- **Multimodal Contextualization:** By layering attention blocks, the model builds a shared understanding of layout, structure, and content across the entire document.

This multimodal correction step enhances robustness, especially for noisy scans, hand-written elements, or complex tables without visible borders.By leveraging VLMs with attention based reasoning, the system not only recovers from earlier stage errors but also produces outputs that closely mirror human expectations of table structure and readability.

4.2 Proposed Approaches

4.2.1 Generic solution pipeline

In this section, we present our final generic solution pipeline, designed to address the various challenges encountered during the tables extraction process from PDFs or image-based documents. This solution is considered **generic** because it is adaptable to a wide range of table types, including those with varying languages, text layouts, and complexities. The pipeline is engineered to handle diverse formats, from simple tables to complex, noisy images, and it ensures accurate data extraction **regardless of language**, structural intricacies, or image quality. By incorporating flexible solutions at each stage, the pipeline ensures that it can be applied to different scenarios, making it versatile and efficient for a variety of document processing tasks.

We present our generic pipeline by first introducing the key issues encountered in the data extraction process, and then describing the solutions implemented to solve these challenges effectively using our approach.

4.2.1.1 Tables Detection

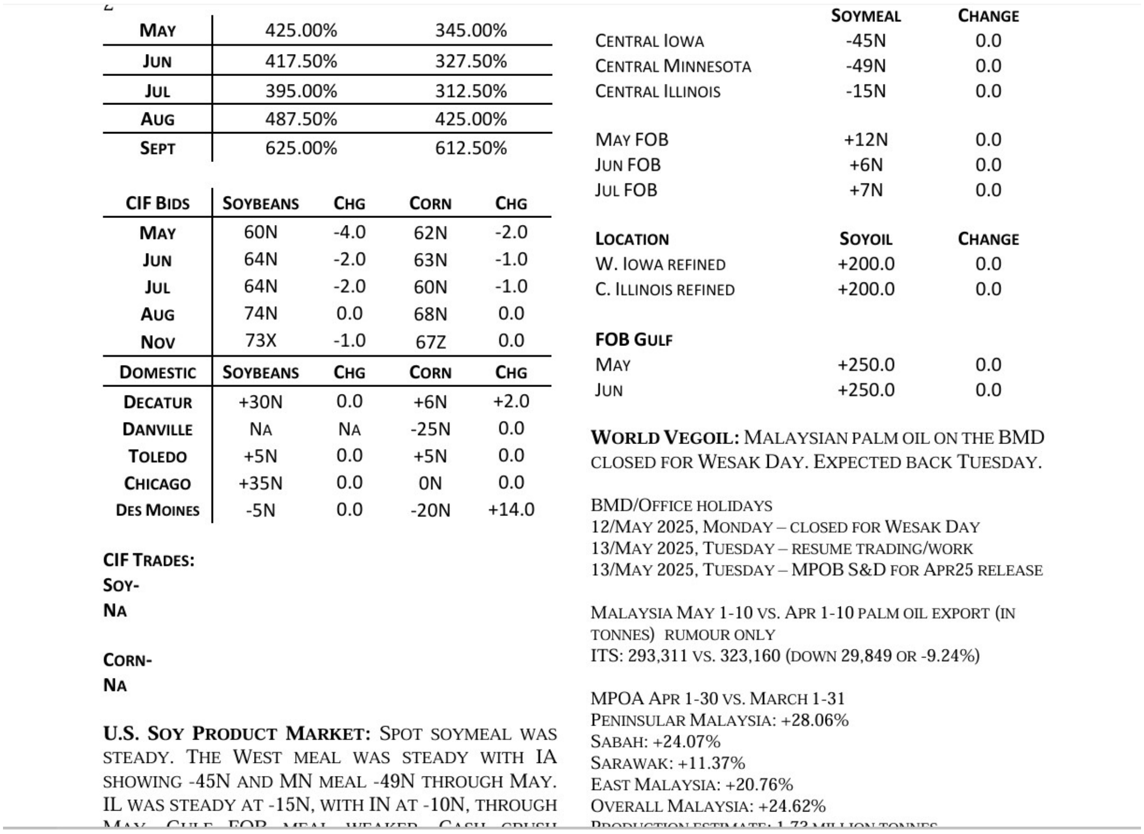
Issues:

Table detection in documents faces several challenges :

- **Complex Layouts:** Borderless tables, tables embedded within text, or irregular table structures (e.g: merged cells , or nested tables) make detection difficult.
- **Document Quality:** Low resolution scans,and noise can obscure table boundaries and content, complicating detection.
- **Inconsistent Formatting:** Variations in row/column spacing, tables spanning multiple pages, or complex table formatting can hinder accurate identification.
- **Contextual Ambiguities:** Tables within paragraphs or with mixed formatting (like images or links) may be misidentified.
- **PDF Conversion Issues:** When converting documents, table structures may be lost or distorted, especially in textless PDFs.

- **Algorithmic Limitations:** Many table detection algorithms struggle with non-standard layouts and complex table formats.

The figure 4.5 illustrates issues in table detection, such as borderless tables and tables embedded within text, which complicate the accurate extraction and understanding of table structures.



May	425.00%	345.00%
Jun	417.50%	327.50%
Jul	395.00%	312.50%
Aug	487.50%	425.00%
Sept	625.00%	612.50%

CIF Bids	Soybeans	Chg	Corn	Chg
May	60N	-4.0	62N	-2.0
Jun	64N	-2.0	63N	-1.0
Jul	64N	-2.0	60N	-1.0
Aug	74N	0.0	68N	0.0
Nov	73X	-1.0	67Z	0.0
Domestic	**Soybeans**	**Chg**	**Corn**	**Chg**
Decatur	+30N	0.0	+6N	+2.0
Danville	Na	Na	-25N	0.0
Toledo	+5N	0.0	+5N	0.0
Chicago	+35N	0.0	0N	0.0
Des Moines	-5N	0.0	-20N	+14.0

CIF Trades:
Soy-
Na

Corn-
Na

U.S. Soy Product Market: Spot soymeal was steady. The West meal was steady with IA showing -45N and MN meal -49N through May. IL was steady at -15N, with IN at -10N, through

	Soymeal	Change
Central Iowa	-45N	0.0
Central Minnesota	-49N	0.0
Central Illinois	-15N	0.0
May FOB	+12N	0.0
Jun FOB	+6N	0.0
Jul FOB	+7N	0.0
Location	**Soyoil**	**Change**
W. Iowa refined	+200.0	0.0
C. Illinois refined	+200.0	0.0
FOB Gulf		
May	+250.0	0.0
Jun	+250.0	0.0

World Vegoil: Malaysian palm oil on the BMD closed for Wesak Day. Expected back Tuesday.

BMD/Office holidays
12/May 2025, Monday – closed for Wesak Day
13/May 2025, Tuesday – resume trading/work
13/May 2025, Tuesday – MPOB S&D for Apr25 release

Malaysia May 1-10 vs. Apr 1-10 palm oil export (in tonnes) rumour only
ITS: 293,311 vs. 323,160 (down 29,849 or -9.24%)

MPOA Apr 1-30 vs. March 1-31
Peninsular Malaysia: +28.06%
Sabah: +24.07%
Sarawak: +11.37%
East Malaysia: +20.76%
Overall Malaysia: +24.62%

Figure 4.5: Tables detection Issue

To address these challenges, more advanced and flexible algorithms are needed for robust table detection across diverse document types.

Solution:

To tackle these table detection issues, we trained YOLOv12 on our customized dataset **(You can see the dataset details in section 5.1.2)** , which contains a wide variety of table formats and layouts.This dataset includes both borderless and traditional tables, as well as tables embedded within text, and tables that span multiple pages. By leveraging YOLOv12's ability to handle complex visual data, we can effectively detect tables with varying structures, document qualities, and contexts.

4.2.1.2 Data Recognition and Extraction

Issues:

Data recognition within tables can be equally challenging, especially when dealing with documents that have inconsistent formatting, varying data types, or noisy content. The key issues in this stage include:

- **Unstructured Data:** Data within tables can vary in format, such as mixed text and numeric data, dates, and other specialized fields. This lack of structure makes it hard for traditional recognition systems to process and interpret the content accurately.
- **OCR Errors:** When dealing with scanned or image-based documents, Optical Character Recognition (OCR) may introduce errors, such as misrecognized characters, incorrect formatting, or missing values, which can compromise data accuracy.
- **Missing Data:** Tables may have missing or incomplete values, particularly in the case of scanned documents where parts of the table might be obscured, causing gaps in the data.
- **Irregular Formatting:** Some tables may have irregular layouts, such as merged cells, empty cells, or multi-row/column headers, which complicate the task of accurately extracting the data.
- **Language and Character Variations:** Tables may contain different languages, special characters, or symbols that are difficult for the recognition systems to interpret, especially when multiple languages or non-standard characters are involved.

The figure 4.6 illustrates a table that demonstrates several data recognition challenges. It includes unstructured data with mixed text and numerical and dates information. The table's complex structure, featuring multi-line entries and varying formats, could lead to OCR errors and missing data. Additionally, the use of multiple languages,including Chinese and English, increases the difficulty of accurately processing and interpreting the content. This highlights the difficulties faced in extracting clean and structured data from scanned or image-based documents.

Vsls Line-up and Berthing Prospects at Machong Port 东莞深赤湾码头船期表											4/1/25
No.	Ship Name 船名	LOA 船长	Cargos 货物	QUANTITY 数量(MTS)	ETA 预抵时间	ETB 靠泊时间	Cargo Balance 余货未卸	ETD 预计完货	Berth No. 泊位名称	Receiver 收货人	Draft 吃水(M)
1	CHANG SHENG HUI 昌晟晖	102	CORN 玉米	20790	2000/30/03	0900/01/04	20790	Rvtg	2#	内贸	9.1
2	BAO NENG 16 保能16	225	CORN 玉米	67000	2000/31/03	0430/01/04	67000	Rvtg	3#	内贸	13.5
3	BEST TRADER 佳运	225	TAPIOCA 泰木薯	49000	2000/21/03	1600/25/03	25640	Rvtg	4a#	厦门港务 (外贸)	10.9
4	TINOS 帝诺斯	229	化肥	23655	1500/27/03	0100/28/03	8905	0830/02/04	5#	外贸	13
5	DRAFTZILLA 逸漂	190	MOP 以钾	18000	2338/29/03	1430/02/04	18000	Rvtg	5#	上海农资 (外贸)	12.73
6	YONG AN 永安	179	TAPIOCA 泰木薯	20000	1600/27/03	Rvtg		Rvtg	4#/5#	建发（外贸）	8.7
7	TRUONG LONG 03 特隆03	229	TAPIOCA 泰木薯	61250.458	2210/28/03	Rvtg		Rvtg	4#/5#	厦门港务 (外贸)	12
8	FEDERAL TIBER 泰伯	190	NPK 挪复	43023	0100/30/03	Rvtg		Rvtg	4#/5#	外贸	11
9	CASTOR 卡斯托	225	SFMS 葵花籽粕	60500	0400/30/03	Rvtg		Rvtg	2#/3#/4#/5#	建发+象屿+托福 (外贸)	12.5
10	MILAS 米拉斯	190	SFMS 乌葵粕	41578.512	0600/30/03	Rvtg		Rvtg	2#/3#/4#/5#	蔚特（外贸）	11.1
11	MARCO 马可	225	WHEAT+CMP 加小麦+菜粕	67559.233	0200/02/04	Rvtg		Rvtg	2#/3#/4#/5#	明穗+建发 (外贸)	13.0
12	MATER 马特	210	TAPIOCA 泰木薯	52000	/03/04	Rvtg		Rvtg	4#/5#	明穗（外贸）	10.9
13	GUO JIA NENG YUAN 609 国家能源609	200	CORN 玉米	64000	/03/04	Rvtg		Rvtg	2#/3#	内贸	11.7
14	DARYA RASHMI 达雅拉西米	229	WHEAT 加小麦	33000	0100/08/04	Rvtg		Rvtg	2#/3#	益海（外贸）	12.95
15	EAGLE EXPRESS 雄鹰快航	210	TAPIOCA 泰木薯	52000	/09/04	Rvtg		Rvtg	4#/5#	建发（外贸）	
16	MCQUEEN 海洋皇后	200	TAPIOCA 泰木薯	50000	/11/04	Rvtg		Rvtg	4#/5#	厦门港务 (外贸)	
17	AQUAHOLIC 艾克	229	SOYABEAN 巴西大豆	30000	/13/04	Rvtg		Rvtg	2#/3#	邦基（外贸）	

Figure 4.6: Data Recognition Issue

Solution:

The solution leverages **PaddleOCR**, a highly versatile and robust OCR tool designed to detect and extract data from various tables types, handling mixed text, numbers, dates, and specialized fields with ease. It supports multiple languages, including English and Chinese, making it effective for processing complex documents in any language. It can also manage irregular formatting, misrecognized characters, and missing values, ensuring accurate data extraction even from documents with varying layouts. At its core, PaddleOCR utilizes a **Convolutional Recurrent Neural Network (CRNN)**, which combines **Convolutional Neural Networks (CNNs)** to extract spatial features and **Recurrent Neural Networks (RNNs)** to capture the sequential dependencies of the text. **The Connectionist Temporal Classification (CTC)** loss function enables the model to predict text sequences efficiently without needing explicit alignment between the input and output, ensuring precise and reliable recognition of text within images.

4.2.1.3 Redundancy Elimination

Issues:

The extraction process, particularly with OCR tools like PaddleOCR, can sometimes result in redundant data due to overlapping bounding boxes or duplicated information, As the system may incorrectly identify the same piece of data multiple times or detect overlapping regions within the table, causing the same content to be extracted more than once. This redundancy can lead to inconsistencies in the final dataset.

Solution:

To address the redundancy issue, we used **Non-Maximum Suppression (NMS)**, a technique commonly used in object detection tasks. NMS helps eliminate overlapping bounding boxes by selecting the most relevant one and discarding others that overlap significantly. In the context of OCR, NMS works by comparing the bounding boxes around the detected text and removing duplicates, keeping only the most accurate representation of each piece of data. This process ensures that the final extracted data is clean and free from redundancy, making it easier to process and analyze.

4.2.1.4 Tables Review and correction

Issues:

Even though **PaddleOCR** efficiently extracts data, errors can still occur, especially in extra complex or noisy table images. Issues like low image quality, overlapping text, or intricate table structures can lead to mismatches during extraction.Therefore, a review and correction phase is essential to detect any mismatches or anomalies ,and ensure the data aligns accurately with the original content.

Solution

To address extraction errors, we leverage the capabilities of the **Mistral VLM (Vision-Language Model)** during the correction phase. By combining visual and linguistic understanding, Mistral VLM analyzes the extracted table and compares it with the original table image. It detects and corrects errors such as misrecognized characters, formatting inconsistencies, and misaligned rows or columns.

This process relies heavily on **prompt engineering**, which guides the model to interpret the table structure and content accurately. Carefully designed prompts help the model focus on structural alignment, cell matching, and semantic correction. The model ensures the corrected output closely matches the intended format by cross-referencing the visual structure with the extracted content.

To better understand the rationale behind selecting Mistral and to explore the detailed prompt engineering strategies used, please refer to the annex section 3. That section also presents a benchmark comparison of different models, highlighting the proven effectiveness of Mistral VLM in this different tasks.

4.2.1.5 Tables Traduction

Issues:

OCR can sometimes produce incorrect translations when converting non-English text, such as Chinese or Spanish, into English.This mistranslation can lead to inaccuracies, misinterpretations, or a loss of context, particularly when dealing with specialized terms, idiomatic expressions, or cultural nuances that may not have a direct equivalent in English.
For example, Chinese characters for quantities or dates may be mistranslated, or a Spanish term might be interpreted incorrectly in the context of a technical or business document which can compromise the overall accuracy of the extracted data.

Solution:

To address translation inaccuracies, we use **Mistral VLM** for translating non-English text into English.It is highly effective because it combines both visual and linguistic understanding, allowing it to capture the context and nuances of the original text more accurately than traditional OCR translation tools. By leveraging Mistral VLM, we can ensure that the translated data aligns with the original content's meaning, reducing the risk of misinterpretations.

4.2.1.6 Complete Process

YOLO, **PaddleOCR**, and **Mistral VLM** work together as **complementary** components to optimize table data extraction and correction. **YOLO** detects and locates the relevant table areas within images, while **PaddleOCR** efficiently extracts the textual data. **Mistral VLM** then ensures the accuracy of the extracted information by identifying and correcting any errors. This combined approach results in the most accurate and structured table data output.

Figure 4.7 illustrates the detailed final solution pipeline.

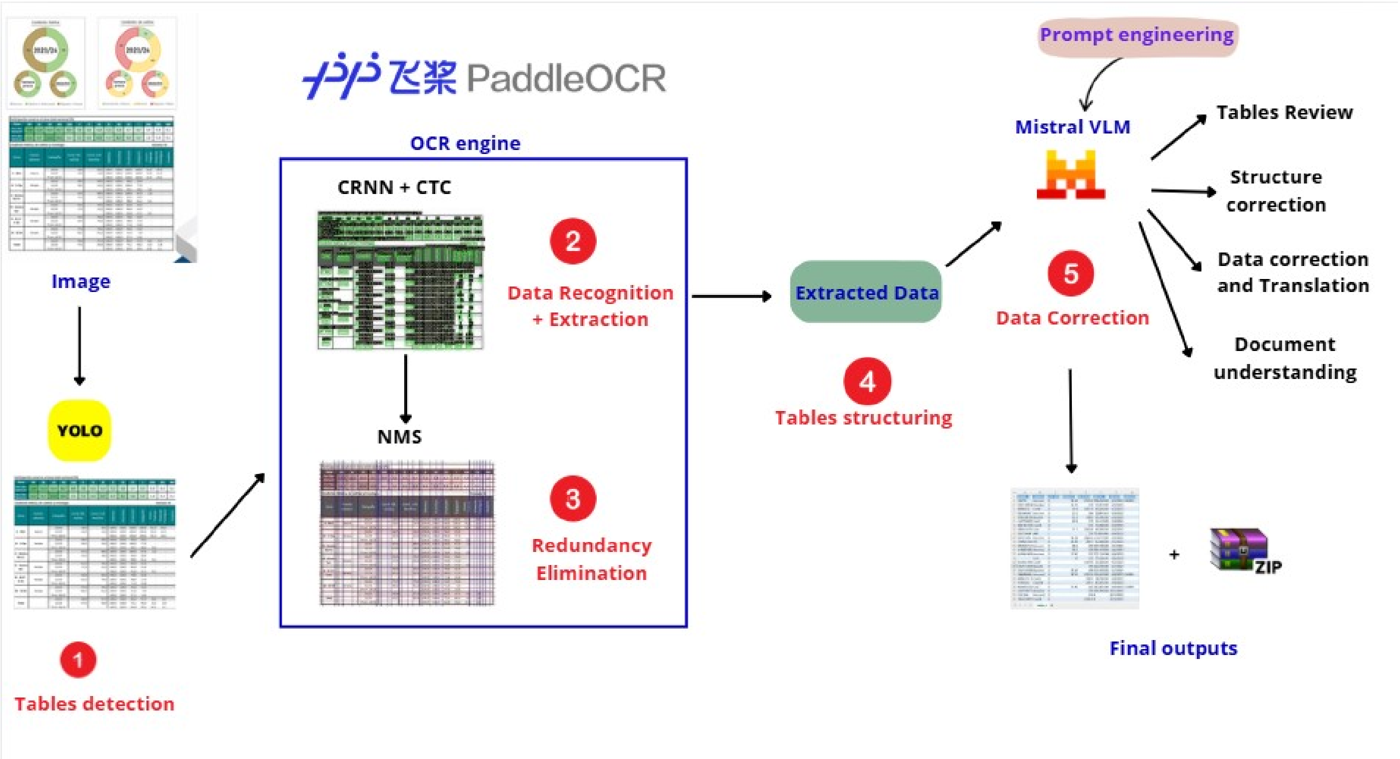


Figure 4.7: Generic Solution Pipeline

This Final Generic Solution was selected for its strong balance between performance, flexibility, and resource efficiency. It delivers accurate results across diverse and complex table formats while remaining lightweight and capable of running in low-resource environments even **without GPU support**. Its speed and reliability make it well-suited for real-time or large-scale document processing.

More than just a table extraction tool, our solution offers a complete, **AI-powered platform** designed to support analysts throughout their entire workflow. It empowers users not only to extract data, but also to manipulate it, reshape tables, ask complex questions in natural language, and automatically generate comprehensive reports with statistics, visualizations, and summaries **(see section 6.2.8)**.
With real-time table preview, interactive querying, and advanced automation features, this **All-in-one** system provides analysts with everything they need to efficiently transform unstructured documents into actionable insights.

4.2.2 Other Solutions

In this section, we present specialized solutions developed during the internship, tailored to meet the specific needs of different teams and table languages. These solutions are categorized by language, addressing the unique challenges posed by each language to ensure effective data extraction and processing. Nevertheless, our generic solution pipeline remains a language-independent generalized solution and presents the best performances **(see section 5.3.4)**.

4.2.2.1 English Tables pipeline

The primary team involved in this solution was the Sugar Analysts team. This team is responsible for scraping and analyzing sugar prices, market trends, and other related data. Their work is crucial for providing insights into the global sugar market, supporting decisions regarding pricing, purchasing, and sales strategies.

The solution was designed using **PaddleOCR** (as shown in figure 4.8), a powerful OCR tool. PaddleOCR was chosen due to its strong performance in detecting and recognizing text in images, even for complex and unstructured tables. This tool was employed to detect, recognize, and extract the tables directly from the images of documents containing sugar market data.

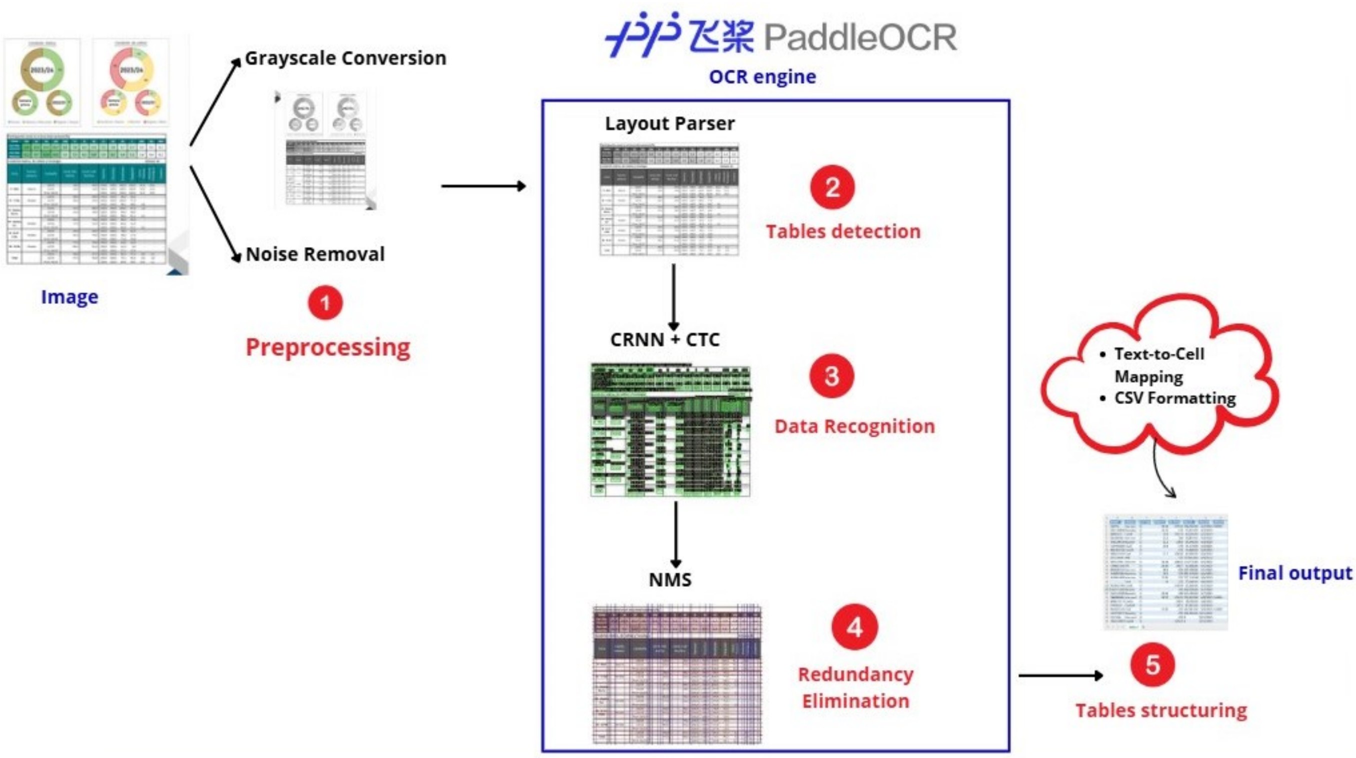


Figure 4.8: English Solution Pipeline

This solution differs from the final generic solution primarily in the **preprocessing step**, which has been adjusted to better handle the specific data characteristics. Overall, this solution significantly improved the productivity and accuracy of the Sugar Analysts team, allowing them to process large volumes of data with minimal human intervention **(See section 5.3.4 for the accuracy results and 5.1.2 for the ground truth dataset)**. From this solution onward, **we kept PaddleOCR** in **the final solution** as it proved its effectiveness in accurately recognizing and extracting data from complex table formats.

4.2.2.2 Chinese Solution Pipeline

This solution is designed for teams like **Customer Success** and **Linepus**, which work with complex Chinese tables in images. The Customer Success team analyzes feed prices and trends, while the Linepus team tracks vessels and manages product data. Both teams deal with intricate tables containing Chinese characters, which are difficult to extract and translate using traditional methods.

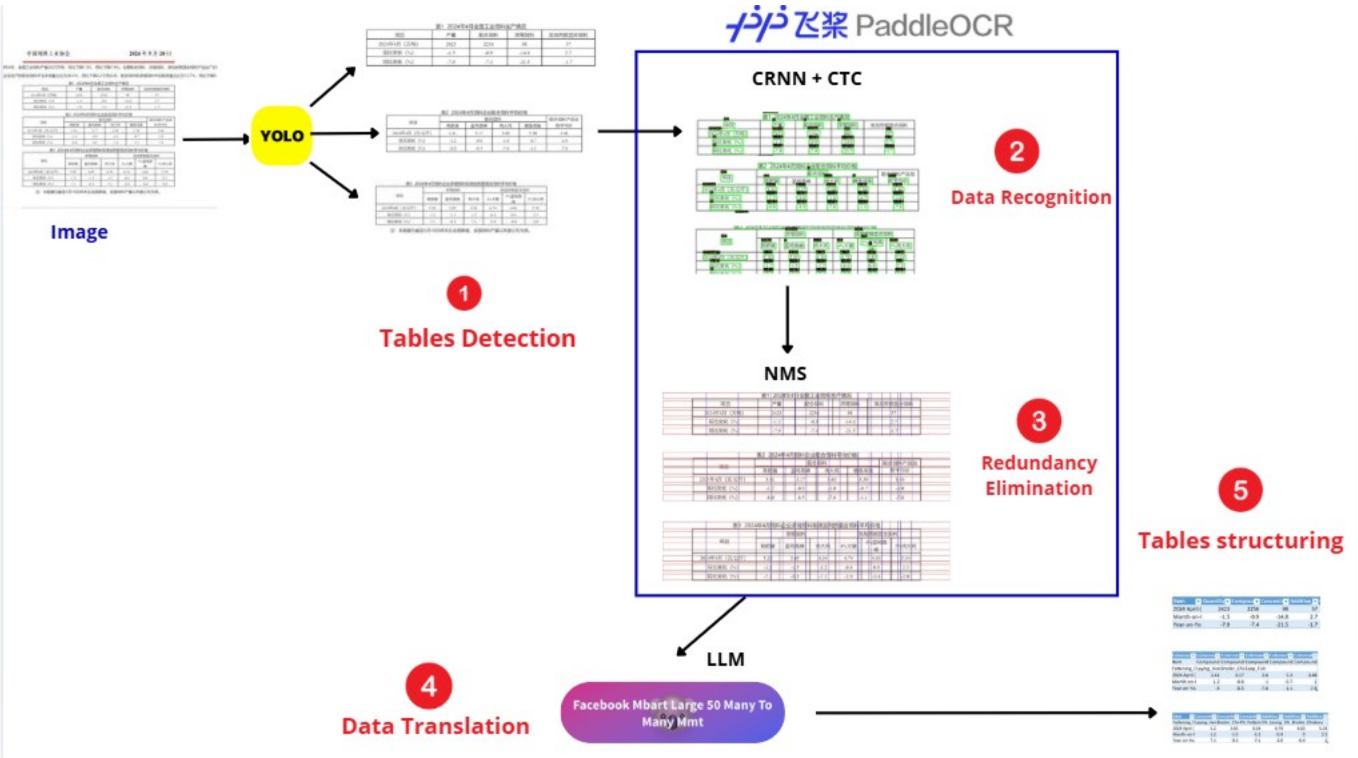


Figure 4.9: Chinese Solution Pipeline

Dataset Creation and Annotation: For this solution, we trained YOLO on our dataset **(You can see the dataset details in section 5.1.4)**, which was created using the team's data to ensure it was tailored to the specific requirements of the task.We also created a ground truth consisting of image-text pairs for accurate annotation and training. This ground truth base was crucial for testing the accuracy and effectiveness of the solution.

Data Translation using LLM: Before experimenting with **VLMs**, we tested **LLMs** to ensure the translated data was accurate and comprehensible. Since the data was originally in Chinese, we incorporated an LLM to ensure that the extracted information was understandable to the teams. **PaddleOCR** alone was insufficient for extracting the Chinese text accurately. To address this, we employed Facebook's **mBART-Large-50-Many-to-Many-MMT** (as shown in figure 4.9), a large multilingual model, to translate the extracted CSV data into English. This made it possible for both the **Customer Success** team and the **Linepus team** to work with the data seamlessly, facilitating better analysis and decision-making.

From this solution architecture we kept **YOLO** in the final solution due to its proven effectiveness in table detection.

The solution enhanced efficiency and accuracy **(See section 5.3.4 for the accuracy results**) by automating table detection and data extraction.

4.2.2.3 Spanish Solution Pipeline

This solution specifically supports teams like the **Wheat Team** and **Crop Analysis Team**, who work with Spanish language agricultural data. It automates the extraction process, improving efficiency and accuracy in handling complex, embedded table content.

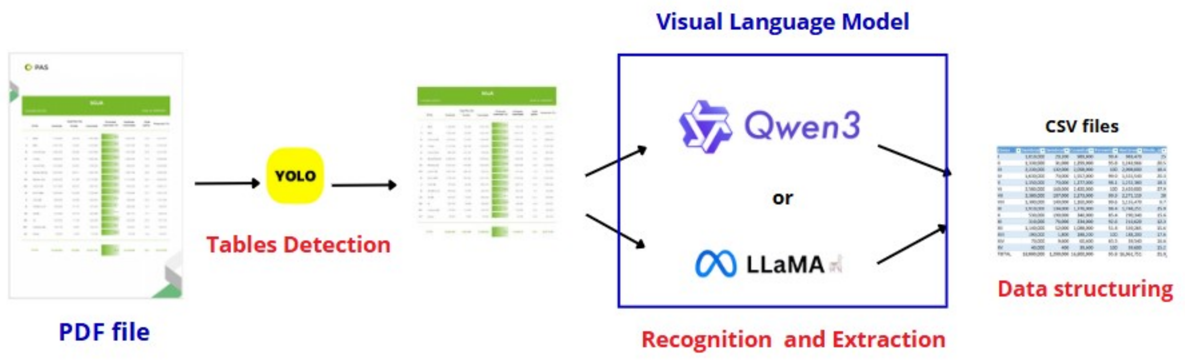


Figure 4.10: Spanish Solution Pipeline

Dataset Creation and Annotation:
For this solution, we trained YOLO on our dataset **(see section 5.1.4 for the dataset details)**, which was created using the team's data to ensure it was tailored to the specific requirements of the task.We also created a ground truth consisting of image-text pairs for accurate annotation and training. This ground truth base was crucial for testing the accuracy and effectiveness of the solution.

Table(s) Extraction: To extract tables from the detected regions, we employed **VLMs** (as shown in figure 4.10), which are well-suited for understanding both the layout and the data within tables. These models are able to interpret the visual structure and the semantic relationships between elements such as headers, rows, and columns, making them highly effective for extracting tables from complex documents.

We benchmarked multiple **VLMs** and fine-tuned two models, **QWEN** [16] and **LLAMA** [17] (see Annex section 4) , to improve their accuracy for extracting tables from Spanish documents. The fine-tuning process showed promising results, significantly enhancing the models performance. However, due to **GPU and resource limitations** in **Colab Notebook**, we could not retain the fine-tuned VLM approach in the final solution.

To overcome these limitations, **we kept the VLM** component for table extraction but replaced the original model with a more resource-efficient alternative, **the Mistral-small VLM**. This allowed us to maintain the table extraction functionality while optimizing performance within our computational constraints. **(See section 5.3.4 for the accuracy results**).

Conclusion

This chapter detailed the development of our AI-based solution for table extraction, covering its core components from detection and recognition to correction and translation. We also introduced specialized pipelines designed to handle different use cases and document types.

Chapter 5

Experimental Protocol

Introduction

This chapter presents the experimental protocol used to develop and evaluate our table detection and recognition solution. It covers the datasets, evaluation metrics, parameters configurations, followed by an overview of the results obtained from various experiments.

5.1 Datasets

This section defines and presents the custom-built datasets used in our solution. These datasets were specifically created to support the training, fine-tuning, and evaluation of the model, with a focus on understanding and extracting information from complex tabular data, particularly in multilingual contexts.

5.1.1 Dnexr-generic-solution Dataset

Table 5.1 illustrates the detailed characteristics of the Dnexr-generic-solution dataset, which is used by the generic solution.

Characteristic	**Description**
Dataset name	`Dnexr-generic-solution`
Total number of Images	2,009 images
Split	1,407 training / 408 validation / 194 test
Image format	JPG
Resolution	640 $\times$ 640 pixels
Source	PDF and image-based documents
Language	Multiple languages (English, Chinese, and others)
Number of classes	2 (Tables, Graphs)
Document age	Mix of recent and historical documents
Image type	Numerical and scanned images

Table 5.1: Dnexr-generic-solution Dataset

Figure 5.1 shows a sample from the Dnexr-generic-solution Dataset , featuring various images extracted from different PDF documents.

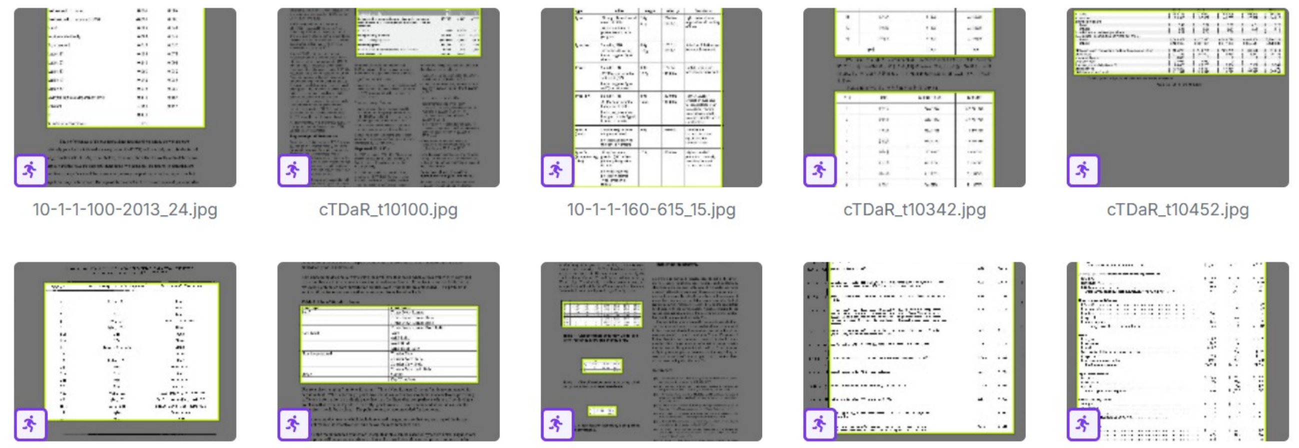


Figure 5.1: Dnexr-generic-solution Dataset Sample

5.1.2 Dnexr-Sugar-English Dataset

Table 5.2 illustrates the detailed characteristics of the Dnexr-Sugar-English, which is used by the English solution.

Characteristic	**Description**
Dataset name	`Dnexr-Sugar-English`
Total number of Images	164 images
Split	115 training / 33 validation / 16 test
Format	Image and text
Image Resolution	640 × 640 pixels
Language	English
Document age	Recent images
Image type	Numerical and scanned images

Table 5.2: Dnexr-Sugar-English Dataset

Figure 5.2 shows a sample from the ground truth Dnexr-Sugar-English Dataset, featuring various images extracted from different PDF documents.

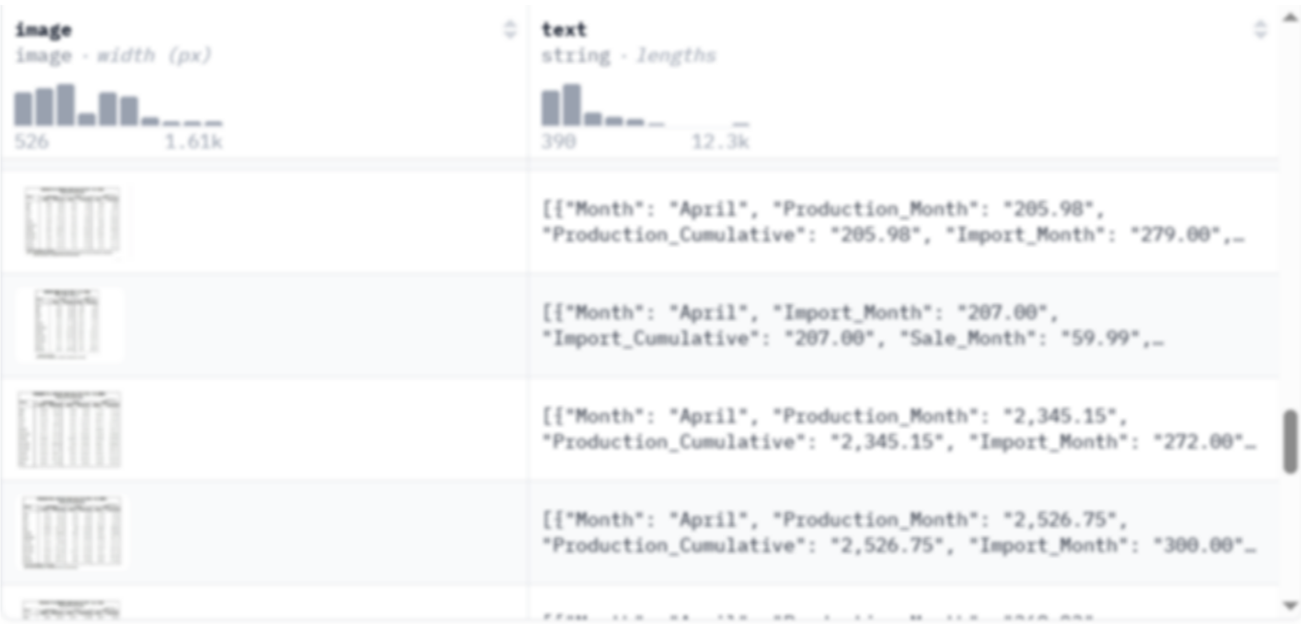


Figure 5.2: Dnexr-Sugar-English Dataset Sample

5.1.3 Dnexr-lienups-Chinese Dataset

Table 5.3 illustrates the detailed characteristics of the Dnexr-lienups-Chinese dataset, which primarily contains Chinese text but may also include some English text integrated within the Chinese tables, and is used by the Chinese solution.

Characteristic	**Description**
Dataset name	`Dnexr-lienups-Chinese`
Total number of documents	200 images
Split	140 training / 40 validation / 20 test
Image format	JPG
Resolution	640 $\times$ 640 pixels
Source	PDF and image-based documents
Language	Chinese and English
Number of classes	1 (Tables)
Document age	Mix of recent and historical documents
Image type	Numerical images

Table 5.3: Dnexr-lienups-Chinese Dataset

Figure 5.3 shows a sample from the Dnexr-lienups-Chinese dataset, featuring various images extracted from different PDF documents.

Figure 5.3: Dnexr-lienups-Chinese Dataset Sample

5.1.4 Dnexr-Wheat-Spanish Dataset

Table 5.4 illustrates the detailed characteristics of the Dnexr-Wheat-Spanish dataset, which is used by the Spanish solution.

Characteristic	**Description**
Dataset name	`Dnexr-Wheat-Spanish`
Total number of documents	342 images
Split	239 training / 69 validation / 34 test
Image format	JPG
Resolution	640 × 640 pixels
Source	PDF and image-based documents
Language	Spanish
Number of classes	3 (Borderless, Bordered, Cell)
Document age	Mix of recent and historical documents
Image type	Numerical and scanned

Table 5.4: Dnexr-Wheat-Spanish Dataset

Figure 5.4 shows a sample from the Dnexr-Wheat-Spanish dataset, featuring various images extracted from different PDF documents.

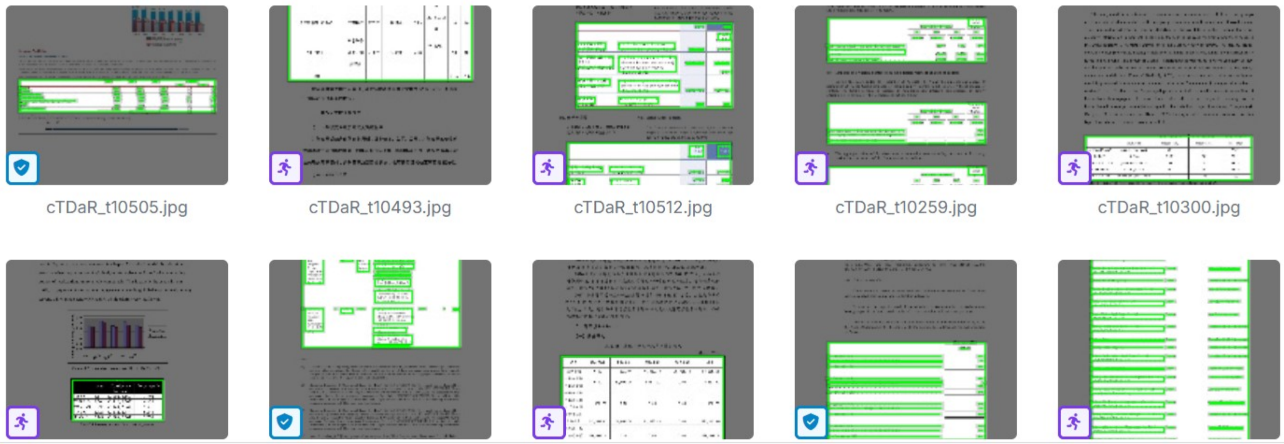


Figure 5.4: Dnexr-Wheat-Spanish Dataset Sample

5.2 Evaluation Metrics for Deep Learning Algorithms

To evaluate the performance of a deep learning-based recognition system, especially in object detection and classification tasks, we use several widely accepted evaluation metrics. These help compare models and select the most optimal one for our pipeline.

5.2.1 Classification Metrics

Accuracy — Refers to the proportion of correctly predicted instances out of the total number of predictions made. It is the most commonly used metric for evaluating classification problems.

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \tag{5.1}$$

Precision — Indicates the proportion of true positive predictions among all predicted positives. It helps evaluate how accurate the positive predictions are.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.2}$$

In multiclass classification, precision is computed for each class i, and then the overall average is calculated across all n classes as follows:

$$\text{Precision}_t = \frac{\sum_{i=1}^{n} \text{precision}_i}{n} \tag{5.3}$$

Recall (or sensitivity) —Measures the proportion of actual positives that were correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5.4}$$

5.2.2 Object Detection Metrics

In object detection tasks, the evaluation metrics are slightly different as they combine both classification and localization aspects. Below are some important metrics used to assess object detection models:

- **Intersection over Union (IoU)** — IoU is a measure of the overlap between the predicted bounding box and the ground truth box. It is calculated as:

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \tag{5.5}$$

A higher IoU indicates better localization of the predicted bounding box.

- **Loss Functions for Object Detection**

  - **Box Loss**: Box loss measures the error in the predicted bounding box coordinates, and it is often calculated using the Mean Squared Error (MSE) between the predicted and ground truth coordinates. This helps the model learn better localization of the objects. The formula for box loss is:

$$\text{box-loss} = \frac{1}{N}\sum_{i=1}^{N}\left((\hat{x}_i - x_i)^2 + (\hat{y}_i - y_i)^2 + (\hat{w}_i - w_i)^2 + \left(\hat{h}_i - h_i\right)^2\right) \tag{5.6}$$

    where $\hat{x}_i, \hat{y}_i, \hat{w}_i, \hat{h}_i$ are the predicted bounding box coordinates, and x_i, y_i, w_i, h_i are the ground truth bounding box coordinates. The loss measures the error between the predicted and actual bounding box coordinates, typically using Mean Squared Error (MSE).

- **Mean Average Precision(mAP)** mAP is a popular evaluation metric used in object detection. It is deioud as the average of precision values at different recall thresholds, providing a comprehensive assessment of both precision and recall across multiple classes.

  - **mAP at IoU 50** (mAP50) — This specific metric evaluates the precision at an IoU threshold of 50%. It is a commonly used metric in object detection tasks and can be calculated as:

$$mAP_{50} = \frac{1}{N}\sum_{i=1}^{N}\text{Precision}_i \tag{5.7}$$

Where N is the number of categories, and Precision_i is the precision for the i-th class at an IoU threshold of 50%. The value typically ranges from 0 to 1, with higher values indicating better detection performance.

5.2.3 Characters Recognition Metrics

- **WER (Word Error Rate)** measures the accuracy at the word level and is calculated as:

$$WER = \frac{S + D + I}{N}$$

where S is substitutions, D is deletions, I is insertions, and N is the total number of words in the reference.

  - S (Substitutions): The number of words in the output that are incorrect and need to be replaced by the correct words from the reference.
  - D (Deletions): The number of words in the output that are missing compared to the reference.
  - I (Insertions): The number of extra words in the output that aren't in the reference.

- **CER (Character Error Rate)** measures accuracy at the character level and is calculated as:

$$CER = \frac{S + D + I}{N}$$

where S is substitutions, D is deletions, I is insertions, and N is the total number of characters in the reference.

5.3 Experiments

5.3.1 Detection performance

- **Parameters Configuration**

Table 5.1 presents the configuration of a set of parameters for the YOLOv12 learning model used in the tables detection experiment.

Parameter	Value
Batch Size	16
Learning Rate	0.01
Number of Classes	2

Table 5.5: Training parameters of detection models

- **Results Evaluation** Once the training is started, we can monitor the variation of generated metrics such as mean average Precision (mAP) and the Loss function.

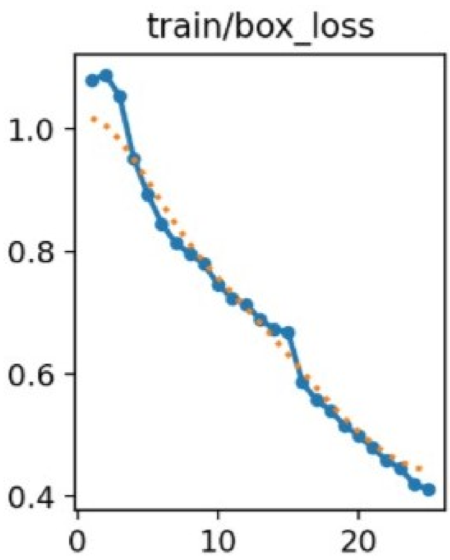


Figure 5.5: train box loss

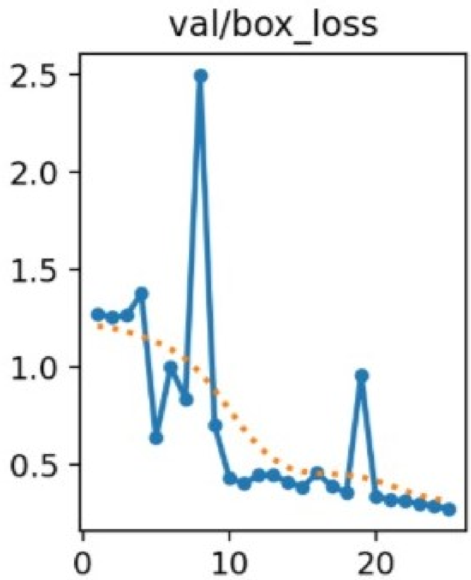


Figure 5.6: validation box loss

Figure 5.5 and 5.6 illustrates the loss curves for box, class, and dfl during both training and validation. Initially, the loss values fluctuate as the model adjusts to complex data. However, as training progresses, the loss values stabilize, indicating that the model has learned the essential patterns in the data. Once stabilized, the model's performance improves at a slower rate, suggesting it has effectively reached an optimal state. This stabilization signifies that the model is now more accurate in tasks like table detection, demonstrating improved reliability for table extraction from documents.

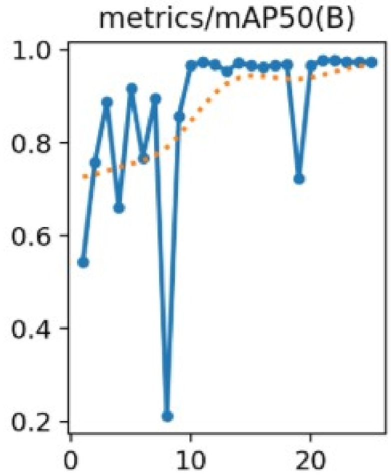


Figure 5.7: Mean Average Precision (mAP)

- **Evaluation of Mean Average Precision (mAP):** The figure illustrates the evolution of the mAP50 metric **(see section 5.2.2)** throughout the training process. The curve shows a rapid increase during the initial epochs, stabilizing after approximately 20 epochs and reaching a high value of **0.985**.

  This performance trend indicates that the model quickly learned to detect and localize objects with high precision, achieving strong accuracy and demonstrating effective generalization on the training data.

YOLO Models Performance comparison on Dnexr-generic-solution dataset

Model Name	**mAP50**	**Precision**	**Recall**
YOLOv4	92.1%	76%	74%
YOLOv8	96.1%	84%	80.6%
YOLOv11	97.2%	87%	82.5%
YOLOv12	**98.5%**	**93%**	**87%**

Table 5.6: YOLO Models Performance comparison on Dnexr-generic-solution dataset

In this comparison table 5.6 , we evaluated the performance of four YOLO models: YOLOv4, YOLOv8, YOLOv11, and YOLOv12 on the pdf-table-extraction dataset. **YOLOv12** stands out with the highest mAP50 of **98.5%**, precision of **93%**, and recall of **87%**, demonstrating superior accuracy in detecting tables. While YOLOv11 performs well with good balance in precision and recall, it offers even better results, making it the ideal choice for our table detection task.

We opted for **YOLOv12** due to its exceptional performance in both detection accuracy and speed, ensuring optimal results in real time environments.

Visual Test Results with YOLOv12

Figures 5.10 and 5.11 show the visual detection results of tables using the YOLOv12 . These figures highlight the model's ability to accurately detect and classify the "table" class in images.

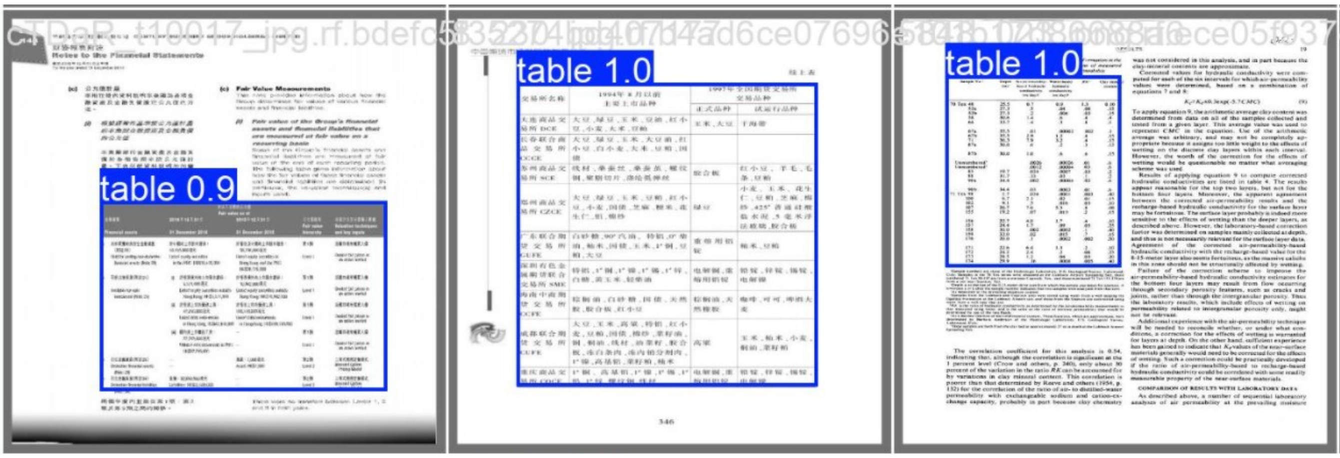


Figure 5.8: Yolov12 Detection results 1

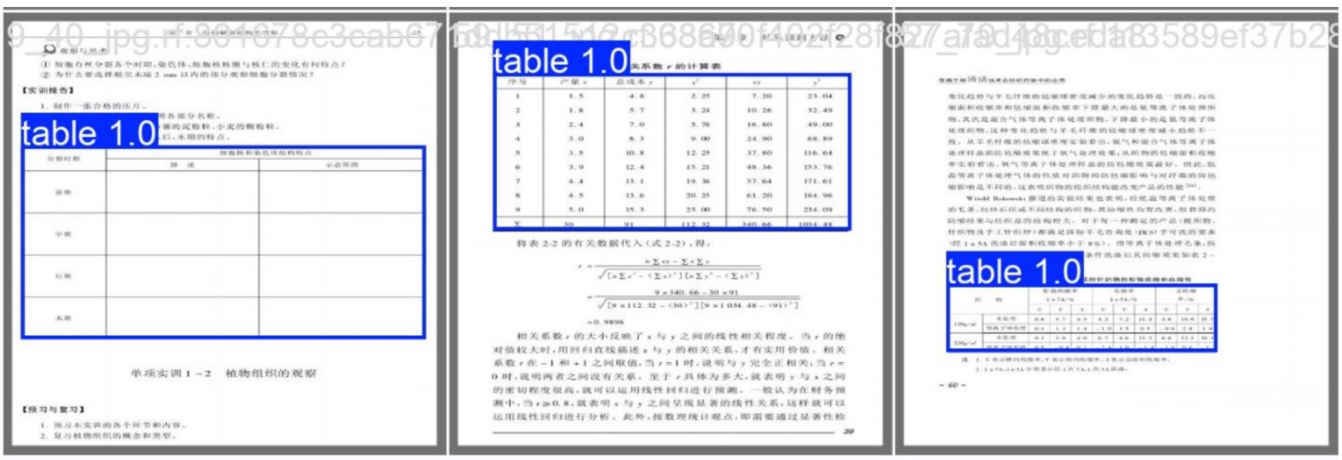


Figure 5.9: Yolov12 Detection results 2

5.3.2 Recognition Performance

This section evaluates the performance of the OCR systems used, including EasyOCR, PPOCR, and Tesseract, assessing their accuracy in text recognition. We will then present the performance of our final solution, integrating PaddleOCR, and compare its results with the other models.

Figure 5.10 represents a scanned photo of a table from the **1970s**, which is complex and challenging to detect due to the quality of the image, In this context, we evaluate the results of various OCR models in detecting and recognizing the text. The performance of each model is assessed based on its ability to handle such a challenging image.

UPLAND COTTON

STATE	AREA PLANTED			
	1977	1978	IND 1979	1979/1978
		1,000 ACRES		PERCENT
ALA	420.0	335.0	380.0	113
ARIZ	517.0	540.0	600.0	111
ARK	950.0	820.0	840.0	102
CALIF	1,400.0	1,480.0	1,650.0	111
GA	230.0	120.0	140.0	117
KY	.9	.3	.2	67
LA	545.0	515.0	490.0	95
MISS	1,380.0	1,180.0	1,300.0	110
MO	270.0	210.0	235.0	112
N C	87.0	45.0	75.0	167
OKLA	535.0	600.0	600.0	100
S C	170.0	105.0	125.0	119
TENN	325.0	250.0	260.0	104
TEX	6,650.0	6,950.0	7,200.0	104
VA	1.0	.2	.5	250
TOTAL-STATES SURVEYED	13,480.9	13,150.5	13,895.7	105.7
STATES NOT SURVEYED	138.5	141.1		
U S	13,619.4	13,291.6		

SUGARBEETS

STATE	AREA PLANTED			
	1977	1978	IND 1979	1979/1978
		1,000 ACRES		PERCENT
ARIZ 1/	12.9	15.7	11.8	75
CALIF 1/	227.0	207.0	215.0	104
COLO	77.0	89.0	80.0	90
IDAHO	115.4	136.3	100.0	73
KANS	26.0	28.0	16.0	57
MICH	92.3	93.0	93.0	100
MINN	264.0	265.0	254.0	96
MONT	[illegible]	[illegible]	[illegible]	[illegible]

Figure 5.10: Test Image

Figure 5.11 shows the results of data detection and recognition using PaddleOCR. As illustrated in the figure, PaddleOCR excels in both detecting and recognizing text, providing very accurate results even in complex or degraded images. The model effectively handles various challenges such as distortions, low resolution, and unusual fonts, ensuring high-quality output.

To see the results of EasyOCR and PaddleOCR, refer to the annex in section 2.

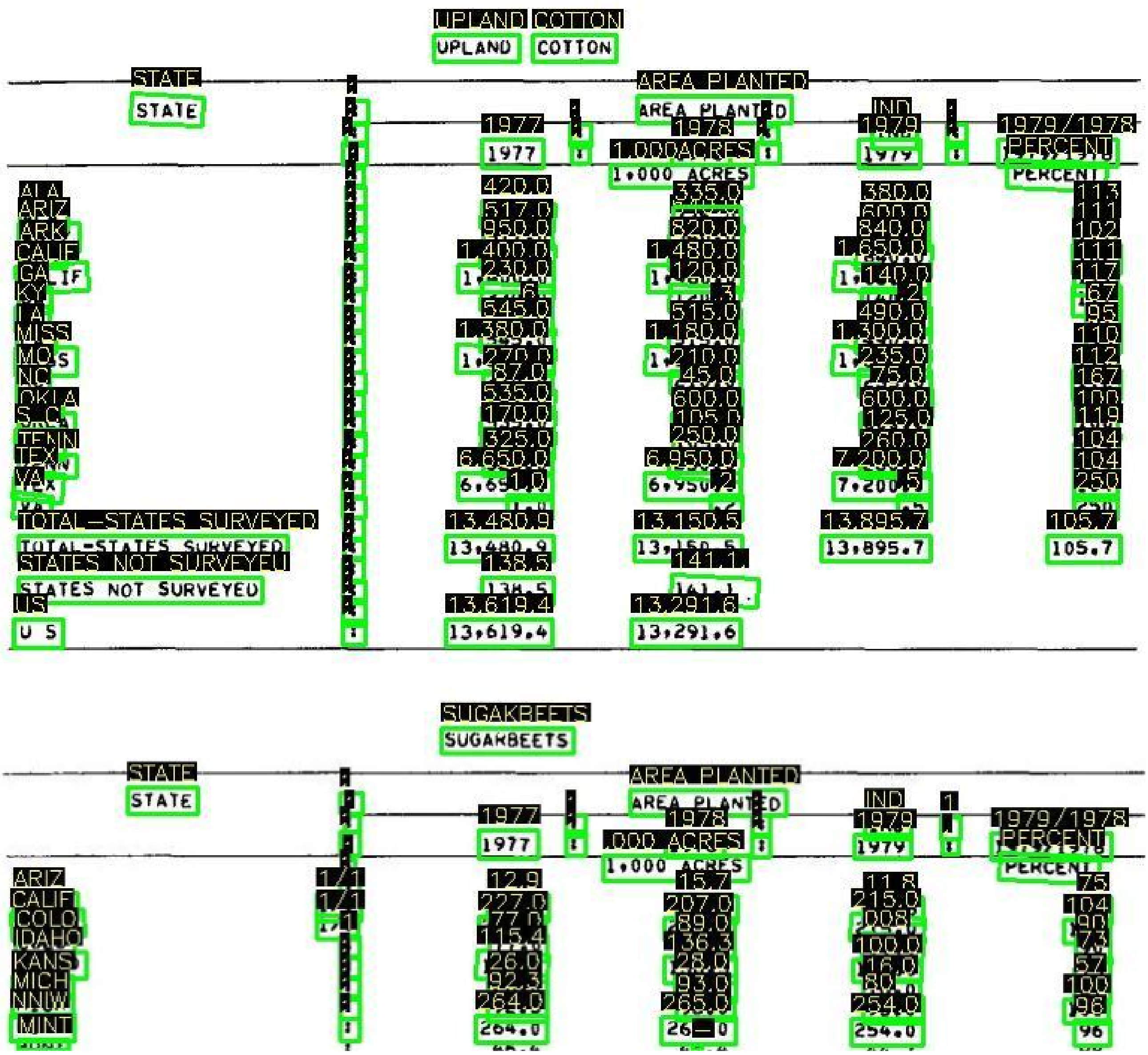


Figure 5.11: PaddleOCR Detection and Recognition results

Figure 5.12 presents the **confidence scores** associated with the text recognition results from multiple models. These scores indicate the level of certainty the model has in its recognition, with higher percentages signifying greater accuracy.

As seen in the table, **PaddleOCR** consistently achieves the highest **confidence scores**, outperforming other models. This demonstrates its superior ability in both text detection and recognition, as it effectively handles challenging conditions such as low image quality, distortions, and complex fonts.

The higher confidence values suggest that **PaddleOCR** is more reliable and accurate in delivering correct results, making it the most effective model for our solustion, as illustrated by the results.

Detected Area	EasyOCR	PaddleOCR	Tesseract
UPLAND COTTON	UpLanu COTTON(69.23%)	UPLAND COTTON (100%)	UPLAND COTTON (100%)
STATE	STATE (100%)	STATE (100%)	STATE (100%)
1977	1971 (75%)	1977 (100%)	1977 (100%)
AREA PLANTED	Area PLANIED(66.67%)	AREA PLANTED (100%)	Area PLANTED(75%)
IND 1979	IND979 (75.0%)	IND1979 (85.0%)	1 1979 (62.5%)
420.0	Not Detected	420.0 (100%)	420 (60%)
TOTAL-STATES SURVEYED	Tutal-StaTES SURVEYED(60.67%)	TOTAL-STATES SURVEYED (100%)	TOTAL=STATES SUKVEYED(90.48%)
6,950.0	950 (42.86%)	6.950.0 (100%)	619500 (71.43%)
STATES NOT SURVEYED	StATEs NOT surveyev (47.37%)	STATES NOT SURVEYED (100%)	Not Detected
MO	Not Detected	MO (100%)	"O (50%)

Figure 5.12: OCR models confidence score comparison

For further evaluation we used the **WER** (Word Error Rate) and **CER** (Character Error Rate) metrics to evaluate the performance of three OCR engines on our reference dataset **(see section 5.2.3)**. They represent the error rates at the word and character levels respectively.

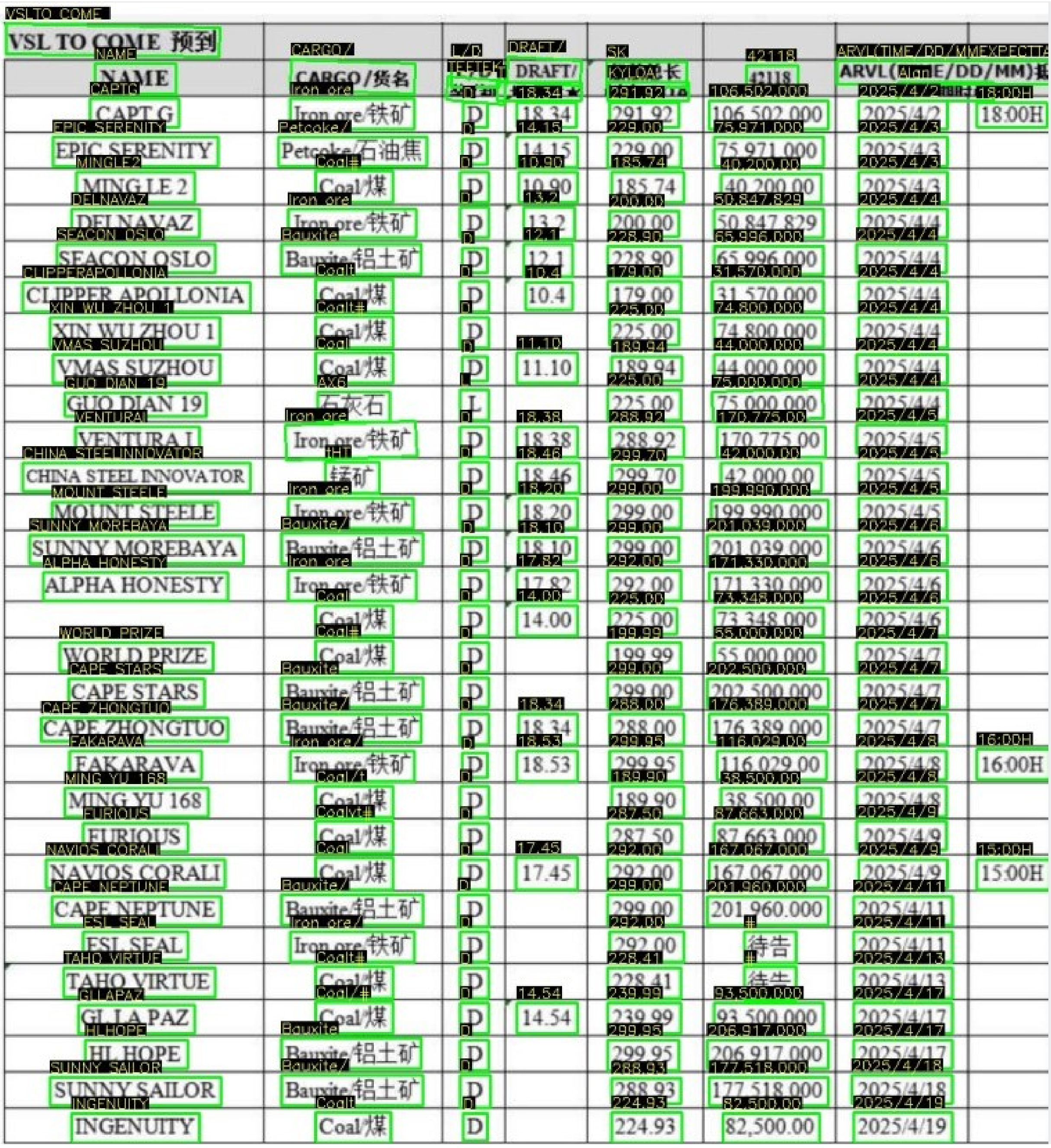
Engine	**WER**	**CER**
EasyOCR	0.65	0.52
Tesseract	0.56	0.49
PaddleOCR	**0.36**	**0.29**

Table 5.7: WER and CER scores for OCR engines on Dnexr-generic-solution dataset

- **EasyOCR** has a WER of 0.65 and a CER of 0.52, indicating a relatively high error rate compared to the others.
- **Tesseract** shows a WER of 0.56 and a CER of 0.49, which is slightly better than EasyOCR in terms of word error rate but worse in terms of character error rate.
- **PaddleOCR** performs the best overall, with a WER of 0.36 and a CER of 0.29, showing the lowest error rates for both word and character levels.

As the table 5.7 indicates, **PaddleOCR** provides the most accurate results among the three engines, both at the word and character levels.

Figure 5.13 provides another example of complex English and Chinese data tables, successfully detected and recognized using our chatbot solution. This demonstrates the system's ability to handle mixed languages and numerical data, showcasing its effectiveness in extracting accurate information from complex document layouts. The ability to process such multi-lingual content emphasizes the robustness and versatility of our solution in real-world applications.

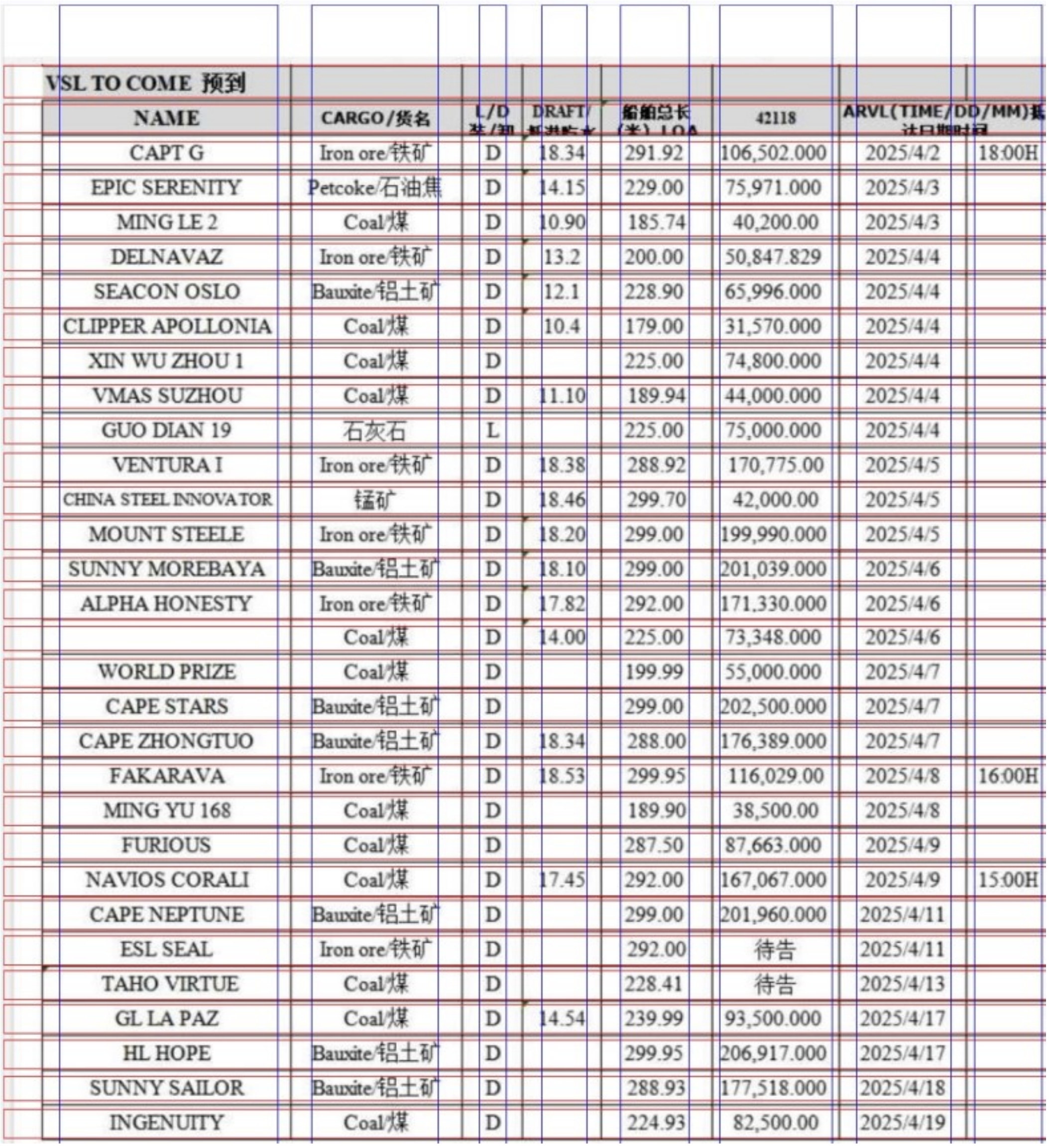


VSL TO COME 预到

NAME	CARGO/货名	[illegible]	DRAFT/	[illegible]长	[illegible]	ARVL(TIME/DD/MM)	
CAPT G	Iron ore/铁矿	D	18.34	291.92	106,502.000	2025/4/2	18:00H
EPIC SERENITY	Petcoke/石油焦	D	14.15	229.00	75,971.000	2025/4/3	
MING LE 2	Coal/煤	D	10.90	185.74	40,200.00	2025/4/3	
DELNAVAZ	Iron ore/铁矿	D	13.2	200.00	50,847.829	2025/4/4	
SEACON OSLO	Bauxite/铝土矿	D	12.1	228.90	65,996.000	2025/4/4	
CLIPPER APOLLONIA	Coal/煤	D	10.4	179.00	31,570.000	2025/4/4	
XIN WU ZHOU 1	Coal/煤	D		225.00	74,800.000	2025/4/4	
VMAS SUZHOU	Coal/煤	D	11.10	189.94	44,000.000	2025/4/4	
GUO DIAN 19	石灰石	L		225.00	75,000.000	2025/4/4	
VENTURA I	Iron ore/铁矿	D	18.38	288.92	170,775.00	2025/4/5	
CHINA STEEL INNOVATOR	锰矿	D	18.46	299.70	42,000.00	2025/4/5	
MOUNT STEELE	Iron ore/铁矿	D	18.20	299.00	199,990.000	2025/4/5	
SUNNY MOREBAYA	Bauxite/铝土矿	D	18.10	299.00	201,039.000	2025/4/6	
ALPHA HONESTY	Iron ore/铁矿	D	17.82	292.00	171,330.000	2025/4/6	
	Coal/煤	D	14.00	225.00	73,348.000	2025/4/6	
WORLD PRIZE	Coal/煤	D		199.99	55,000.000	2025/4/7	
CAPE STARS	Bauxite/铝土矿	D		299.00	202,500.000	2025/4/7	
CAPE ZHONGTUO	Bauxite/铝土矿	D	18.34	288.00	176,389.000	2025/4/7	
FAKARAVA	Iron ore/铁矿	D	18.53	299.95	116,029.00	2025/4/8	16:00H
MING YU 168	Coal/煤	D		189.90	38,500.00	2025/4/8	
FURIOUS	Coal/煤	D		287.50	87,663.000	2025/4/9	
NAVIOS CORALI	Coal/煤	D	17.45	292.00	167,067.000	2025/4/9	15:00H
CAPE NEPTUNE	Bauxite/铝土矿	D		299.00	201,960.000	2025/4/11	
ESL SEAL	Iron ore/铁矿	D		292.00	待告	2025/4/11	
TAHO VIRTUE	Coal/煤	D		228.41	待告	2025/4/13	
GL LA PAZ	Coal/煤	D	14.54	239.99	93,500.000	2025/4/17	
HL HOPE	Bauxite/铝土矿	D		299.95	206,917.000	2025/4/17	
SUNNY SAILOR	Bauxite/铝土矿	D		288.93	177,518.000	2025/4/18	
INGENUITY	Coal/煤	D		224.93	82,500.00	2025/4/19	

Figure 5.13: Tables Recognition results

Figure 5.14 demonstrates the impact of **Non-Maximum Suppression (NMS)** on OCR results. NMS effectively eliminates overlapping bounding boxes by retaining only the most confident detections, leading to clearer text representation and reduced false positives. The confusion matrix further shows the improvement in detection accuracy, particularly in complex layouts with closely spaced or overlapping text.



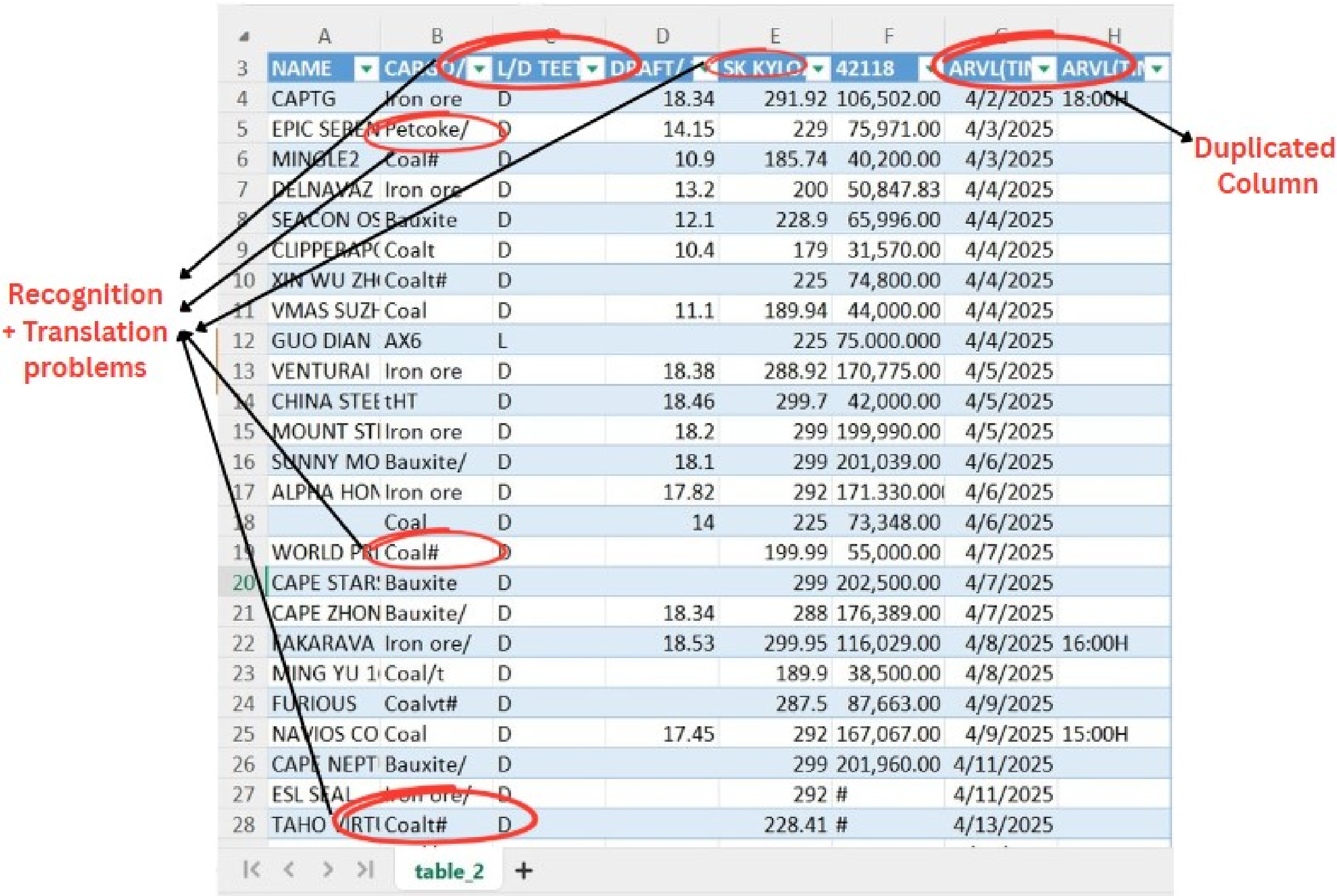
VSL TO COME 预到							
NAME	CARGO/货名	L/D 装/卸	DRAFT/抵港吃水	船舶总长(米) LOA	42118	ARVL(TIME/DD/MM)抵达日期时间	
CAPT G	Iron ore/铁矿	D	18.34	291.92	106,502.000	2025/4/2	18:00H
EPIC SERENITY	Petcoke/石油焦	D	14.15	229.00	75,971.000	2025/4/3	
MING LE 2	Coal/煤	D	10.90	185.74	40,200.00	2025/4/3	
DELNAVAZ	Iron ore/铁矿	D	13.2	200.00	50,847.829	2025/4/4	
SEACON OSLO	Bauxite/铝土矿	D	12.1	228.90	65,996.000	2025/4/4	
CLIPPER APOLLONIA	Coal/煤	D	10.4	179.00	31,570.000	2025/4/4	
XIN WU ZHOU 1	Coal/煤	D		225.00	74,800.000	2025/4/4	
VMAS SUZHOU	Coal/煤	D	11.10	189.94	44,000.000	2025/4/4	
GUO DIAN 19	石灰石	L		225.00	75,000.000	2025/4/4	
VENTURA I	Iron ore/铁矿	D	18.38	288.92	170,775.00	2025/4/5	
CHINA STEEL INNOVATOR	锰矿	D	18.46	299.70	42,000.00	2025/4/5	
MOUNT STEELE	Iron ore/铁矿	D	18.20	299.00	199,990.000	2025/4/5	
SUNNY MOREBAYA	Bauxite/铝土矿	D	18.10	299.00	201,039.000	2025/4/6	
ALPHA HONESTY	Iron ore/铁矿	D	17.82	292.00	171,330.000	2025/4/6	
	Coal/煤	D	14.00	225.00	73,348.000	2025/4/6	
WORLD PRIZE	Coal/煤	D		199.99	55,000.000	2025/4/7	
CAPE STARS	Bauxite/铝土矿	D		299.00	202,500.000	2025/4/7	
CAPE ZHONGTUO	Bauxite/铝土矿	D	18.34	288.00	176,389.000	2025/4/7	
FAKARAVA	Iron ore/铁矿	D	18.53	299.95	116,029.00	2025/4/8	16:00H
MING YU 168	Coal/煤	D		189.90	38,500.00	2025/4/8	
FURIOUS	Coal/煤	D		287.50	87,663.000	2025/4/9	
NAVIOS CORALI	Coal/煤	D	17.45	292.00	167,067.000	2025/4/9	15:00H
CAPE NEPTUNE	Bauxite/铝土矿	D		299.00	201,960.000	2025/4/11	
ESL SEAL	Iron ore/铁矿	D		292.00	待告	2025/4/11	
TAHO VIRTUE	Coal/煤	D		228.41	待告	2025/4/13	
GL LA PAZ	Coal/煤	D	14.54	239.99	93,500.000	2025/4/17	
HL HOPE	Bauxite/铝土矿	D		299.95	206,917.000	2025/4/17	
SUNNY SAILOR	Bauxite/铝土矿	D		288.93	177,518.000	2025/4/18	
INGENUITY	Coal/煤	D		224.93	82,500.00	2025/4/19	

Figure 5.14: NMS results

5.3.3 Correction Performance

The first table, shown in Figure 5.15, was extracted using **Paddle OCR**, which can still make errors due to its lack of contextual understanding. While NMS helps reduce text detection redundancies, **Paddle OCR** can struggles with very complex layouts, merged headers, and translation inaccuracies, as it lacks the intelligence to interpret the data logically.

Despite these limitations, Paddle OCR can still facilitate the work of more advanced systems like Mistral, providing a preliminary structure for the data.



	A	B	C	D	E	F	G	H
3	NAME	CARGO/	L/D TEET	DRAFT/	SK KYLO	42118	ARVL(TIN	ARVL(T
4	CAPTG	Iron ore	D	18.34	291.92	106,502.00	4/2/2025	18:00H
5	EPIC SEREN	Petcoke/	D	14.15	229	75,971.00	4/3/2025	
6	MINGLE2	Coal#	D	10.9	185.74	40,200.00	4/3/2025	
7	DELNAVAZ	Iron ore	D	13.2	200	50,847.83	4/4/2025	
8	SEACON OS	Bauxite	D	12.1	228.9	65,996.00	4/4/2025	
9	CLIPPERAP(	Coalt	D	10.4	179	31,570.00	4/4/2025	
10	XIN WU ZH(	Coalt#	D		225	74,800.00	4/4/2025	
11	VMAS SUZH	Coal	D	11.1	189.94	44,000.00	4/4/2025	
12	GUO DIAN	AX6	L		225	75.000.000	4/4/2025	
13	VENTURAI	Iron ore	D	18.38	288.92	170,775.00	4/5/2025	
14	CHINA STEE	tHT	D	18.46	299.7	42,000.00	4/5/2025	
15	MOUNT STI	Iron ore	D	18.2	299	199,990.00	4/5/2025	
16	SUNNY MO	Bauxite/	D	18.1	299	201,039.00	4/6/2025	
17	ALPHA HON	Iron ore	D	17.82	292	171.330.00(	4/6/2025	
18		Coal	D	14	225	73,348.00	4/6/2025	
19	WORLD PR	Coal#	D		199.99	55,000.00	4/7/2025	
20	CAPE STAR:	Bauxite	D		299	202,500.00	4/7/2025	
21	CAPE ZHON	Bauxite/	D	18.34	288	176,389.00	4/7/2025	
22	TAKARAVA	Iron ore/	D	18.53	299.95	116,029.00	4/8/2025	16:00H
23	MING YU 1(	Coal/t	D		189.9	38,500.00	4/8/2025	
24	FURIOUS	Coalvt#	D		287.5	87,663.00	4/9/2025	
25	NAVIOS CO	Coal	D	17.45	292	167,067.00	4/9/2025	15:00H
26	CAPE NEPT	Bauxite/	D		299	201,960.00	4/11/2025	
27	ESL SEAL	Iron ore/	D		292	#	4/11/2025	
28	TAHO VIRT(	Coalt#	D		228.41	#	4/13/2025	

Figure 5.15: Initial table by Paddleocr

The second table, shown in Figure 5.16, was corrected by the Mistral system and showcases the advantages of using advanced language modeling (VLMs). Unlike **Paddle OCR**, Mistral can better understand the data's structure and context, providing accurate translations and properly handling complex tabular layouts, correcting errors such as misaligned rows, incorrect text formatting, and incomplete data, resulting in a clearer and more accurate table.

	A	B	C	D	E	F	G
1	Source Document: third						
2	NAME	CARGO	L/D	LOA	LENGTH	QTTV(MT)	ARV[illegible]DD/MM)
3	CAPT G	Iron ore	D	18.34	291.92	106,502.00	2025/4/2 18.00H
4	EPIC SEREN	Petcoke	D	14.15	229	75,971.00	4/3/2025
5	MING LE 2	Coal	D	10.9	185.74	40,200.00	4/3/2025
6	DELNAVAZ	Iron ore	D	13.2	200	50,847.83	4/4/2025
7	SEACON OS	Bauxite	D	12.1	228.9	65,996.00	4/4/2025
8	CLIPPER AP	Coal	D	10.4	179	31,570.00	4/4/2025
9	XIN WU ZH	Coal	D		225	74,800.00	4/4/2025
10	VMAS SUZI	Coal	D	11.1	189.94	44,000.00	4/4/2025
11	GUO DIAN	Limestone	L		225	75,000.00	4/4/2025
12	VENTURA I	Iron ore	D	18.38	288.92	170,775.00	4/5/2025
13	CHINA STEE	Rare earth	D	18.46	299.7	42,000.00	4/5/2025
14	MOUNT ST	Iron ore	D	18.2	299	199,990.00	4/5/2025
15	SUNNY MO	Bauxite	D	18.1	299	201,039.00	4/6/2025
16	ALPHA HOI	Iron ore	D	17.82	292	171,330.00	4/6/2025
17		Coal	D	14	225.00	10/25/2100	4/6/2025
18	WORLD PRI	Coal	D		199.99	55,000.00	4/7/2025
19	CAPE STAR	Bauxite	D		299	202,500.00	4/7/2025
20	CAPE ZHON	Bauxite	D	18.34	288	176,389.00	4/7/2025
21	FAKARAVA	Iron ore	D	18.53	299.95	116,029.00	2025/4/8 16.00H
22	MING YU 1	Coal	D		189.9	38,500.00	4/8/2025
23	FURIOUS	Coal	D		287.5	87,663.00	4/9/2025
24	NAVIOS CO	Coal	D	17.45	292	167,067.00	2025/4/9 15.00H
25	CAPE NEPT	Bauxite	D		299	201,960.00	4/11/2025
26	ESL SEAL	Iron ore	D		292	To be advisec	4/11/2025
27	TAHO VIRT	Coal	D		228.41	To be advisec	4/13/2025
28	GL LA PAZ	Coal	D	14.54	239.99	93,500.00	4/17/2025
29	HL HOPE	Bauxite	D		299.95	206,917.00	4/17/2025
30	SUNNY SAIL	Bauxite	D		288.93	177,518.00	4/18/2025
31	INGENUITY	Coal	D		224.93	82,500.00	4/19/2025

table_3_mistral (2)

Figure 5.16: Final Corrected table by Mistral

This demonstrates how Paddle OCR and **Mistral** can be **complementary**: while Paddle OCR helps in extracting and structuring the data, Mistral can easily review and correct them, ultimately leading to more accurate and readable results.

5.3.4 Comparison of Generic and Language-Specific Solutions Performance

Language	Solution	Accuracy (%)
English	Generic Solution	**98.2**
	English Solution	95.0

Table 5.8: Accuracy comparison of the Generic and English solution on Dnexr-Sugar-English dataset

Table 5.8 showcases the remarkable performance of the Generic Solution, achieving an impressive accuracy of 98.2%, compared to the specialized English Solution, which reached 95.0%, all evaluated on the ground truth Dnexr-Sugar-English dataset **(see dataset details 5.1.2)**. The Generic Solution proves to be a powerful approach.

Language	Solution	Accuracy (%)
Chinese	Generic Solution	**96.6**
	Chinese Solution	92.0

Table 5.9: Accuracy comparison of the Generic and Chinese solution on Dnexr-lienups-Chinese dataset

Table 5.9 demonstrates the exceptional accuracy of the Generic Solution, achieving 96.6% when evaluated on the ground truth Dnexr-lienups-Chinese dataset **(see dataset details 5.1.3)**, far surpassing the Chinese Solution own accuracy of 92.0%. The Generic Solution proves its superiority across different datasets.

Language	Solution	Accuracy (%)
Spanish	Generic Solution	**97.0**
	Spanish Solution	93.1

Table 5.10: Accuracy comparison of the Generic and Spanish solution on Dnexr-Wheat-Spanish datase

Table 5.10 highlights the remarkable performance of the Generic Solution, with an outstanding accuracy of 97.0%, evaluated on the ground truth Dnexr-Wheat-Spanish dataset **(see dataset details 5.1.4)**, compared to the Spanish Solution 93.1%.
This further showcases the impressive power of the Generic Solution in outperforming specialized solutions on their own datasets.

5.3.5 Exceeding Industry Standards

One of the most notable achievements of our solution is its ability to accurately extract tables that leading AI platforms like **ChatGPT Pro and DeepSeek** often struggle with. These models fail to handle heavily distorted images, misaligned tables, complex or nested table structures, and data with mixed languages or noisy backgrounds. We tested both models by passing the same table with identical prompts, but **ChatGPT 4.0** was unable to recognize or extract the data correctly, highlighting a significant difference in performance between the two models in processing the same information.

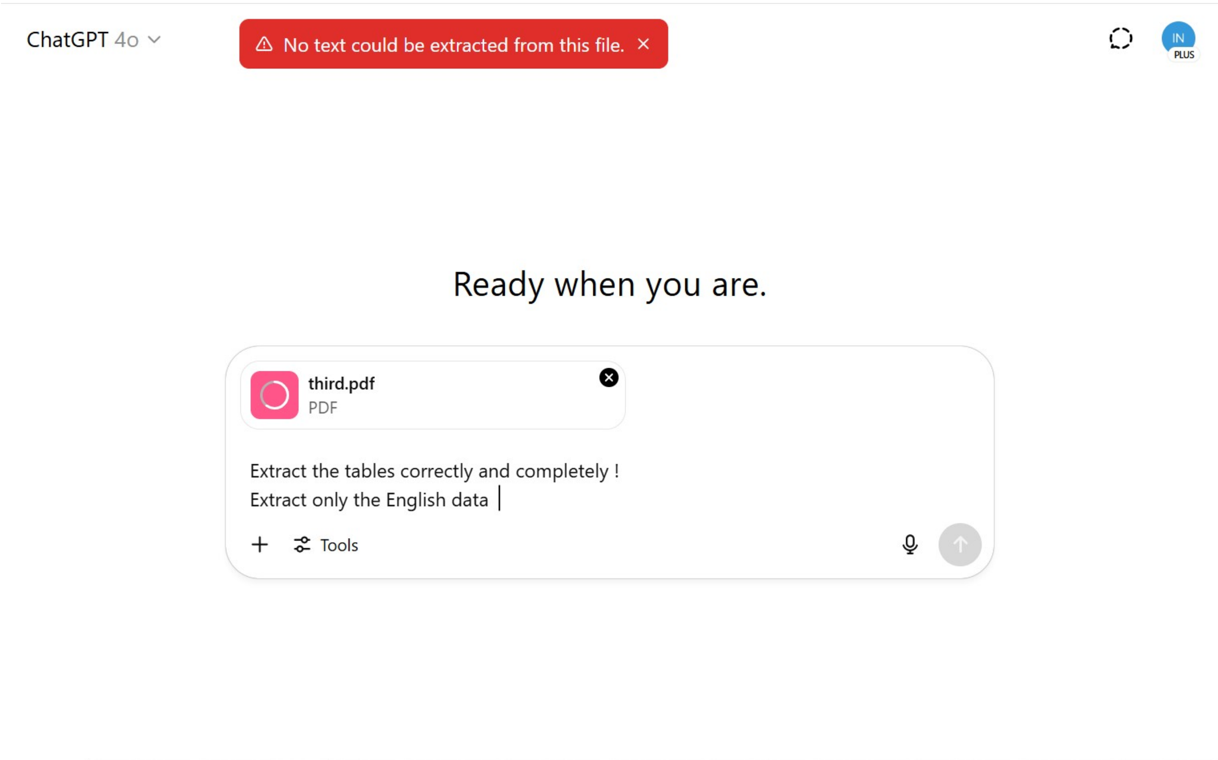


Figure 5.17: Chatgpt Failure

Conclusion

This chapter presented the datasets used for training and evaluation, along with the applied evaluation metrics. We also conducted experiments to assess the performance of various models. The results highlighted the superior performance of the Generic Solution over language-specific models, demonstrating its effectiveness in handling complex data, including multilingual and challenging document layouts. These experiments validate the robustness and versatility of our approach.

Chapter 6

Technologies and Deployment

Introduction

This chapter covers the used technologies and deployment model of our solution. We first describe the working environment and then provide an overview of the deployment architecture and the web interfaces that facilitate user interaction with the system.

6.1 Working Environment

The working environment refers to the collection of tools, hardware, and technical resources necessary for the development and deployment of the proposed solution.

6.1.1 Technologies

Figure 5.1 displays the logos of various tools and frameworks used in the development of the project.

Figure 6.1: Development tools

- **GitHub[18]:** A web-based platform for version control and collaboration, enabling developers to store and manage code, track changes, and collaborate on projects.
- **Google Colab[19]:** A free cloud-based environment for Python programming, providing access to GPUs for data science, machine learning, and AI development. It is the most practical approach. It is a notebook for running code. However, since its resources are free, it imposes GPU usage limits (12 hours). We used this tool to implement our solution and to test several models before selecting which ones to adopt.
- **Gradio[20]:** An open-source Python library for building easy-to-use web interfaces for machine learning models, enabling quick testing and deployment. It was used to develop the front end of the platform, including the chatbot interface, allowing users to interact with the system seamlessly.
- **Visual Studio Code [21]:** A popular, lightweight source code editor with support for many programming languages, debugging, and version control integration.
- **Roboflow[22]:** A platform that helps with image processing, especially for building and deploying computer vision models, with tools for dataset management and model training.
- **Hugging Face[23]:** A leading company in Natural Language Processing (NLP) that provides pre-trained models, datasets, and tools for machine learning, primarily in NLP.

6.1.2 Implementation Tools

We present, in Table 5.1, the characteristics of Google Colab notebook used during the implementation of our solution.

Components	**Description**
Processor	Xeon CPU
RAM	12.7 GB
GPU RAM	15 GB
Disk Space	112.6 GB
Operating System	Linux

Table 6.1: Google Colab Machine Specifications

6.1.3 Deployment Diagram

The deployment diagram illustrates the combination of a software's logical and physical architecture. It shows how software components are deployed on physical resources , providing a comprehensive view of the system's hardware and software composition.

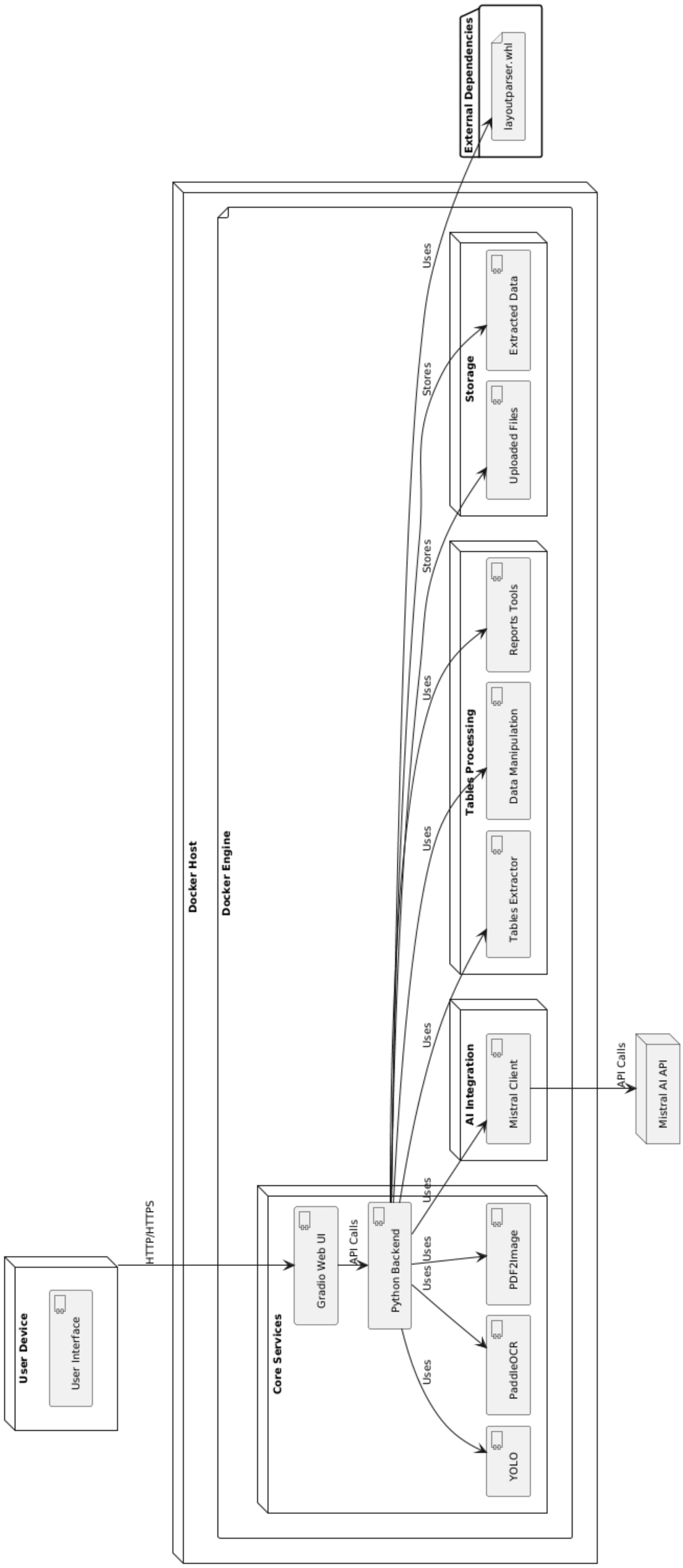


Figure 6.2: Deployment Diagram

The physical architecture of our software follows a containerized deployment approach using Docker. Each component of the system is deployed in its own container to ensure efficient management and scalability.

- **User Device:** The user's device communicates with the system via a web interface using HTTP/HTTPS.
- **Docker Host:** The Docker host manages all the services and ensures smooth communication between containers. The Docker engine is responsible for containerizing and managing all the components.
- **Core Services:**
  - **Gradio Web UI:** The Gradio web UI allows users to interact with the system.
- **Python Backend:** The Python backend orchestrates all services and integrates various AI models and processing tools, such as:
  - **YOLO:** For object detection in images.
  - **PaddleOCR:** For optical character recognition.
  - **PDF2Image:** For converting PDFs to images for further processing.
- **AI Integration:** The Python backend also integrates with the **Mistral Client**, which interacts with the **Mistral AI API** for additional AI-based processing such as correction and translation.
- **Tables Processing:** This module handles the processing of tables extracted from documents:
  - **Tables Extractor:** Extracts tables from documents.
  - **Data Manipulation:** Performs operations on the extracted table data, such as data correction, merging CSV files, reshaping (melting/pivoting) tables, and other preprocessing tasks.
  - **Reports Tools:** Handles the generation and manipulation of reports.
- **Storage:** The system stores two main categories of data:
  - **Uploaded Files:** Contains the raw uploaded files from users.

   - **Extracted Data:** Stores the processed and extracted data and tables from the files.

- **External Dependencies:** The system depends on an external package **layout-parser.whl** for parsing layouts of documents.

All components are interconnected within the Docker Engine,interacting via API calls. Data is processed, stored, and visualized within the system.

6.2 Graphic interfaces

6.2.1 Files Processing Interface

The Files Processing Interface is a component of the chatbot-driven data analysis system that allows an analyst to upload files and optionally specify advanced settings, such as custom prompts or selected pages and page ranges for processing. Once the input is submitted, the chatbot returns processed results, including CSV files and other outputs generated by the pipeline. Finally, the analyst has the option to clear the session to reset the environment for a new analysis.

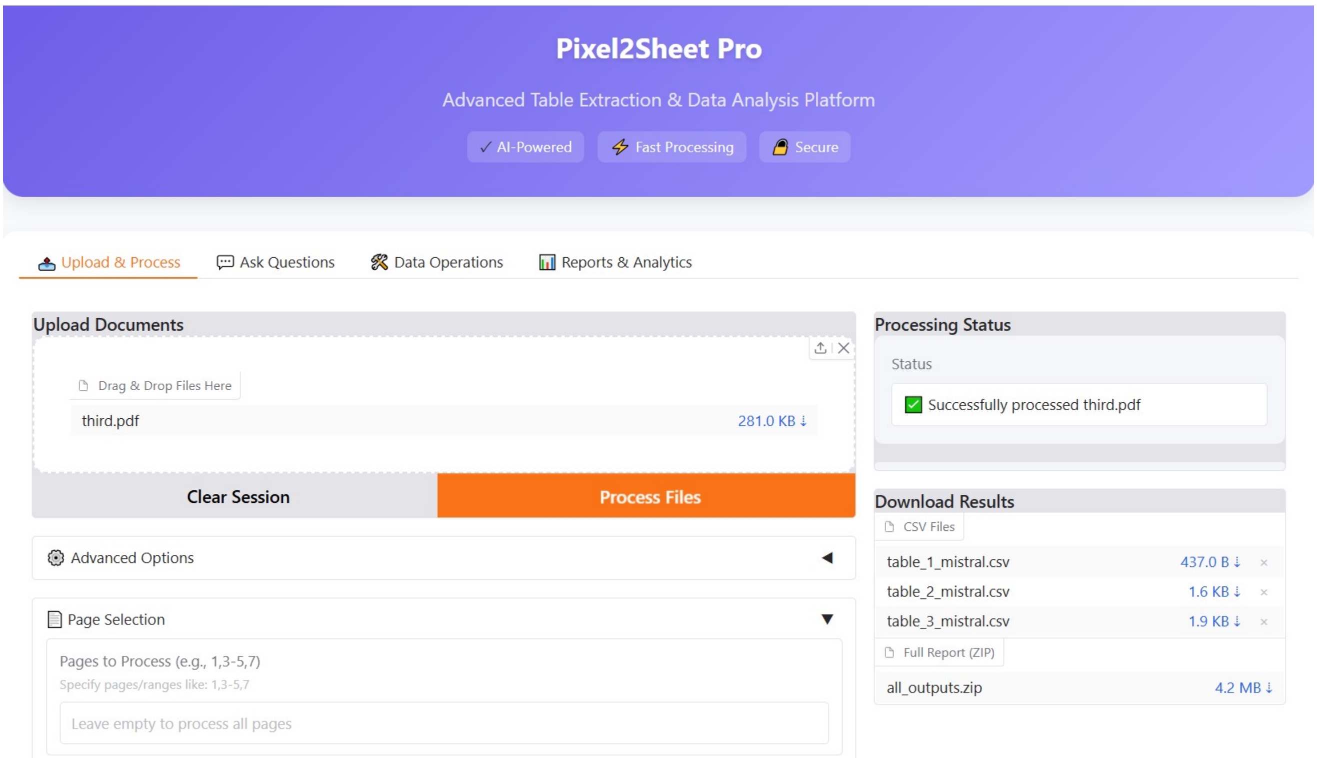


Figure 6.3: Files Processing Interface

6.2.2 Preview and Edit Tables Interface

The Preview and Edit Tables interface is designed for **Previewing and Editing Extracted Tables**, where users can:

- Select the table to preview (example: *Table 1*).
- View a fully editable preview of the extracted table, allowing users to inspect and make necessary adjustments to the data.
- Modify the table structure, such as sorting columns in ascending or descending order, clearing sort settings, adding new columns to the left or right, deleting columns, adding new rows, and modifying cell values etc...
- Save changes made to the table or download it in CSV formats for further use.
- Refresh the preview to see any updates or changes made to the table.

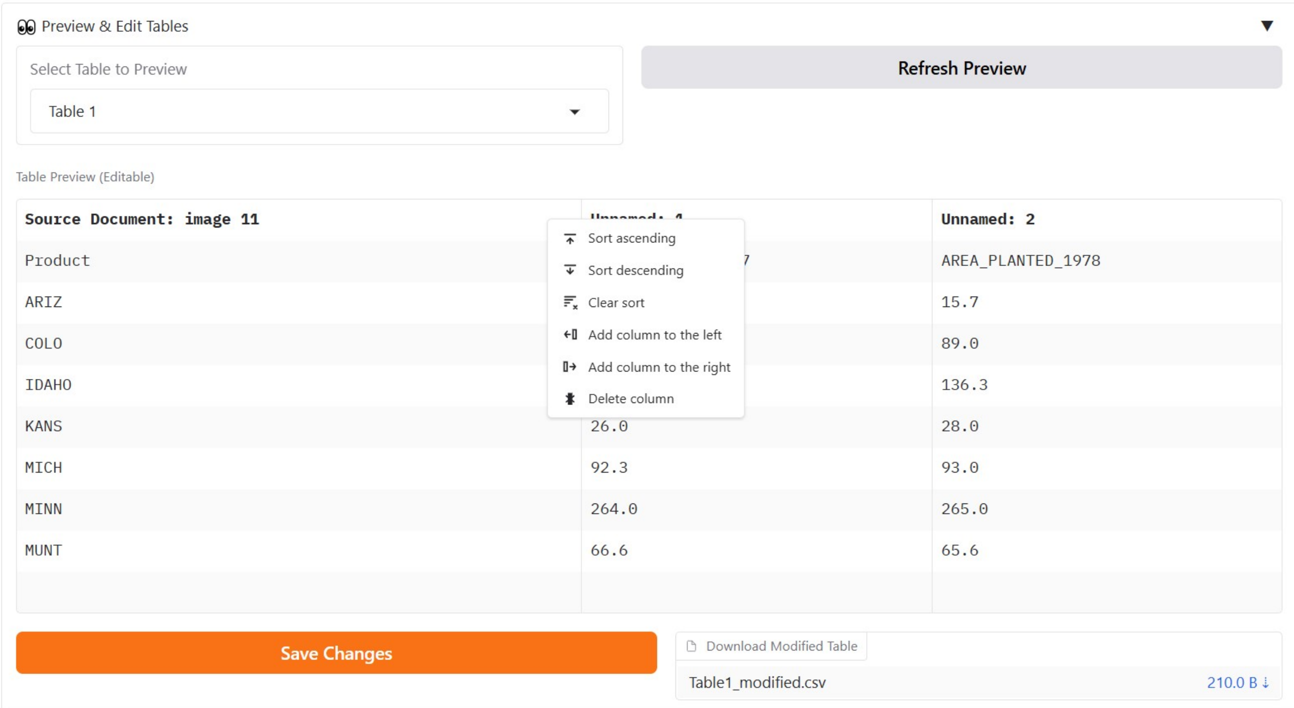


Figure 6.4: Preview and Edit Tables Interface

6.2.3 Ask Questions Interface

The interface allows users to ask specific questions about the uploaded data, such as comparing tables, extracting insights, and identifying trends or anomalies.

Users can input custom questions or use example questions, and the system processes the data to provide answers based on the queries made, offering detailed comparisons or relevant insights.

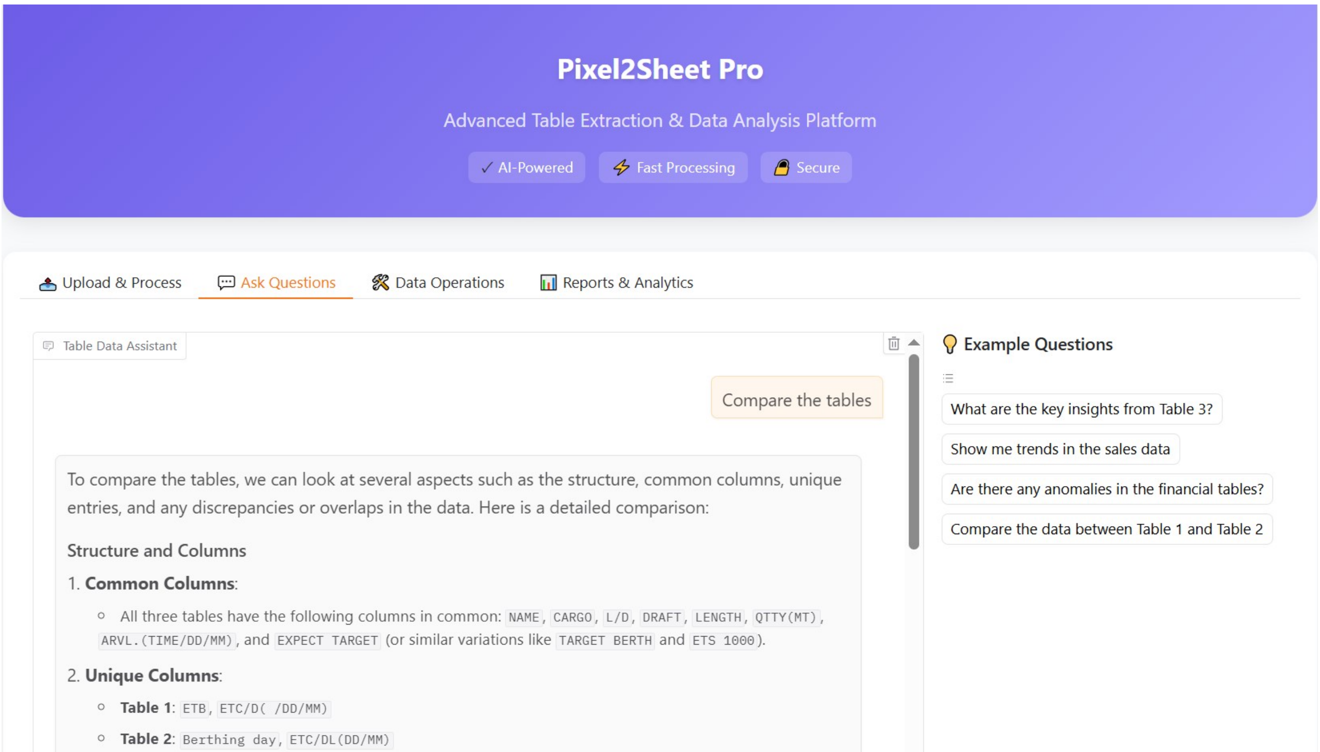


Figure 6.5: Ask Questions Interface

6.2.4 Data Operations Interface

The interface presents data operations options, allowing users to:

- **Correct Data**: Fix or modify data entries.
- **Merge Tables**: Combine multiple tables into one.
- **Regenerate Tables**: Recreate tables from the existing data.

These operations enable users to perform essential data adjustments and transformations directly within the platform, streamlining the data analysis process.

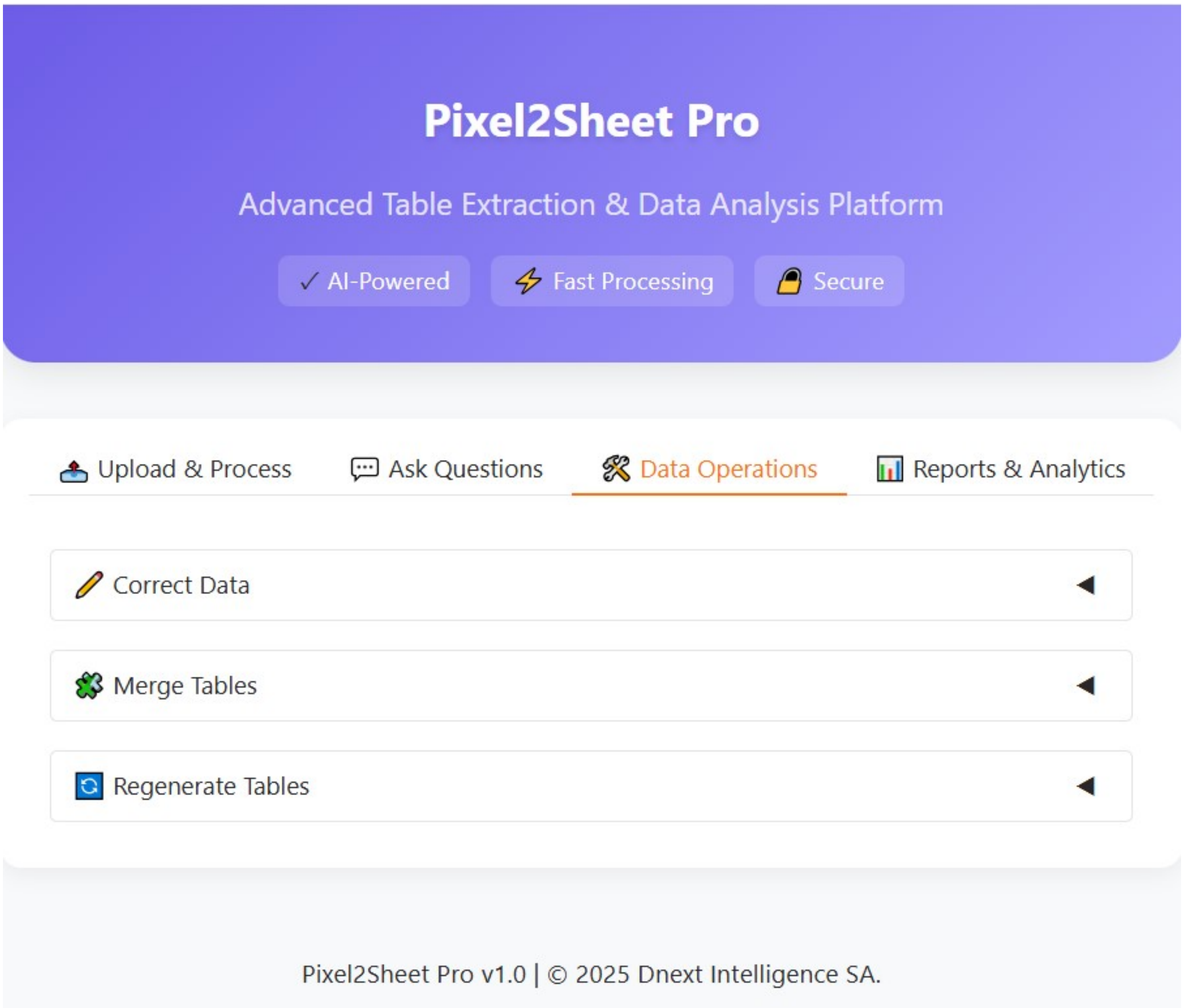


Figure 6.6: Data Operations Interface

6.2.5 Correct Data Interface

The interface is for the **Correct Data** operation, where users can:

- Enters the tables number to correct.
- Provide specific correction instructions (e.g., *Convert all dates to YYYY-MM-DD format*, *Fix the currency columns*).
- Apply the necessary corrections to the data.

After the corrections are applied, the corrected files can be downloaded.

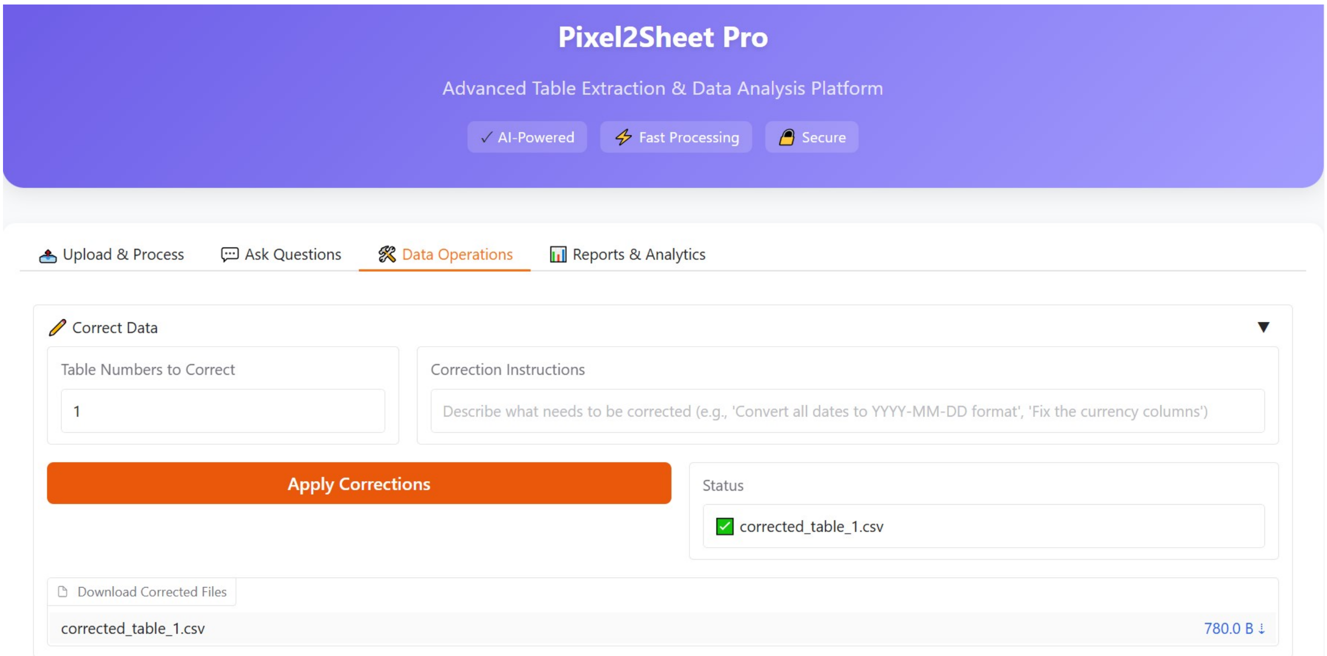


Figure 6.7: Correct Data Interface

6.2.6 Merge Tables Interface

The interface is for the **Merge Tables** operation, where users can:

- Enters the tables number to merge.
- Provide merge instructions, such as how to combine the data (e.g., *Merge by customer ID*, *Combine all sales data by date*).
- After performing the merge operation, users can download the merged table.

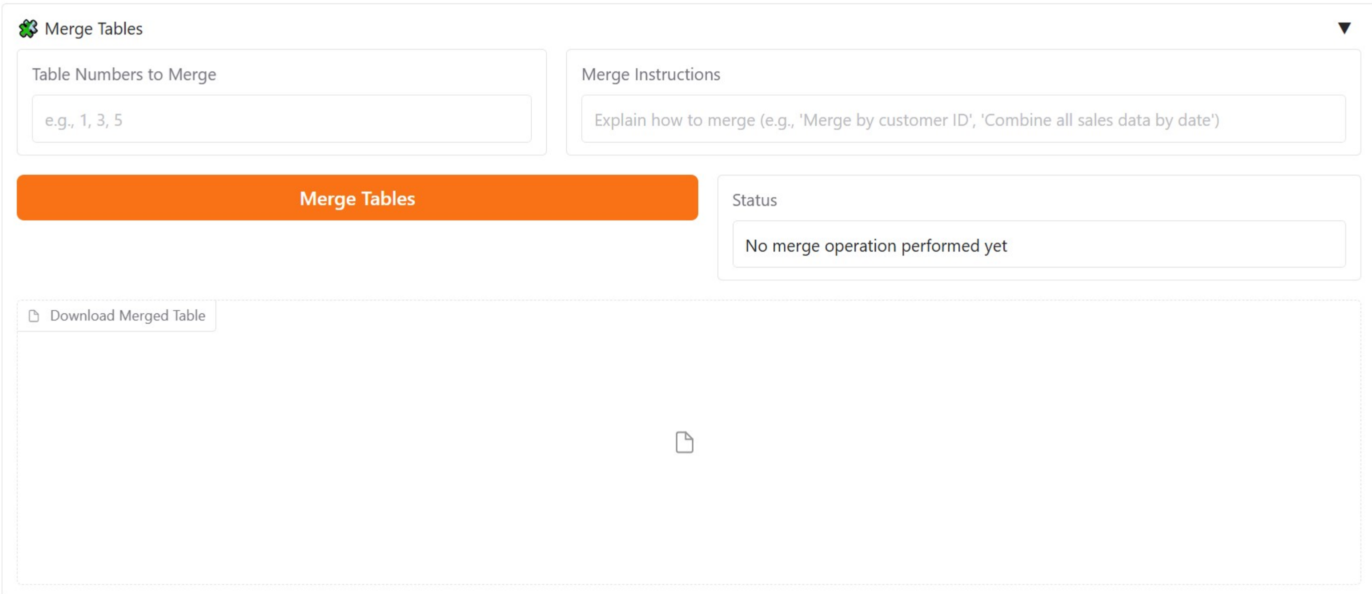


Figure 6.8: Merge Tables Interface

6.2.7 Regenerate Tables Interface

This interface enables analysts to select one of the extracted tables by specifying its number. The system then performs a reshaping operation typically by melting the table into a more appropriate format for analysis. Once the reshaping is complete, users can view the status message confirming success and download the resulting CSV file.

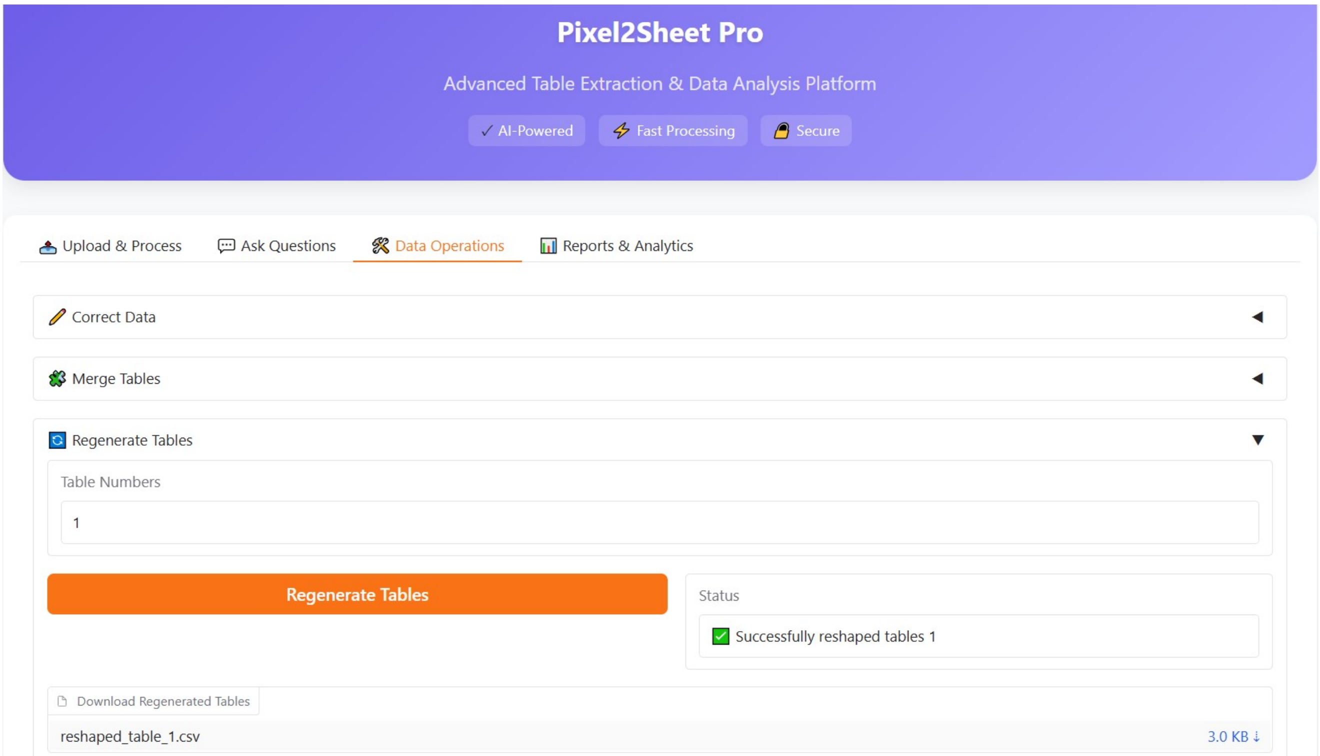


Figure 6.9: Regenerate Tables Interface

6.2.8 Reports and Analytics Interface

The interface is for **Generating Comprehensive Reports**, where users can:

- Select the table to analyze (example: *table_1*).
- View an overview of the selected table, including the number of columns and rows.
- Generate a table summary , Key insights, Data Quaity , Suggestions for Further Analysis , and visualization.
- Generate and download a full report package, which can be refreshed as needed.

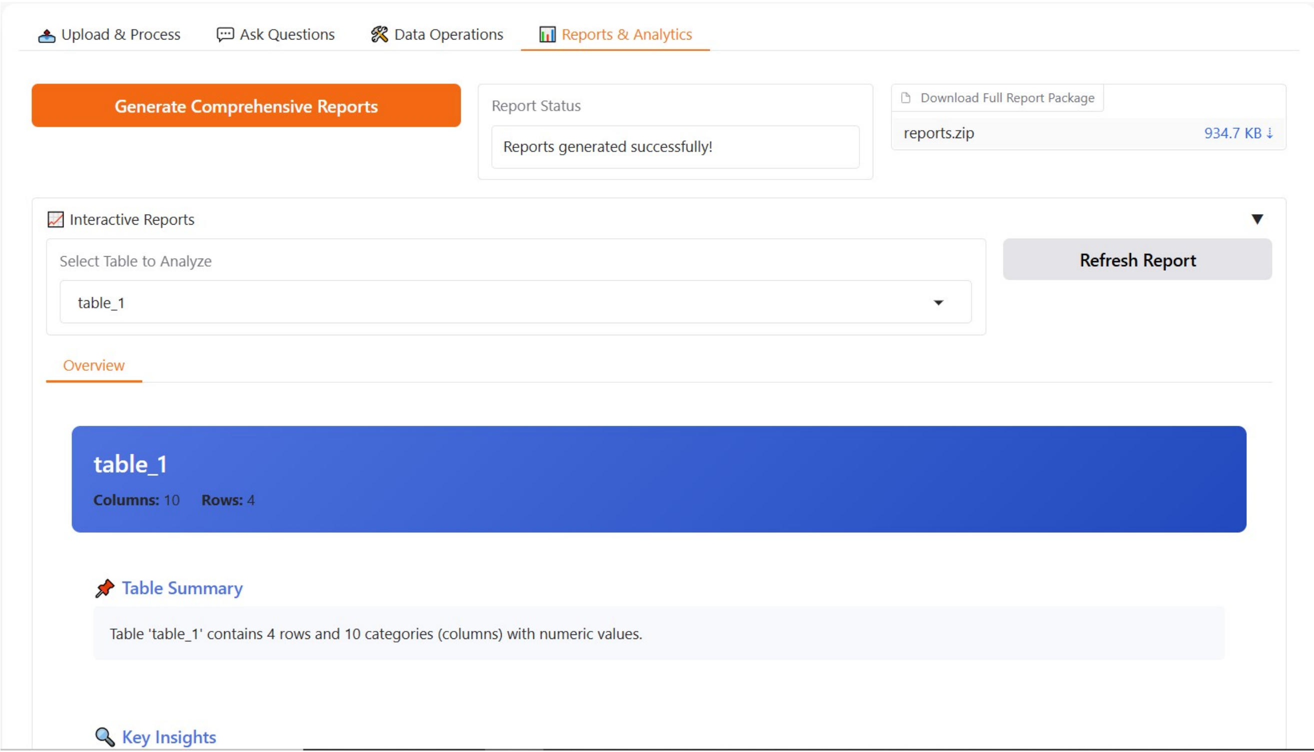


Figure 6.10: Reports and Analytics Interface

Examples of generated visualizations:

Correlation Heatmap

Figure 6.11 illustrates a correlation heatmap.A correlation heatmap visually represents the linear relationships between pairs of numerical variables using a color scale. Dark red indicates strong positive correlation (close to +1), dark blue represents strong negative correlation (close to -1), and values near 0 indicate weak or no linear correlation. This type of plot helps analysts quickly identify which variables move together and which are inversely related.

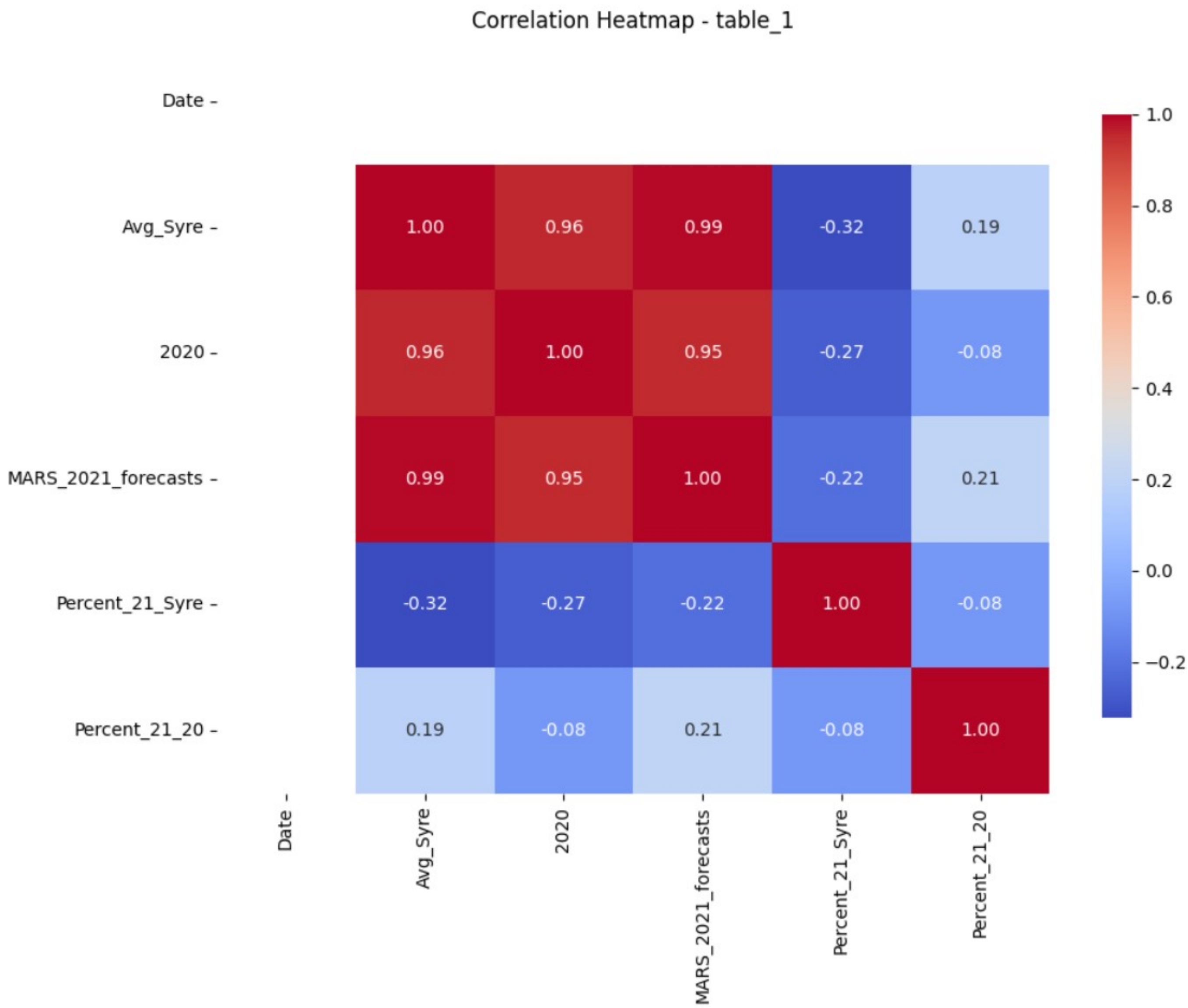


Figure 6.11: Correlation Heatmap

Pairplot of Top Categories

Figure 6.12 illustrates a pairplot of the top numerical categories. Which is a grid of scatter plots and histograms that reveals the pairwise relationships and distributions between variables. It helps analysts detect correlations, trends, clusters, and outliers. Strong linear relationships are visible as diagonal lines in scatter plots, while histograms on the diagonal show the distribution of individual features.

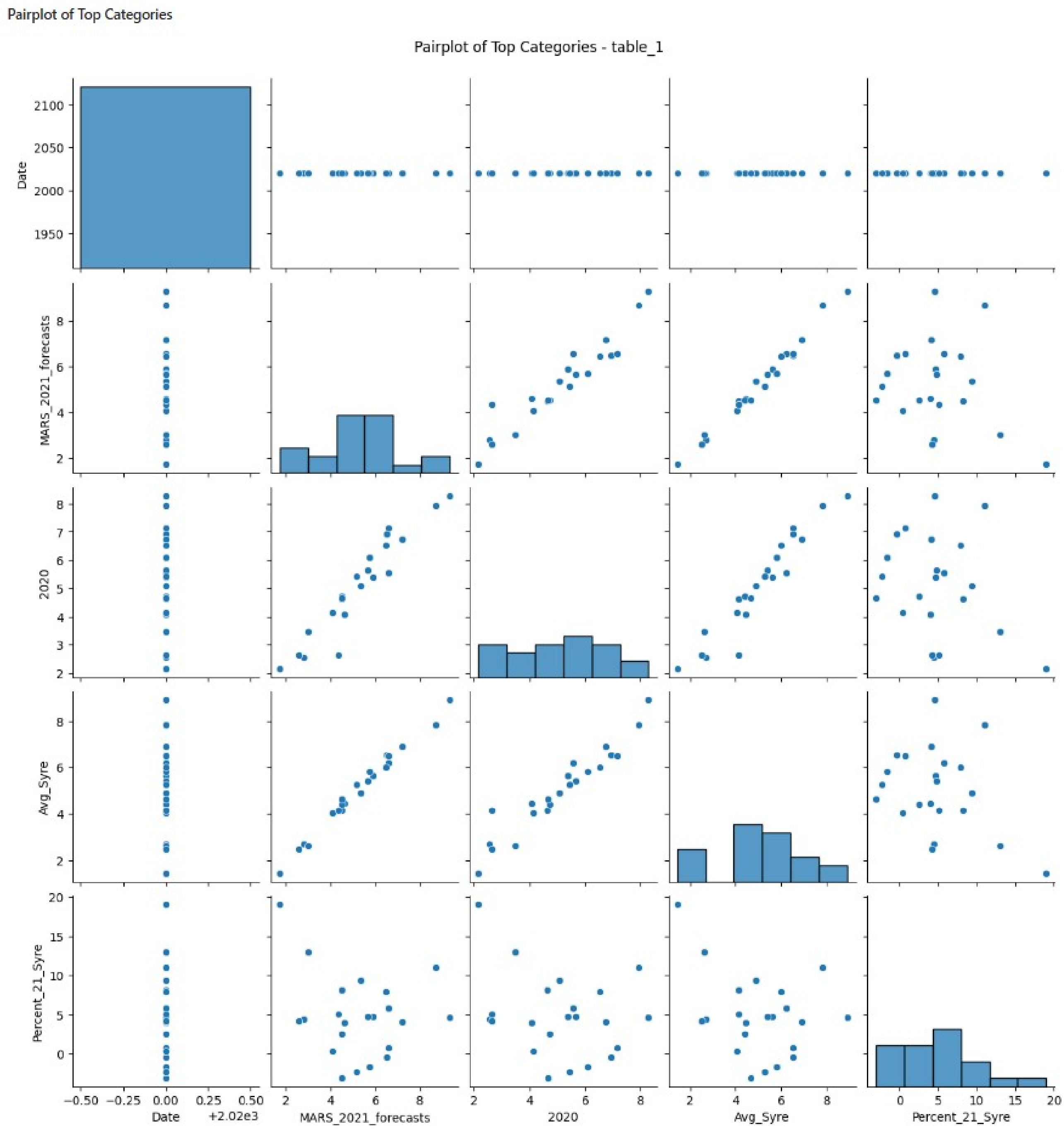


Figure 6.12: Pairplot of Top Categories

KDE and Cumulative Contribution Plots for Top Numerical Categories

Figure 6.13 includes two visualizations. The first is a Kernel Density Estimation (KDE) plot, which displays the smoothed distribution of top numerical categories. KDE plots are useful for understanding the probability density and distribution shape of variables. The overlapping curves show how each variable spreads across its range and how they compare.

The second plot shows the cumulative contribution of the same categories, sorted by total value. This type of plot helps identify how much each variable adds to the overall sum, and whether a small number of categories dominate the dataset a principle commonly used in Pareto analysis.

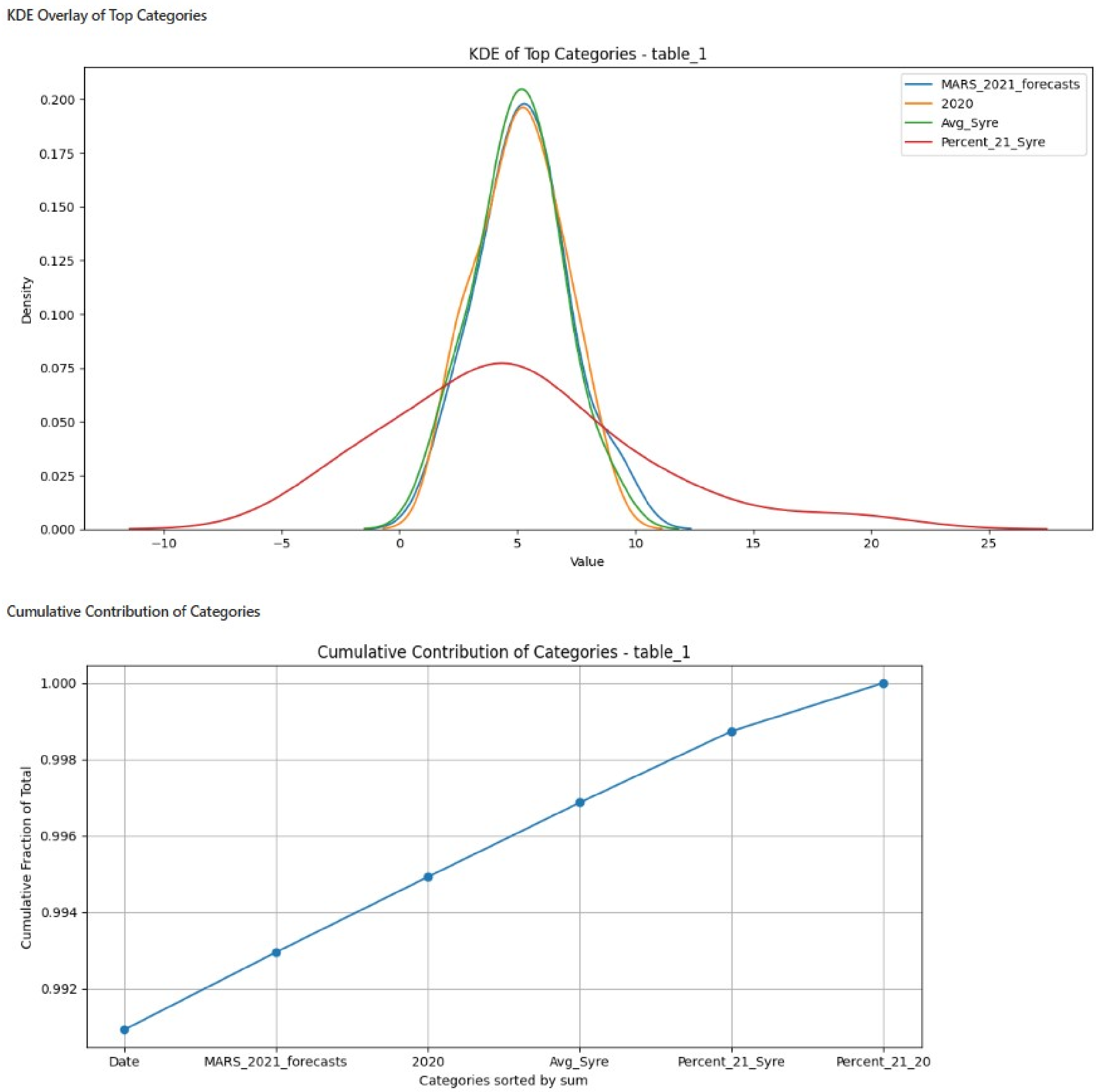


Figure 6.13: KDE and Cumulative Contribution Plots for Top Numerical Categories

Statistics:

Figure 6.9 presents a descriptive statistics table summarizing key numerical variables. It includes fundamental metrics such as count, mean, standard deviation (std), minimum, maximum, and quartiles (25%, 50%, 75%). These statistics provide a quick overview of each variable's distribution, central tendency, and spread, helping analysts assess data quality, detect outliers, and prepare for further analysis.

Statistics

Statistic	Date	Avg_Syre	2020	MARS_2021_forecasts	Percent_21_Syre	Percent_21_20
count	28.0	22.0	22.0	22.0	22.0	22.0
mean	2020.0	5.041818181818182	5.111363636363637	5.259090909090909	4.845454545454546	3.2681818181818185
std	0.0	1.8025712755570038	1.7517319197339296	1.8816176148423056	5.220356425774126	16.48232491586168
min	2020.0	1.44	2.16	1.72	-3.0	-21.0
25%	2020.0	4.1325	4.0875	4.385	1.225	-5.0
50%	2020.0	5.085	5.23	5.255000000000001	4.5	-1.45
75%	2020.0	6.1575	6.4275	6.5	7.375	9.55
max	2020.0	8.9	8.28	9.31	19.0	64.0

Figure 6.14: Statistics Table

Conclusion

In this chapter, we discussed the technologies, tools, and deployment architecture used in our solution. The system's design and user interfaces were presented, highlighting the seamless interaction between components to deliver an efficient and user-friendly experience.

General Conclusion

This report has outlined the development of *Pixel2Sheet Pro*, an AI-powered platform designed to streamline the extraction, structuring, and analysis of unstructured market data, such as that found in PDFs, images, and scanned documents. The core objective was to create an efficient and accurate solution for converting complex tables into structured, machine-readable formats like CSV, while offering an interactive, intuitive interface for data manipulation and analysis. This system aims to automate the extraction and processing of trading data, supporting faster, more informed decision-making in fast-paced trading environments.

The project began with an exploration of existing solutions and the identification of their limitations. We identified significant challenges in handling unstructured data, such as the complexity of document layouts, the high volume of data to be processed, and the risk of human errors during manual transcription. The solution, *Pixel2Sheet Pro*, leverages a modular architecture and advanced AI techniques using **YOLOv12** for table detection, **PaddleOCR** for text recognition, and **Mistral-small** VLM for validation, correction, and translation. The system ensures high accuracy, speed, and efficiency in converting images and PDFs into structured data.

The results demonstrate the system's ability to accurately detect tables in a variety of document formats, extract relevant data, and convert it into usable, editable CSV files. Additionally, the interactive querying and data manipulation features allow analysts to ask questions about the data and perform operations like merging tables, generating reports, and conducting analyses all through natural language commands due to our interactive Chatbot interfaces. These capabilities significantly reduce the time and effort required for manual data handling, enabling analysts to focus on higher value tasks such as strategy development and market trend analysis.

However, despite its success in the current implementation, there are several opportunities for further research and development to extend the system's capabilities.

Future work could include:

- **Expanding Data Formats:** The system will evolve to handle not just tables, but a wider range of data formats, including data embedded in emails, charts, and graphs. This will further enhance our solution versatility and allow analysts to extract and analyze data from diverse sources.
- **Cloud Solutions for Resource Optimization:** Given the resource-intensive nature of AI-powered systems, future versions of *Pixel2Sheet Pro* will integrate cloud solutions to handle scaling and computational requirements more effectively. This will ensure that the system remains accessible and operational even in resource-constrained environments.
- **Multilingual and Cross-Domain Data Handling:** The system will be expanded to better handle multilingual data, enabling analysts to interact with documents in various languages. Additionally, cross-domain support will allow the system to adapt to different industries and datasets, ensuring broad applicability across sectors.
- **Real-time Performance and User Feedback Integration:** Continuous optimization will be pursued to enhance the real-time performance of the system, especially in high-volume trading scenarios. Integrating user feedback and conducting advanced error analysis will help refine the system's accuracy and adapt to specific user needs and document formats.

In conclusion, this report has presented a foundational AI-powered solution to automate the extraction, structuring, and analysis of unstructured data in the trading industry. By extending the platform's capabilities to handle various data formats and integrating scalable cloud solutions, the system will continue to evolve, providing trading analysts with powerful, real-time tools to enhance their decision-making processes. Through ongoing development and innovation, *Pixel2Sheet Pro* aims to redefine the standard for intelligent data analysis automation, making it an indispensable tool for the future of trading.

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Annex

1. Tested Deep Learning Methods for Detection

YOLOv4

YOLOv4 (2020) , significantly improved over its predecessors in both speed and detection accuracy. It follows a three part architecture:

- **Backbone (CSPDarknet53)**: Extracts visual features from input images using a 53-layer convolutional network enhanced with Cross Stage Partial connections
- **Neck (SPP + PANet)**: The neck enhances and combines features from different scales of the image.
  - **Spatial Pyramid Pooling (SPP)**: Increases the receptive field of the model, enabling it to capture both small and large objects effectively.
  - **Path Aggregation Network (PANet)**: Improves the flow of features between different layers of the network, which enhances the quality of the predictions.
- **Head** : Produces final object detections, including bounding box coordinates, confidence scores, and class labels.

YOLOv4 served as our initial benchmark, offering robust performance in detecting complex table layouts, especially in low-quality or noisy scanned documents.

YOLOv8

YOLOv8 (2023) [13], developed by Ultralytics, introduced several important advancements that significantly improved the accuracy and efficiency of object detection tasks. These innovations include the use of C2f modules, task-specific detection heads, and anchor-free detection. The key components of YOLOv8 are:

- **C2f Modules**: The introduction of C2f modules enhanced feature fusion and gradient flow, allowing the model to combine features from different layers more effectively.

- **Task Specific Detection Heads**: YOLOv8 includes specialized detection heads tailored to specific tasks. This allows the model to perform more efficiently across a variety of detection tasks, including complex object detection and semantic segmentation, by optimizing the head for each particular task.

- **Anchor-Free Detection**: By eliminating the need for predefined anchor boxes, YOLOv8 simplifies the detection process. This change leads to a more flexible model that can detect objects without being constrained by predefined box sizes, improving its generalization and adaptability.

These enhancements make YOLOv8 more adaptable and efficient, especially in detecting objects with complex structures, such as tables in scanned documents. YOLOv8's improved architecture ensures faster processing and better accuracy across a wide range of detection tasks.

YOLOv11

YOLOv11 (2024) [13] Represents a significant step forward in the YOLO series, focusing on the C3K2 module, feature aggregation, and optimized training pipelines. These innovations improve both feature extraction and training efficiency. Key components of YOLOv11 include:

- **C3K2 Module**: This module enhances feature extraction by introducing a more effective architecture for processing image data, improving the model's ability to detect smaller and more complex objects.

- **Feature Aggregation**: The model integrates multi-scale features more effectively, improving its ability to handle objects of various sizes and orientations. This allows YOLOv11 to perform well in challenging environments, such as low-light or cluttered scenes.

- **Optimized Training Pipelines**: YOLOv11 reduces training time with more efficient training processes, enabling faster deployment while maintaining high performance across different tasks.

2. OCR Tool Evaluation and Selection

During the research phase, several OCR tools and frameworks were evaluated to determine the most effective solution for extracting structured tabular data. After testing various options, we ultimately opted for **PaddleOCR** due to its superior performance in detecting and recognizing text within tables, especially in cases involving curved, rotated, or low-resolution text regions. PaddleOCR's modular architecture, integration of state-of-the-art models, and high accuracy in handling complex layouts made it the most effective choice for our structured data extraction needs.

However, we also explored other OCR tools to ensure a comprehensive evaluation:

- **Tesseract OCR** [33]: An open-source OCR engine developed by Google. While Tesseract provides reasonable performance on clean, printed documents, it struggled with layout understanding and complex table structures, particularly in scanned or noisy documents.
- **EasyOCR** [31]: A deep learning-based OCR tool that supports multiple languages and various text orientations. EasyOCR demonstrated better performance than Tesseract for irregular layouts but had limitations in text region accuracy and structured extraction.

Despite promising results with other OCR tools, none of them matched the accuracy and robustness of PaddleOCR in extracting structured tabular data, especially under challenging conditions like noise, low resolution, and complex layouts.

Detection Performance

This figure 6.15 shows the results of data detection and recognition using EasyOCR. While the model performs adequately in many scenarios, it struggles with detecting certain data, particularly in complex or degraded images. In such cases, some text is missed or misinterpreted due to issues like low image quality, distortions, or unusual fonts.

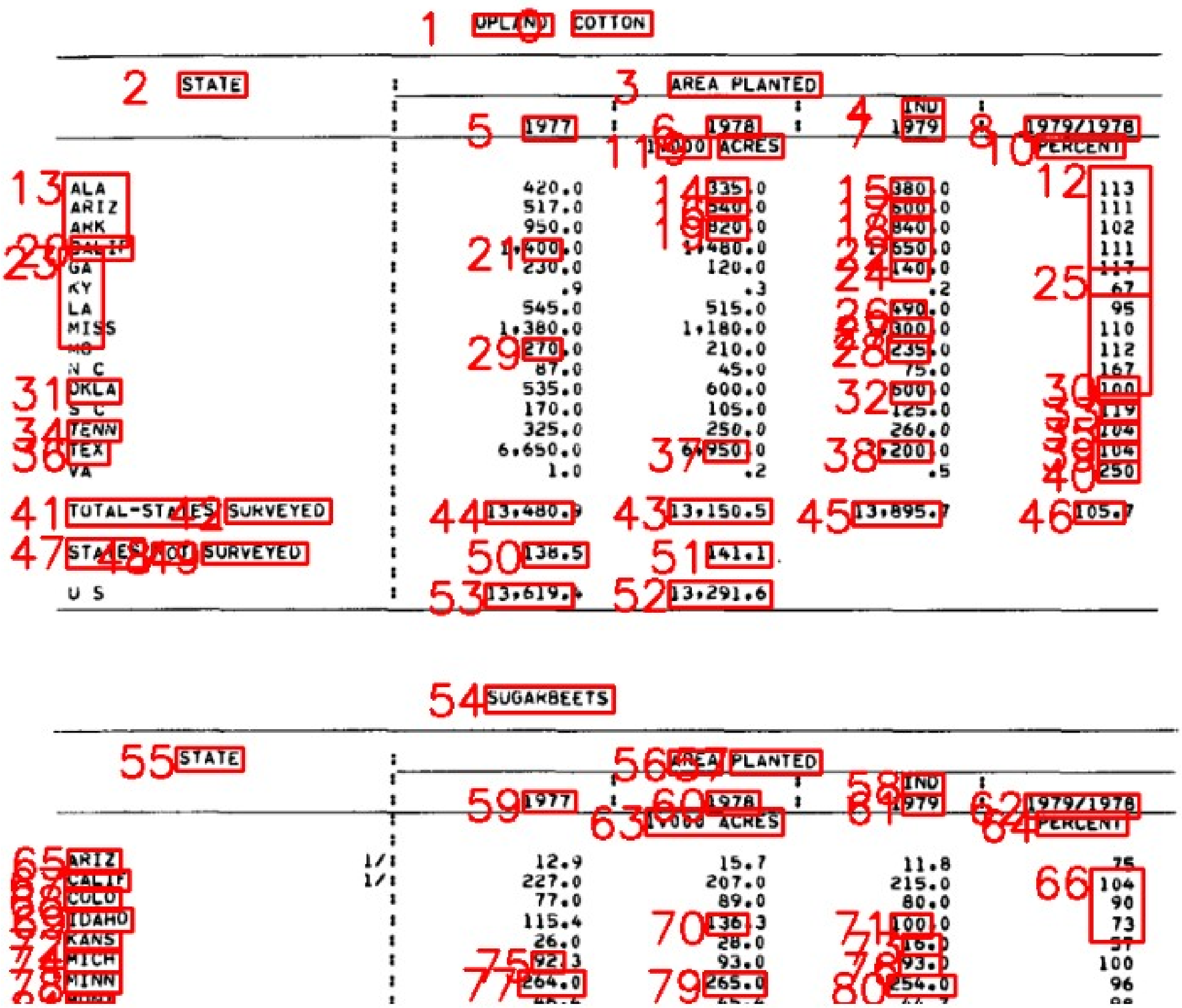
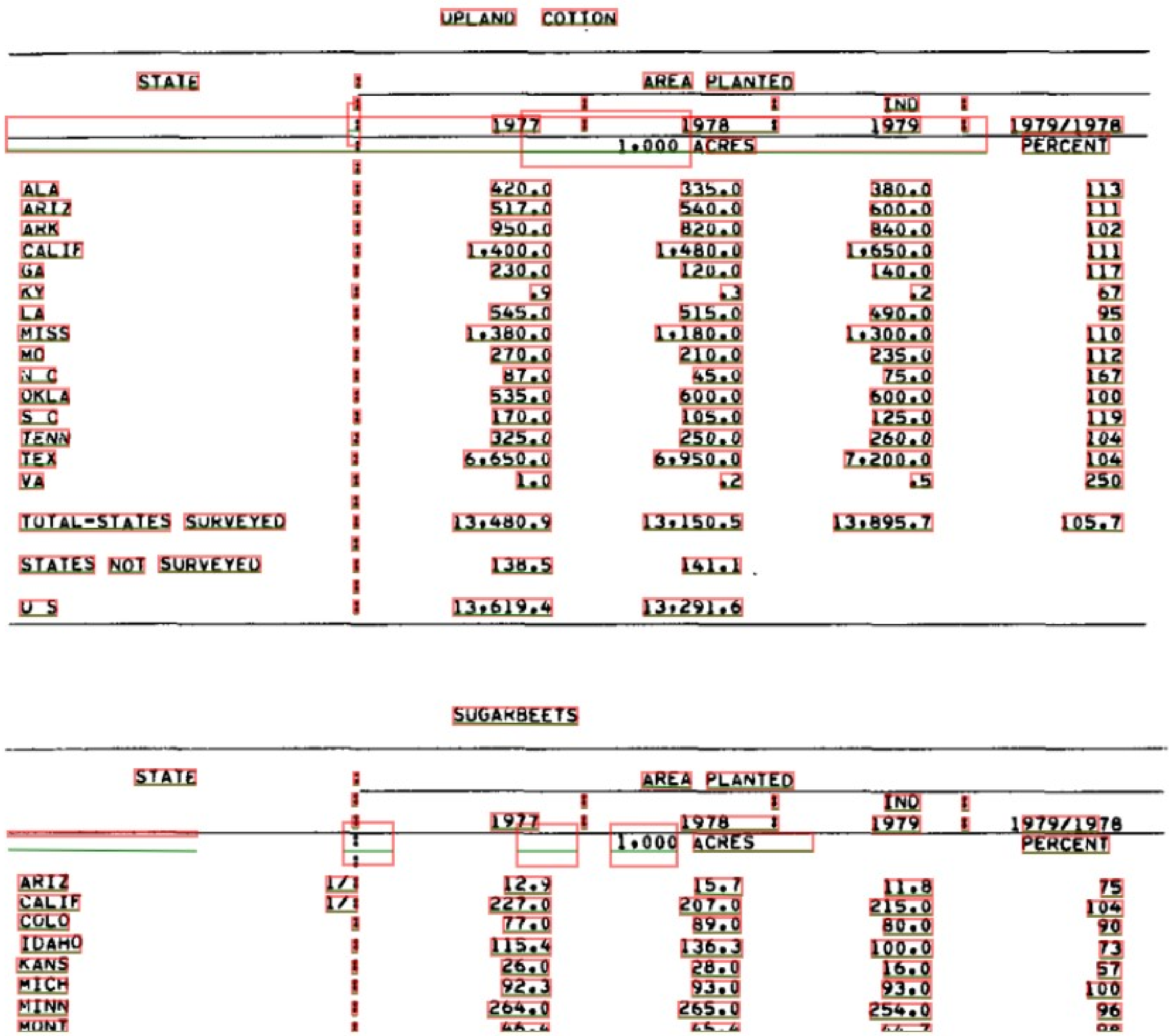


Figure 6.15: EasyOCR Detection results

This figure 6.16 shows the results of data detection and recognition using Tesseract OCR. While the model provides good detection results, it struggles with accurate recognition, especially in cases involving complex layouts or degraded image quality.



UPLAND COTTON

STATE	AREA PLANTED 1977	1978	IND 1979	1979/1978 PERCENT
	1,000 ACRES			
ALA	420.0	335.0	380.0	113
ARIZ	517.0	540.0	600.0	111
ARK	950.0	820.0	840.0	102
CALIF	1,400.0	1,480.0	1,650.0	111
GA	230.0	120.0	140.0	117
KY	.9	.3	.2	67
LA	545.0	515.0	490.0	95
MISS	1,380.0	1,180.0	1,300.0	110
MO	270.0	210.0	235.0	112
N C	87.0	45.0	75.0	167
OKLA	535.0	600.0	600.0	100
S C	170.0	105.0	125.0	119
TENN	325.0	250.0	260.0	104
TEX	6,650.0	6,950.0	7,200.0	104
VA	1.0	.2	.5	250
TOTAL-STATES SURVEYED	13,480.9	13,150.5	13,895.7	105.7
STATES NOT SURVEYED	138.5	141.1		
U S	13,619.4	13,291.6		

SUGARBEETS

STATE	AREA PLANTED 1977	1978	IND 1979	1979/1978 PERCENT
	1,000 ACRES			
ARIZ 1/	12.9	15.7	11.8	75
CALIF 1/	227.0	207.0	215.0	104
COLO	77.0	89.0	80.0	90
IDAHO	115.4	136.3	100.0	73
KANS	26.0	28.0	16.0	57
MICH	92.3	93.0	93.0	100
MINN	264.0	265.0	254.0	96
MONT	[illegible]	[illegible]	[illegible]	[illegible]

Figure 6.16: Tesseract OCR Detection results

3. Mistral Role and Prompt Strategy

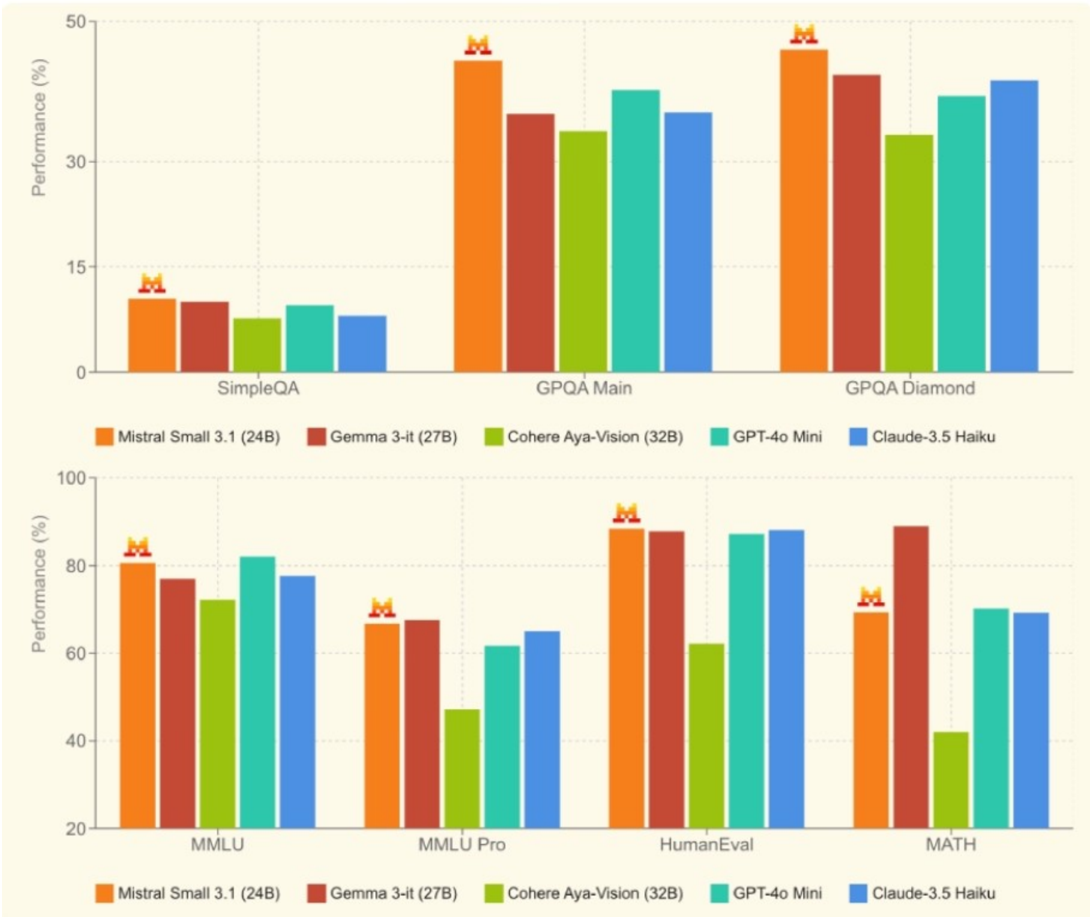


Figure 6.17: Performance comparison of various models across multiple benchmarks.

We integrated **Mistral Small**, an efficient and high performing open-weight language model, for the tasks of reviewing, validating, translating, and correcting table outputs. The selection of Mistral was driven by its excellent balance between reasoning capabilities, multilingual understanding, and computational efficiency, making it an ideal candidate for real-time, production grade systems.

As illustrated in the performance comparison (Figure 6.17) adapted from [34], Mistral Small 3.1 demonstrates superior performance across various benchmarks, including **GPQA Diamond**, **MMLU**, and **HumanEval**. In these metrics, it consistently outperforms other models, such as **Gemma 3-it (27B)** and **Cohere Aya-Vision (32B)**.

The combination of its strong performance and efficient use of computational resources made Mistral Small 3.1 the ideal choice for the tasks at hand. Its ability to handle multilingual understanding and complex reasoning tasks ensures its versatility and effectiveness in real-world applications, particularly when operating in real-time production environments.

Consequently, Mistral Small 3.1 was selected for its ability to deliver high-quality output while maintaining operational efficiency.

Prompt Engineering

Prompt engineering is the technique of crafting well-structured, strategic inputs (prompts) to guide a language model's behavior. Rather than relying on generic queries, we create tailored instructions that leverage the model's reasoning, formatting, and domain-specific capabilities.

In our context, prompt engineering ensures that the language model understands:

- The structure of the desired table outputs.
- How to detect and fix inconsistencies or missing data.
- When to translate content, validate semantic correctness, or infer missing entries.

To fully unlock its capabilities, we implemented a robust prompt engineering strategy:

1. **Contextual Initialization:** Introduce the task clearly (e.g :"You are reviewing a table extracted via OCR and preparing it in valid table format.").
2. **Instructional Prompts:** Provide explicit instructions on how to correct misread values, standardize formatting, handle translations, and output clean, structured data.
3. **Validation Layer:** Ask the model to double-check its output and flag potential inconsistencies or uncertainties in the data.

A key innovation in our system is how these prompts are sourced and managed in a real-world environment. **Analysts using the chatbot in production actively contribute to building a dynamic prompt base**, providing tailored prompts for specific table formats and domains. This evolving repository of proven, context-specific prompts ensures continuous improvement and adaptability.

For users who do not have specific formatting requirements or domain constraints, the system also offers a set of **standard, pre engineered prompts** that cover the most common table structures. This ensures that even without custom inputs, the chatbot can deliver reliable, high-quality results out of the box.

This dual-mode prompt strategy combining domain-specific intelligence with general-purpose robustness makes our system uniquely flexible and powerful in real-world applications.

4. Fine-tuning

The fine-tuning process adapted both **QWEN** and **LLAMA** vision models for table extraction using a custom dataset. The dataset included images of tables and their text-based annotations, detailing headers, rows, columns, and structural relationships. This allowed the models to learn complex table structures and improve extraction accuracy. The following sections focus specifically on the fine-tuning and performance of the **QWEN** model in this task.

QWEN Finetuning

For **QWEN**, the fine-tuning process involved training the model on our custom dataset. The primary goal was to enable QWEN to learn how to extract tables in our required format, accurately identifying and structuring the table elements during the extraction process.

The following section provides details about the training setup, including key hyper-parameters, the loss progression during training, and the evolution of **BLEU** (Bilingual Evaluation Understudy) and **ROUGE** (Recall-Oriented Understudy for Gisting Evaluation) scores **(see section 6.2.8)**, which offer further insight into the model's learning behavior and are widely used metrics for evaluating text generation and translation tasks.

Fine tuning Metrics

Loss Functions and Training Loss

In machine learning, loss functions are fundamental tools used to evaluate how well a model's predictions align with the actual target values.

Training loss specifically refers to the error calculated on the training dataset. It reflects how well the model is learning during training.The training loss is given by the following formula:

$$\mathcal{L}_{\text{train}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M} y_{ij}\log(\hat{y}_{ij}) \tag{6.1}$$

where N is the number of training examples, M is the number of classes, y_{ij} is the true label indicator, and $\hat{y}_{ij}$ is the predicted probability for class j of example i.

BiLingual Evaluation Understudy The BLEU score [28] is a widely used metric for assessing the quality of machine-translated text by comparing it to one or more human-generated reference texts. It ranges from 0 to 1, with higher scores indicating closer alignment with the reference.

BLEU is calculated as:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{6.2}$$

where w_n represents the weight of n-gram precision (typically $\frac{1}{N}$), and p_n is the n-gram precision. BP is the brevity penalty, penalizing overly short translations:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ \exp\left(1 - \frac{r}{c}\right) & \text{if } c \leq r \end{cases} \tag{6.3}$$

where c is the length of the candidate translation and r is the length of the closest reference. In our context, BLEU is used to evaluate how accurately the model translates tabular data containing non-English content by comparing its output to our reference base.

Recall-Oriented Understudy for Gisting Evaluation The ROUGE score [29] is a set of metrics for evaluating automatic text generation and machine translation, with a strong focus on recall. It assesses the similarity between generated and reference texts by measuring overlapping units such as n-grams, subsequences, and word sequences.

Key ROUGE metrics include:

- **ROUGE-N**: Measures the recall of n-gram overlaps between the candidate and reference texts. It is defined as:

$$\text{ROUGE-N} = \frac{\sum_{S \in \{\text{Reference}\}} \sum_{gram_n \in S} \text{Count}_{match}(gram_n)}{\sum_{S \in \{\text{Reference}\}} \sum_{gram_n \in S} \text{Count}(gram_n)} \tag{6.4}$$

  where $\text{Count}_{match}(gram_n)$ is the number of n-grams in the candidate that match the reference, and $\text{Count}(gram_n)$ is the total number of n-grams in the reference. This captures how much of the reference content is correctly generated.

- **ROUGE-L**: Based on the Longest Common Subsequence (LCS), this metric evaluates the sequence similarity between the candidate and reference. It is given by:

$$\text{ROUGE-L} = \frac{LCS(C, R)}{|R|} \tag{6.5}$$

where $LCS(C, R)$ is the length of the longest common subsequence between the candidate (C) and reference (R), and $|R|$ is the length of the reference. This reflects how well the overall sentence structure is preserved.

- **ROUGE-W**: A weighted version of ROUGE-L that emphasizes longer and more meaningful subsequences while penalizing shorter ones. It provides a more nuanced evaluation of sequence similarity.

Fine-tuning dataset

Characteristic	**Description**
Dataset name	`Fine-tuning`
Total number of documents	170 image-text pairs
Split	136 training / 34 test
Image format	JPG
Text format	Plain string (markdown)
Resolution	640 × 640 pixels (for images)
Source	PDF documents
Language	English and Spanish
Document age	Mix of recent and historical documents
Content type	Table images with corresponding structured text

Table 6.2: Detailed characteristics of the `Fine-tuning dataset`

Key Training Setup and Hyperparameters

The fine-tuning process emphasized memory efficiency and model generalization through the careful selection of hyperparameters and training strategies, as summarized in Table 6.3. The fine-tuning was performed using the **Unsloth framework** [35], which simplifies model training and optimizes memory usage. Unsloth is designed to simplify working with vision-language models by providing tools for efficient model loading, training, and dataset handling.

Parameter	**Value**	**Description**
Batch size (per device)	2	Small batch size to fit GPU memory constraints
Gradient accumulation steps	4	Simulates effective batch size of 8 by accumulating gradients
Maximum training steps	60	Limited steps to avoid overfitting on the small Tables dataset
Learning rate	2×10^{-4}	Moderate learning rate for stable convergence
Optimizer	AdamW (8-bit precision)	Efficient optimizer with reduced memory footprint
Mixed precision training	FP16 enabled	Uses 16-bit floating point format to reduce memory and speed up training
BF16 training	Enabled	Uses Brain Floating Point 16-bit format optimized for deep learning hardware
Weight decay	0.01	Regularization to reduce overfitting
Learning rate scheduler	Linear decay	Gradual reduction of learning rate across training steps
Warmup steps	5	Stabilizes initial training phase by gradually increasing LR
Maximum sequence length	2048 tokens	Supports long conversational context with image + instruction
Seed	3407	Ensures reproducibility
Logging frequency	Every 5 steps	Allows monitoring training progress frequently

Table 6.3: Hyperparameters and Training Configuration for Fine-Tuning Qwen2.5-VL

Due to GPU memory constraints, a small batch size of **2** per device was used, with gradient accumulation over **4** steps to simulate an effective batch size of **8**, enabling stable weight updates. The model was trained using the AdamW optimizer with **8-bit** precision and a learning rate of 2×10^{-4} for efficient convergence. Mixed precision **(FP16)** and **BF16** training were enabled to optimize speed and memory usage on compatible hardware.

To prevent overfitting with the small dataset and limited training steps **(50 steps)**, a weight decay of **0.01** was applied. A warmup phase of **5 steps** stabilized the learning rate early in training. The maximum sequence length was set to **2048 tokens** to accommodate complex table data and long instructions. Logging was performed every **5 steps** for close monitoring of training progress.

To evaluate model behavior over time, training and validation metrics were plotted. Figure 6.18 illustrates the model's convergence and stability throughout the training process.

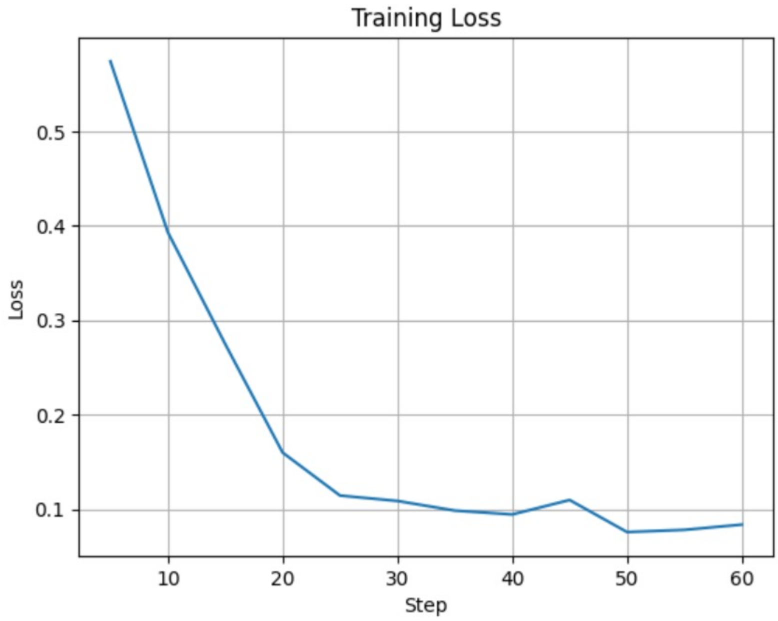


Figure 6.18: Training Loss vs. Step

The training loss curve in Figure 6.18 shows a clear downward trend, indicating successful learning over time. Initially, the loss decreases sharply, reflecting rapid learning in the early training steps. As training progresses beyond **step 20**, the rate of improvement slows, and the loss converges to a lower bound around **0.1**. Minor fluctuations after **step 40** suggest that the model has reached a plateau and is making only marginal gains. Overall, the loss curve indicates effective training with no signs of overfitting or instability.

Before Fine-Tuning

Before fine-tuning, the BLEU score originally developed for evaluating machine translation was used to assess the quality of the model's output. In our case, this choice is justified because the input tables are often in languages other than English, and the model is expected to translate and generate their English equivalents. Therefore, BLEU provides a reasonable measure of how closely the generated text matches the reference English version. The model achieved a BLEU score of **0.18**, as shown in Figure 6.19, which reflects its initial difficulty in accurately translating and generating structured outputs from multilingual inputs.

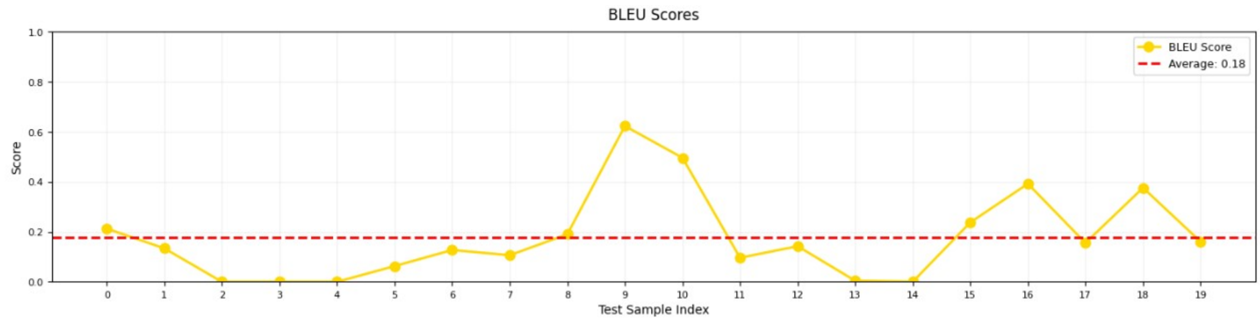


Figure 6.19: BLEU score evaluation

Furthermore, the ROUGE score was calculated to provide additional insights into the model's performance. The results shown in Figure 6.20 show that ROUGE-1 F-Measure has a score of **0.36**, ROUGE-2 F-Measure has a score of **0.19**, and ROUGE-L F-Measure has a score of **0.26**. These ROUGE scores reflect the model's difficulties in producing text that closely aligns with the reference outputs.

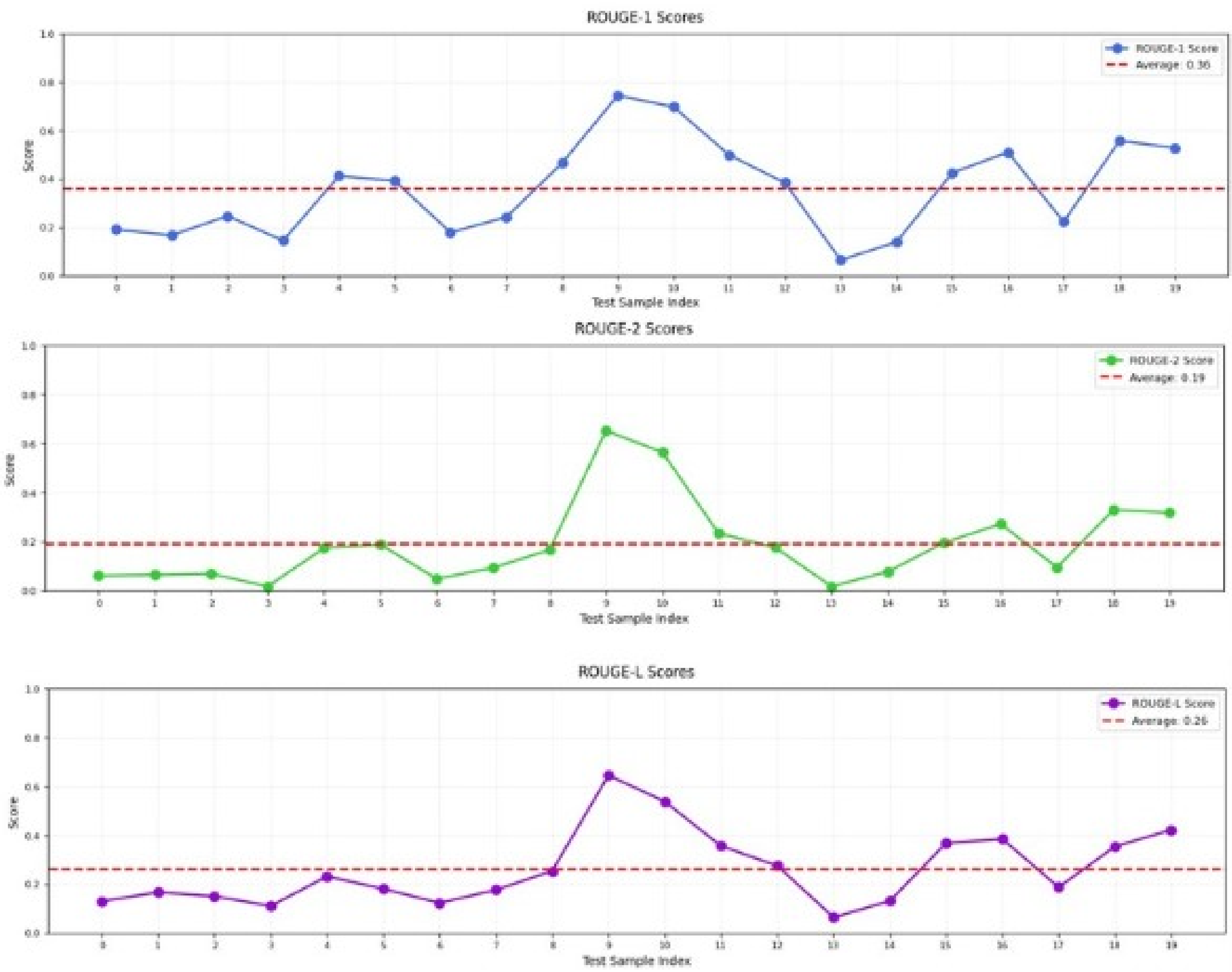


Figure 6.20: ROUGE score evaluation

After fine-tuning

After fine-tuning the model, we observed only a slight improvement in performance, but not to the extent we expected ,indicating that the fine-tuning process did not lead to the anticipated increase in accuracy. This suggests that despite the adjustments made, the model still struggles to accurately extract and generate tables following the natural language prompts.

Additionally, the BLEU score, which evaluates the quality of text generation, showed a small increase to **0.25**, as shown in Figure 6.21. While this improvement suggests some positive change, it was not as significant as we hoped, highlighting that the model still has substantial room for improvement in generating precise and meaningful tables.

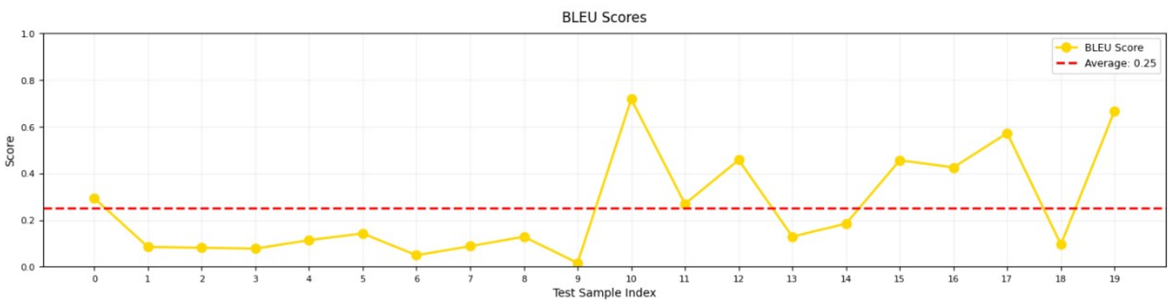


Figure 6.21: BLEU score evaluation

Additionally, the ROUGE score was assessed to provide further insights into the model's performance. The results showed a ROUGE-1 F-Measure score of **0.42**, a ROUGE-2 F-Measure score of **0.22**, and a ROUGE-L F-Measure score of **0.34**, as shown in Figure 4.13. These scores indicate a slight improvement in the model's ability to produce text that aligns with the reference outputs, though the improvements were not as significant as anticipated, highlighting that the model still requires further fine-tuning or larger dataset to enhance its performance in generating more accurate and meaningful outputs.

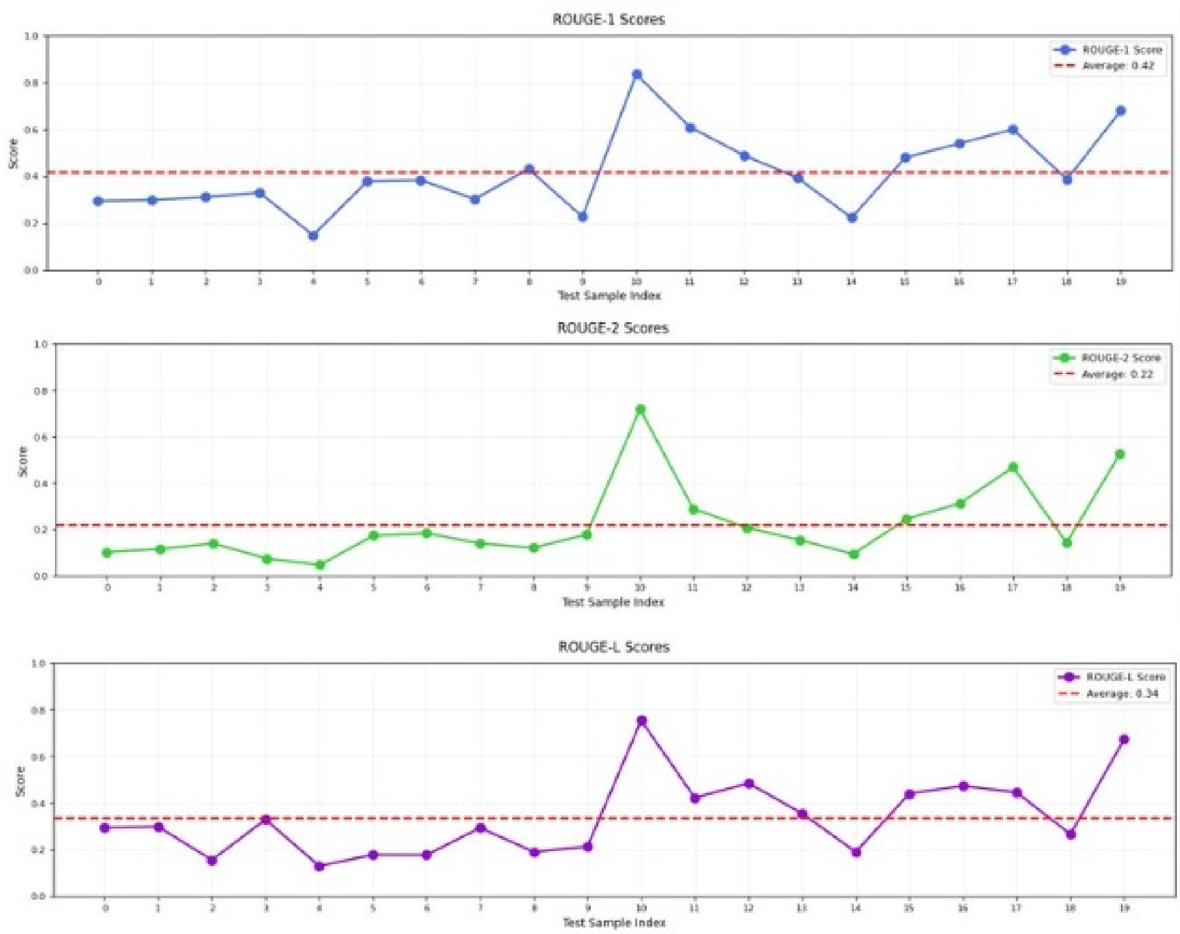


Figure 6.22: ROUGE score evaluation

Comparative Analysis

To clearly demonstrate the improvements achieved through fine-tuning, we present a comparative analysis of the model's performance before and after fine-tuning using key evaluation metrics. Table 6.4 summarizes the main results, highlighting gains in accuracy and text quality.

Metric	Before Fine-Tuning	After Fine-Tuning
BLEU Score	0.18	**0.25**
ROUGE-1 F-Measure	0.36	**0.42**
ROUGE-2 F-Measure	0.19	**0.22**
ROUGE-L F-Measure	0.26	**0.34**

Table 6.4: Fine-tuning results

The comparative analysis highlights the modest improvements in the model's performance after fine-tuning. There are noticeable increases in the BLEU score and ROUGE score. The BLEU score improved from **0.18** to **0.25**, and the ROUGE-1, ROUGE-2, and ROUGE-L F-Measure scores increased to **0.42**, **0.22**, and **0.34**, respectively. This indicates that the fine-tuned model is slightly more accurate and reliable in extracting markdown tables following natural language prompts, though it still does not achieve the desired level of performance.